# Libertinus and diacritics

## To do

Audit mark anchors to fit roman and italic anchor models; patch as necessary.

Add alt cap dot above for italic and semibold italic.

Revise inherited anchors for letters with bar, hook, or stroke, when reasonable, checking for similar base letterform.

A side finding about the patched fonts. After Ç, the patched fonts render the plain acute (acutecomb), not acute.cap as the originals do. So the rebuilt ccmp seems to have dropped Ç, and possibly other glyphs, from the capital list. I haven't checked how many.

A side finding: semibold U+0351 and U+0357 have mark-anchor X = +100, while regular has −171. That looks wrong, and may come from the copies the original font has in U+0358–U+0363.

## Anchor model

Libertinus has an anchor model where the base anchors are set to locations where the marks would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above the base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark's below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 1 by superimposing their below-anchors.

Every font has five vertical reference lines (with respective $Y$ coordinate values) – ascender line, capline, x-height or meanline, baseline, descender line – within which each glyph lives. These vertical reference lines are the gray vertical lines in Figures 1, 2, 3 and 4. Each glyph has a (tight) bounding box for real minimum and maximum $X$ and $Y$ coordinate values, as well as a so-called ideal bounding box where the glyph seems to live, but accounting for overshoots of bowls and serifs. The entire design goal is that the top and bottom of the ideal bounding box are two vertical reference lines, except for glyphs that are non-full ascenders or non-full descenders.

By clearance, I mean the (vertical) distance between a base and a mark, measured between the ideal bounding boxes of the base and the mark. For some designed fonts, the clearance is consistent for each vertical reference line, so that all glyphs that live at that reference line are separated from all marks with the same clearance; e.g. dots should not dance below the baseline for different letters that sit on the baseline, and acutes and graves sit at about the same height for different letters with similar top vertical reference line.

In roman styles of Libertinus, the clearance is encoded in the base anchor. The $Y$ coordinate of an above-anchor is the top of the base's ideal bounding box plus the clearance of the nearest vertical reference line, and the $Y$ coordinate of a below-anchor is the bottom of the ideal bounding box minus that clearance. For most glyphs that edge of the ideal bounding box is the reference line itself. For a non-full ascender or non-full descender it is not, but the clearance is the same; see the section on vertical reference lines below. The $X$ coordinate of the anchor is at some horizontal optical center of the base glyph, but the meaning of the optical center depends on the glyph letterform, or whatever the designer thinks looks best. For a mostly symmetric letterform in the roman styles, a candidate for this $X$ coordinate is the midpoint of the bounding box of the glyph.

The anchor of a mark is at the horizontal center of the mark, if the mark is symmetric. An asymmetric mark, such as the acute or grave, has its anchor a few units to one side of the center of its outline. Vertically, the anchor of an above-mark is about 140 units above the bottom of the mark outline. That is near the top of a tall mark, such as the circumflex, and above the top of a short mark, such as the macron or dot. Because the above-anchor of a base at the x-height is 216 units above the x-height, the bottom of a tall mark is about 75 units above the base, and a short mark sits a little higher.

Other fonts use a different geometric model for anchors. In Noto, the base's below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark's below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark's below-anchor is at the origin $(0, 0)$, and the mark glyph outline is drawn below the origin. In the middle panel of Figure 2, the mark's below-anchor is the red square at the origin. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors. So in Noto the clearance is encoded in the mark anchor, and in Libertinus it is encoded in the base anchor.

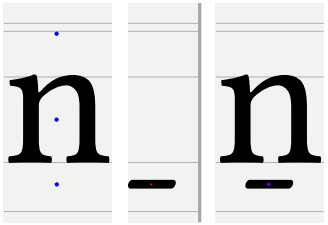


Figure 1: Libertinus n and macron with anchors.

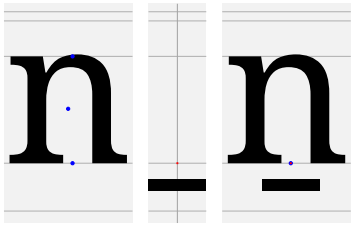


Figure 2: Noto n and macron with anchors.

### Italic anchor model

In italic, the Libertinus anchor model is similar to the roman styles, but it is applied along the italic axis. The italic lane of a base glyph is its advance-width box, slanted by the italic angle $\theta$ (12°). The midline of the italic lane is the line $X = W/2 + Y \tan\theta$, where $W$ is the advance width of the glyph. See the gray band and the dashed blue line in the left panel of Figure 3.

The $Y$ coordinate of an above- or below-anchor is set exactly as in the roman styles: the top or bottom of the glyph's ideal bounding box plus or minus the clearance of the nearest vertical reference line, measured vertically. So marks sit at the same height in roman and italic. The $X$ coordinate is the point of the midline at that height: $ax = W/2 + ay \tan\theta$ and $bx = W/2 + by \tan\theta$.

The height in this formula is the height of the anchor itself, that is, the edge of the ideal bounding box plus or minus the clearance, not the height of the reference line or of the ideal bounding box. So the $Y$ coordinate of the anchor must be determined before the $X$ coordinate can be calculated. In italic, the above-anchor is at $W/2 + 137$ for a letter at the x-height ($ay = 645$), at $W/2 + 181$ for a capital ($ay = 850$), and at $W/2 + 189$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 23$ for a letter on the baseline ($by = -110$) and at $W/2 - 68$ for a descender ($by = -319$). An error of 10 units in the $Y$ coordinate moves the $X$ coordinate by about 2 units.

The anchor of a mark lies on the mark's own slanted axis, through the center of its outline if the mark is symmetric. See the dashed red lines in the middle panel of Figure 3. That is, the mark anchor is at $X = C + Y \tan\theta$, where $Y$ is the height of the anchor, as in the roman styles, and $C$ is where the axis of the mark meets the baseline: the horizontal center of the mark outline after the slant is removed. An asymmetric mark keeps the same offset from its axis as in the roman styles. Superimposing the anchors lays the axis of the mark on the midline of the base, as in the right panel. The mark is displaced from the base along the italic axis, not vertically above or below it.

The midline is a geometric center, not an optical center, but it is a good candidate for the anchor $X$ coordinate of a mostly symmetric letterform. Asymmetric letterforms, and letters that overhang their advance width such as f and j, need curated anchors.

### Semibold italic anchor model

Semibold italic has its own anchor model, which is not the italic anchor model, and the patch keeps it. The outlines are slanted (the italic angle $\theta$ is 11.5°), but the anchors are, almost all, the anchors of regular, copied unchanged: 42 of the 44 above- and below-mark anchors, and 319 of the 384 above- and below-anchors of bases, have exactly the coordinates of regular.

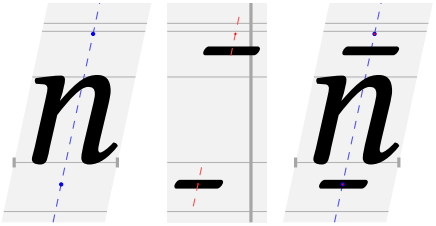


Figure 3: Libertinus italic n, macron and macron below, with anchors. The gray band is the italic lane and the dashed blue line is its midline.

So the anchors do not follow the slant. A slanted outline moves further right the higher it goes, but an anchor copied from regular stays where it was. An above-anchor therefore sits left of where the italic anchor model would put it, by roughly the amount the slant has moved the outline at that height. This is true of the base and of the mark alike: the base's above-anchor is left of the midline of its italic lane, and the mark's anchor is left of the mark's own slanted axis, by about the same amount. When the two anchors are superimposed, the two errors cancel, and the axis of the mark lands near the midline of the base, as it would in italic. See Figure 4, where the filled symbols are the designer's anchors and the hollow symbols are the positions given by the italic anchor model: the above-anchor of n and the anchor of the macron are both far left of the hollow symbols, but by nearly the same distance.

The semibold italic anchor model is therefore the italic anchor model with both anchors shifted left by the same constant $K$: the base anchor is at $X = W/2 + Y \tan\theta - K$ and the mark anchor is at $X = C + Y \tan\theta - K$, with $\tan\theta = 0.2035$ and the same $Y$ coordinates as in the other styles. Above-anchors are high, where the slant has moved the outline far to the right, so their shift is large: $K_a = 150$ (the median of the above-marks; most are between 132 and 168). Below-anchors are just under the baseline, where the slant has moved the outline a little to the left, so their shift is small and negative: $K_b = -12$ (most below-marks are between $-23$ and $-4$). The italic anchor model is the case $K = 0$.

The candidates for a base are $ax = W/2 + ay \tan\theta - 150$ and $bx = W/2 + by \tan\theta + 12$. So the above-anchor is at $W/2 - 19$ for a letter at the x-height ($ay = 645$), at $W/2 + 23$ for a capital ($ay = 850$), and at $W/2 + 31$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 10$ for a letter on the baseline ($by = -110$) and at $W/2 - 53$ for a descender ($by = -319$). For semibold italic n, $W = 606$, so $ax = 284$ and $bx = 293$; the font has $ax = 264$ and $bx = 265$, the values of regular n, and the macron lands 29 units left of the midline.

The cancellation is only approximate, because the designer's anchors do not follow the slant at all, whereas the model's anchors do, apart from the constant $K$. A mark's shift is fixed by the height of its own anchor: for a mark, $K = Y \tan\theta$ plus a constant, where $Y$ is the height of the mark anchor. But the designer's base anchors are near $W/2 - 9$ at every height, so they do not move right for a taller base, such as a capital or an ascender. The error of a mark on a base therefore grows with the difference in height of the two anchors, by 2 units for every 10. The median distance of a designer's base anchor from the candidate is about 20 units for a lowercase letter and 30 units for a capital.

The patch does not move this style to the italic anchor model. It keeps the designer's anchors where they conform to the semibold italic anchor model, and corrects only the anchors that do not, so that they conform to it: a base anchor is moved to the candidate above, and a mark anchor is moved to $X = C + Y \tan\theta - 150$ for an above-mark or $X = C + Y \tan\theta + 12$ for a below-mark. An anchor set by the italic anchor model would be wrong here: a base anchor without the shift would put every above-mark about 150 units too far right, because the marks carry the shift.

The following do not conform to the semibold italic anchor model and probably need to be patched, some of them manually.

- Capitals. The designer's above-anchors are about 25 units left of the candidate, since they are 205 units higher than the anchors of lowercase letters but no further right. The worst are B, D, I, N, O, R, W (40 to 60 units left) and L (159 units left; its anchor is over the stem, as in regular).
- The capital forms of the grave, acute, circumflex and breve (`grave.cap`, `acute.cap`, `circumflex.cap`, `breve.cap`), which the font substitutes over capitals. Their mark anchors were set with the slant ($K$ between $-13$ and 29), but the above-anchors of capitals were not, so these four marks land 160 to 190 units left of the middle of a capital. The other above-marks have no capital form in use and land 20 to 50 units left. The capital forms need mark anchors at $X = C + Y \tan\theta - 150$; `caron.cap` and `dieresis.cap` are not substituted.
- Above-anchors that were set with the slant, as in italic, so that an above-mark lands 130 to 190 units too far right: a, s, h and J.

- Other base anchors more than 40 units from the candidate. Above: c, g, Æ, Ɖ, ƕ, Ƣ, Ƥ, ɳ, ɗ, ɦ, ɟ, Ħ, ɵ, ɧ. Below: A, C, D, J, a, d, f, j, ɋ, ɾ, t, Æ, ĸ, ƕ, Ƣ, Ƥ, ɳ, ɾ, ɑ, ɖ, ɗ, ɚ, ɥ, ɯ, ɻ, Ħ, ɵ. Several of these are copies of the anchors of another glyph. The anchors of ɥ and ɥ are transposed, as in the other styles.
- Marks whose shift is far from the median, so that no base anchor can be right for them and for the other marks: dieresis U+0308 ($K = 111$, so it lands about 40 units left of the other above-marks), comma above U+0313 (125), inverted breve U+0311 (132), hook above U+0309 and double grave U+030F (168, about 18 units right), koronis U+0343 (371), right tack below U+0319 ($-30$) and tilde below U+0330 ($+10$). The double acute U+030B (156) conforms.
- Tilde U+0303, one of the two mark anchors that is not a copy of regular; its shift is 139.
- The Hebrew ligatures U+05F0, U+05F1, U+05F2, which have above- and below-anchors in this lookup table in semibold italic only.
- Letters that overhang their advance width, such as f and j, which need curated anchors in every italic style.

The anchors of precomposed glyphs were not compared with these candidates.

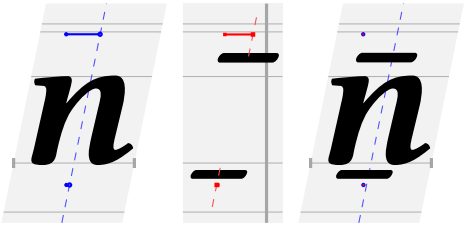


Figure 4: Libertinus semibold italic n, macron and macron below, with the designer's anchors (filled) and the positions given by the italic anchor model (hollow). The above-anchor of n is 170 units left of the midline and the anchor of the macron is 141 units left of its axis.

## Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA letters for Libertinus serif fonts (regular, italic, semibold, semibold italic) in the following tables (Tables 1–11), and the mark anchors of above- and below-marks (Tables 12–15). Recall that Language Science Press uses semibold Libertinus fonts, e.g. `LibertinusSerif-Semibold.otf`, for boldface styles.

### Table row headings

hex: Unicode character codepoint in 4-hex.

cat: Unicode character category. L = letter. Lu = uppercase (capital) letter, Ll = lowercase (small) letter, Lo = other letter. Used to determine expected ideal bounding box. M = mark. Mn = nonspacing mark.

reg, it, sb, si: regular, italic, semibold, semibold italic. In that row, the character is printed alone, with combining dot above, and with combining macron below, to show anchor positioning (via combining grapheme joiner) when the base has an anchor; otherwise HarfBuzz default positioning applies.

patch: Use the patched Libertinus font to print the character alone, with combining dot above, and with combining macron below to show anchor positioning of the patched font (via combining grapheme joiner).

ideal: top and bottom ideal bounding box reference lines, with offset from $Y_{\text{max}}$ or $Y_{\text{min}}$ for non-full descender or non-full ascender. c = capline, x = x-height, a = ascender, b = baseline, d = descender.

a: $X, Y$ coordinates of the above-anchor.

a?: $X, Y$ coordinates of the above-anchor candidate.

a$\delta$: $X, Y$ coordinate differences between above-anchor and above-anchor candidate

b: $X, Y$ coordinates of the below-anchor.

b?: $X, Y$ coordinates of the below-anchor candidate.

b$\delta$: $X, Y$ coordinate differences between below-anchor and below-anchor candidate

In the tables of marks, each column is a mark and the rows are as follows.

reg, it, sb, si, patch: the mark on n and on H (via combining grapheme joiner), in the original and patched fonts.

m: $X, Y$ coordinates of the mark anchor. A class in parentheses is an anchor class other than above (0) or below (2): semibold puts the dot above, diaeresis, ring, hook above and comma above in class 1, whose base anchors are lower (n has 550 in class 1 and 646 in class 0). This is the designer's anchor class `top_punkt`; it places the marks at the same height as class 0, but only 24 bases have it. The patched semibold merges it into class 0.

m?: $X, Y$ coordinates of the mark-anchor candidate. $Y$ is the bottom of the mark outline plus the style's median distance of an above-mark anchor above it (regular 143, italic 142, semibold 157 in class 0 and 60 in class 1, semibold italic 147), or the top of the outline minus the median distance of a below-mark anchor below it (regular 46, italic 48, semibold 54, semibold italic 54). $X$ is on the mark's own axis at that height, $C + Y \tan\theta - K$.

m$\delta$: $X, Y$ coordinate differences between mark anchor and mark-anchor candidate. An asymmetric mark, such as the acute or grave, has an $X$ difference by design.

Table 1: Font metrics for A–Z 1/2

| hex | 0041 | 0042 | 0043 | 0044 | 0045 | 0046 | 0047 | 0048 | 0049 | 004A | 004B | 004C | 004D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu |
| reg | A A̍ A̱ | B B̍ Ḇ | C C̍ C̱ | D D̍ Ḏ | E E̍ E̱ | F F̍ F̱ | G G̍ G̱ | H H̍ H̱ | I I̍ I̱ | J J̍ J̱ | K K̍ Ḵ | L L̍ Ḻ | M M̍ M̱ |
| patch | A A̍ A̱ | B B̍ Ḇ | C C̍ C̱ | D D̍ Ḏ | E E̍ E̱ | F F̍ F̱ | G G̍ G̱ | H H̍ H̱ | I I̍ I̱ | J J̍ J̱ | K K̍ Ḵ | L L̍ Ḻ | M M̍ M̱ |
| ideal | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, d+60 | c, b | c, b | c, b |
| a | 354, 850 | 267, 850 | 340, 850 | 306, 850 | 281, 850 | 284, 850 | 365, 850 | 361, 850 | 150, 850 | 182, 850 | 336, 850 | 151, 850 | 431, 850 |
| a? | 346, 850 | 280, 850 | 324, 850 | 336, 850 | 271, 850 | 244, 850 | 352, 850 | 362, 850 | 148, 850 | 120, 850 | 323, 850 | 260, 850 | 418, 850 |
| a$\delta$ | 8, 0 | -14, 0 | 16, 0 | -30, 0 | 10, 0 | 40, 0 | 14, 0 | -1, 0 | 2, 0 | 62, 0 | 13, 0 | -110, 0 | 13, 0 |
| b | 308, -110 | 268, -110 | 363, -110 | 309, -110 | 287, -110 | 209, -110 | 367, -110 | 362, -110 | 147, -110 | 109, -319 | 342, -110 | 262, -110 | 393, -110 |
| b? | 346, -110 | 280, -110 | 324, -110 | 336, -110 | 271, -110 | 244, -110 | 352, -110 | 362, -110 | 148, -110 | 120, -257 | 323, -110 | 260, -110 | 418, -110 |
| b$\delta$ | -38, 0 | -12, 0 | 39, 0 | -27, 0 | 16, 0 | -36, 0 | 16, 0 | 0, 0 | -1, 0 | -11, -62 | 19, 0 | 2, 0 | -25, 0 |
| it | *A A̍ A̱* | *B B̍ Ḇ* | *C C̍ C̱* | *D D̍ Ḏ* | *E E̍ E̱* | *F F̍ F̱* | *G G̍ G̱* | *H H̍ H̱* | *I I̍ I̱* | *J J̍ J̱* | *K K̍ Ḵ* | *L L̍ Ḻ* | *M M̍ M̱* |
| patch | *A A̍ A̱* | *B B̍ Ḇ* | *C C̍ C̱* | *D D̍ Ḏ* | *E E̍ E̱* | *F F̍ F̱* | *G G̍ G̱* | *H H̍ H̱* | *I I̍ I̱* | *J J̍ J̱* | *K K̍ Ḵ* | *L L̍ Ḻ* | *M M̍ M̱* |
| ideal | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | a-21, d+32 | c, b | c, b | c, b |
| a | 505, 850 | 447, 850 | 500, 850 | 396, 850 | 413, 850 | 406, 850 | 513, 850 | 481, 850 | 316, 850 | 392, 850 | 466, 850 | 389, 850 | 557, 850 |
| a? | 514, 850 | 459, 850 | 489, 850 | 514, 850 | 444, 850 | 409, 850 | 513, 850 | 517, 850 | 321, 850 | 392, 869 | 499, 850 | 440, 850 | 583, 850 |
| a$\delta$ | -9, 0 | -12, 0 | 11, 0 | -118, 0 | -31, 0 | -3, 0 | 0, 0 | -36, 0 | -5, 0 | 0, -19 | -33, 0 | -51, 0 | -26, 0 |
| b | 308, -108 | 238, -108 | 333, -110 | 279, -107 | 269, -107 | 131, -111 | 345, -110 | 320, -107 | 127, -109 | 109, -319 | 292, -110 | 252, -110 | 363, -110 |
| b? | 310, -110 | 255, -110 | 285, -110 | 310, -110 | 240, -110 | 205, -110 | 309, -110 | 313, -110 | 117, -110 | 146, -285 | 295, -110 | 236, -110 | 379, -110 |
| b$\delta$ | -2, 2 | -17, 2 | 48, 0 | -31, 3 | 29, 3 | -74, -1 | 36, 0 | 7, 3 | 10, 1 | -37, -34 | -3, 0 | 16, 0 | -16, 0 |
| sb | **A A̍ A̱** | **B B̍ Ḇ** | **C C̍ C̱** | **D D̍ Ḏ** | **E E̍ E̱** | **F F̍ F̱** | **G G̍ G̱** | **H H̍ H̱** | **I I̍ I̱** | **J J̍ J̱** | **K K̍ Ḵ** | **L L̍ Ḻ** | **M M̍ M̱** |
| patch | **A A̍ A̱** | **B B̍ Ḇ** | **C C̍ C̱** | **D D̍ Ḏ** | **E E̍ E̱** | **F F̍ F̱** | **G G̍ G̱** | **H H̍ H̱** | **I I̍ I̱** | **J J̍ J̱** | **K K̍ Ḵ** | **L L̍ Ḻ** | **M M̍ M̱** |
| ideal | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, d+50 | c, b | c, b | c, b |
| a | | | | | | | | | | 203, 806 | | | |
| a? | 350, 805 | 300, 805 | 332, 805 | 352, 805 | 283, 805 | 268, 805 | 368, 805 | 362, 805 | 160, 805 | 131, 805 | 340, 805 | 267, 805 | 428, 805 |
| a$\delta$ | | | | | | | | | | 72, 1 | | | |
| b | | | | | | | | 362, -110 | | 109, -319 | | | |
| b? | 350, -110 | 300, -110 | 332, -110 | 352, -110 | 283, -110 | 268, -110 | 368, -110 | 362, -110 | 160, -110 | 131, -267 | 340, -110 | 267, -110 | 428, -110 |
| b$\delta$ | | | | | | | | 0, 0 | | -22, -52 | | | |
| si | ***A A̍ A̱*** | ***B B̍ Ḇ*** | ***C C̍ C̱*** | ***D D̍ Ḏ*** | ***E E̍ E̱*** | ***F F̍ F̱*** | ***G G̍ G̱*** | ***H H̍ H̱*** | ***I I̍ I̱*** | ***J J̍ J̱*** | ***K K̍ Ḵ*** | ***L L̍ Ḻ*** | ***M M̍ M̱*** |
| patch | ***A A̍ A̱*** | ***B B̍ Ḇ*** | ***C C̍ C̱*** | ***D D̍ Ḏ*** | ***E E̍ E̱*** | ***F F̍ F̱*** | ***G G̍ G̱*** | ***H H̍ H̱*** | ***I I̍ I̱*** | ***J J̍ J̱*** | ***K K̍ Ḵ*** | ***L L̍ Ḻ*** | ***M M̍ M̱*** |
| ideal | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | a-19, d+68 | c, b | c, b | c, b |
| a | 354, 850 | 267, 850 | 340, 850 | 306, 850 | 273, 850 | 276, 850 | 365, 850 | 361, 850 | 150, 850 | 432, 850 | 336, 850 | 149, 850 | 431, 850 |
| a? | 382, 850 | 319, 850 | 341, 850 | 367, 850 | 292, 850 | 269, 850 | 371, 850 | 396, 850 | 194, 850 | 253, 871 | 369, 850 | 310, 850 | 441, 850 |
| a$\delta$ | -28, 0 | -52, 0 | -1, 0 | -61, 0 | -19, 0 | 7, 0 | -6, 0 | -35, 0 | -44, 0 | 179, -21 | -33, 0 | -161, 0 | -10, 0 |
| b | 308, -110 | 268, -110 | 363, -110 | 279, -110 | 279, -110 | 201, -110 | 367, -110 | 365, -110 | 147, -110 | 109, -319 | 342, -110 | 260, -110 | 393, -110 |
| b? | 349, -110 | 286, -110 | 308, -110 | 334, -110 | 259, -110 | 236, -110 | 338, -110 | 363, -110 | 161, -110 | 187, -249 | 336, -110 | 277, -110 | 408, -110 |
| b$\delta$ | -41, 0 | -18, 0 | 55, 0 | -55, 0 | 20, 0 | -35, 0 | 29, 0 | 2, 0 | -14, 0 | -78, -70 | 6, 0 | -17, 0 | -15, 0 |

Table 2: Font metrics for A–Z 2/2

| hex | 004E | 004F | 0050 | 0051 | 0052 | 0053 | 0054 | 0055 | 0056 | 0057 | 0058 | 0059 | 005A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu |
| reg | N̈N̈N̲ | O̊O̊O̲ | P̊P̊P̲ | Q̊Q̊Q̲ | R̊R̊R̲ | S̊S̊S̲ | T̊T̊T̲ | ŮŮU̲ | V̊V̊V̲ | W̊W̊W̲ | X̊X̊X̲ | Y̊Y̊Y̲ | Z̊Z̊Z̲ |
| patch | N̈N̈N̲ | O̊O̊O̲ | P̊P̊P̲ | Q̊Q̊Q̲ | R̊R̊R̲ | S̊S̊S̲ | T̊T̊T̲ | ŮŮU̲ | V̊V̊V̲ | W̊W̊W̲ | X̊X̊X̲ | Y̊Y̊Y̲ | Z̊Z̊Z̲ |
| ideal | c, b | c, b | c, b | c, d+23 | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b |
| a | 346, 850 | 335, 850 | 277, 850 | 348, 850 | 269, 850 | 232, 850 | 307, 850 | 346, 850 | 336, 850 | 450, 850 | 332, 850 | 295, 850 | 320, 850 |
| a? | 353, 850 | 352, 850 | 260, 850 | 359, 850 | 301, 850 | 236, 850 | 297, 850 | 331, 850 | 326, 850 | 474, 850 | 323, 850 | 288, 850 | 310, 850 |
| a$\delta$ | -7, 0 | -16, 0 | 18, 0 | -11, 0 | -32, 0 | -4, 0 | 10, 0 | 15, 0 | 10, 0 | -24, 0 | 9, 0 | 8, 0 | 10, 0 |
| b | 348, -110 | 350, -110 | 240, -110 |  | 258, -110 | 234, -110 | 300, -110 | 328, -110 | 318, -110 | 479, -110 | 307, -110 | 295, -110 | 287, -110 |
| b? | 353, -110 | 352, -110 | 260, -110 | 359, -294 | 301, -110 | 236, -110 | 297, -110 | 331, -110 | 326, -110 | 474, -110 | 323, -110 | 288, -110 | 310, -110 |
| b$\delta$ | -5, 0 | -2, 0 | -20, 0 |  | -43, 0 | -2, 0 | 3, 0 | -3, 0 | -8, 0 | 5, 0 | -16, 0 | 8, 0 | -22, 0 |
| it | *N̈N̈N̲* | *O̊O̊O̲* | *P̊P̊P̲* | *Q̊Q̊Q̲* | *R̊R̊R̲* | *S̊S̊S̲* | *T̊T̊T̲* | *ŮŮU̲* | *V̊V̊V̲* | *W̊W̊W̲* | *X̊X̊X̲* | *Y̊Y̊Y̲* | *Z̊Z̊Z̲* |
| patch | *N̈N̈N̲* | *O̊O̊O̲* | *P̊P̊P̲* | *Q̊Q̊Q̲* | *R̊R̊R̲* | *S̊S̊S̲* | *T̊T̊T̲* | *ŮŮU̲* | *V̊V̊V̲* | *W̊W̊W̲* | *X̊X̊X̲* | *Y̊Y̊Y̲* | *Z̊Z̊Z̲* |
| ideal | c, b | c, b | c, b | c, d+23 | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b |
| a | 476, 850 | 515, 850 | 426, 850 | 498, 850 | 419, 850 | 402, 850 | 457, 850 | 486, 850 | 481, 850 | 610, 850 | 482, 850 | 435, 850 | 450, 850 |
| a? | 514, 850 | 515, 850 | 430, 850 | 515, 850 | 458, 850 | 408, 850 | 453, 850 | 498, 850 | 479, 850 | 635, 850 | 495, 850 | 457, 850 | 470, 850 |
| a$\delta$ | -38, 0 | 0, 0 | -4, 0 | -17, 0 | -39, 0 | -6, 0 | 4, 0 | -12, 0 | 2, 0 | -25, 0 | -13, 0 | -22, 0 | -20, 0 |
| b | 328, -109 | 310, -113 | 140, -110 |  | 248, -111 | 204, -111 | 277, -109 | 308, -109 | 263, -110 | 439, -107 | 287, -107 | 270, -109 | 267, -107 |
| b? | 310, -110 | 311, -110 | 226, -110 | 272, -294 | 254, -110 | 204, -110 | 249, -110 | 294, -110 | 275, -110 | 431, -110 | 291, -110 | 253, -110 | 266, -110 |
| b$\delta$ | 18, 1 | -1, -3 | -86, 0 |  | -6, -1 | 0, -1 | 28, 1 | 14, 1 | -12, 0 | 8, 3 | -4, 3 | 17, 1 | 1, 3 |
| sb | **N̈N̈N̲** | **O̊O̊O̲** | **P̊P̊P̲** | **Q̊Q̊Q̲** | **R̊R̊R̲** | **S̊S̊S̲** | **T̊T̊T̲** | **ŮŮU̲** | **V̊V̊V̲** | **W̊W̊W̲** | **X̊X̊X̲** | **Y̊Y̊Y̲** | **Z̊Z̊Z̲** |
| patch | **N̈N̈N̲** | **O̊O̊O̲** | **P̊P̊P̲** | **Q̊Q̊Q̲** | **R̊R̊R̲** | **S̊S̊S̲** | **T̊T̊T̲** | **ŮŮU̲** | **V̊V̊V̲** | **W̊W̊W̲** | **X̊X̊X̲** | **Y̊Y̊Y̲** | **Z̊Z̊Z̲** |
| ideal | c, b | c, b | c, b | c, d | c, b | c, b | a-25, b | c, b | c, b | c, b | c, b | c, b | c, b |
| a |  | 363, 802 |  | 363, 802 | 304, 808 | 252, 802 | 298, 802 |  | 331, 805 | 493, 805 | 364, 804 | 316, 805 | 321, 803 |
| a? | 354, 805 | 366, 805 | 278, 805 | 366, 805 | 320, 805 | 238, 805 | 299, 860 | 339, 805 | 318, 805 | 489, 805 | 354, 805 | 308, 805 | 309, 805 |
| a$\delta$ |  | -2, -3 |  | -2, -3 | -16, 3 | 14, -3 | -1, -58 |  | 12, 0 | 4, 0 | 10, -1 | 8, 0 | 12, -2 |
| b |  |  |  |  | 292, -111 | 234, -111 | 301, -107 |  | 303, -108 | 470, -107 | 318, -107 | 316, -109 | 308, -107 |
| b? | 354, -110 | 366, -110 | 278, -110 | 366, -319 | 320, -110 | 238, -110 | 299, -110 | 339, -110 | 318, -110 | 489, -110 | 354, -110 | 308, -110 | 309, -110 |
| b$\delta$ |  |  |  |  | -28, -1 | -4, -1 | 2, 3 |  | -16, 2 | -19, 3 | -36, 3 | 8, 1 | -1, 3 |
| si | ***N̈N̈N̲*** | ***O̊O̊O̲*** | ***P̊P̊P̲*** | ***Q̊Q̊Q̲*** | ***R̊R̊R̲*** | ***S̊S̊S̲*** | ***T̊T̊T̲*** | ***ŮŮU̲*** | ***V̊V̊V̲*** | ***W̊W̊W̲*** | ***X̊X̊X̲*** | ***Y̊Y̊Y̲*** | ***Z̊Z̊Z̲*** |
| patch | ***N̈N̈N̲*** | ***O̊O̊O̲*** | ***P̊P̊P̲*** | ***Q̊Q̊Q̲*** | ***R̊R̊R̲*** | ***S̊S̊S̲*** | ***T̊T̊T̲*** | ***ŮŮU̲*** | ***V̊V̊V̲*** | ***W̊W̊W̲*** | ***X̊X̊X̲*** | ***Y̊Y̊Y̲*** | ***Z̊Z̊Z̲*** |
| ideal | c, b | c, b | c, b | c, d | c, b | c, b | a-25, b | c, b | c, b | c, b | c, b | c, b | c, b |
| a | 346, 850 | 335, 850 | 316, 850 | 348, 850 | 269, 850 | 232, 850 | 307, 850 | 346, 850 | 336, 850 | 450, 850 | 332, 850 | 295, 850 | 320, 850 |
| a? | 394, 850 | 390, 850 | 284, 850 | 390, 850 | 315, 850 | 250, 850 | 312, 865 | 364, 850 | 328, 850 | 501, 850 | 362, 850 | 319, 850 | 322, 850 |
| a$\delta$ | -48, 0 | -55, 0 | 32, 0 | -42, 0 | -46, 0 | -18, 0 | -5, -15 | -18, 0 | 8, 0 | -51, 0 | -30, 0 | -24, 0 | -2, 0 |
| b | 348, -110 | 350, -110 | 240, -110 |  | 258, -110 | 234, -110 | 300, -110 | 328, -110 | 318, -110 | 479, -110 | 307, -110 | 295, -110 | 287, -110 |
| b? | 361, -110 | 357, -110 | 251, -110 | 314, -319 | 282, -110 | 217, -110 | 276, -110 | 331, -110 | 295, -110 | 468, -110 | 329, -110 | 286, -110 | 289, -110 |
| b$\delta$ | -13, 0 | -7, 0 | -11, 0 |  | -24, 0 | 17, 0 | 24, 0 | -3, 0 | 23, 0 | 11, 0 | -22, 0 | 9, 0 | -2, 0 |

Table 3: Font metrics for a–z 1/2

| hex | 0061 | 0062 | 0063 | 0064 | 0065 | 0066 | 0067 | 0068 | 0069 | 006A | 006B | 006C | 006D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | aȧạ | bb̈ḇ | cċc̣ | dd̈ḍ | eėẹ | fff̣ | gġg̣ | hḧḥ | iii̇ | jjj̈ | kk̇ḳ | lll̇ | mm̈ṃ |
| patch | aȧạ | bb̈ḇ | cċc̣ | dd̈ḍ | eėẹ | fff̣ | gġg̣ | hḧḥ | iii̇ | jjj̈ | kk̇ḳ | lll̇ | mm̈ṃ |
| ideal | x, b | a, b | x, b | a, b | x, b | a, b | x+33, d | a, b | a-51, b | a-51, d+9 | a, b | a, b | x, b |
| a | 209, 645 | | 242, 645 | 218, 645 | 243, 645 | | 231, 645 | 299, 645 | 137, 795 | | 306, 784 | 135, 885 | 392, 645 |
| a? | 246, 645 | 232, 885 | 218, 645 | 270, 885 | 222, 645 | 208, 885 | 256, 676 | 270, 885 | 142, 834 | 76, 834 | 261, 885 | 134, 885 | 402, 645 |
| a$\delta$ | -36, 0 | | 24, 0 | -52, -240 | 21, 0 | | -26, -31 | 28, -240 | -5, -39 | | 45, -101 | 1, 0 | -10, 0 |
| b | 196, -110 | 234, -110 | 236, -110 | 221, -110 | 234, -110 | 132, -110 | 229, -319 | 272, -110 | 136, -110 | 109, -319 | 290, -214 | 135, -110 | 408, -110 |
| b? | 246, -110 | 232, -110 | 218, -110 | 270, -110 | 222, -110 | 208, -110 | 256, -319 | 270, -110 | 142, -110 | 76, -308 | 261, -110 | 134, -110 | 402, -110 |
| b$\delta$ | -50, 0 | 2, 0 | 18, 0 | -50, 0 | 12, 0 | -76, 0 | -28, 0 | 2, 0 | -6, 0 | 34, -11 | 29, -104 | 1, 0 | 6, 0 |
| it | *aȧạ* | *bb̈ḇ* | *cċc̣* | *dd̈ḍ* | *eėẹ* | *fff̣* | *gġg̣* | *hḧḥ* | *iii̇* | *jjj̈* | *kk̇ḳ* | *lll̇* | *mm̈ṃ* |
| patch | *aȧạ* | *bb̈ḇ* | *cċc̣* | *dd̈ḍ* | *eėẹ* | *fff̣* | *gġg̣* | *hḧḥ* | *iii̇* | *jjj̈* | *kk̇ḳ* | *lll̇* | *mm̈ṃ* |
| ideal | x, b | a, b | x, b | a, b | x, b | a, d | x, d | a, b | a-80, b | a-80, d+36 | a, b | a, b | x, b |
| a | 341, 647 | | 343, 645 | 218, 645 | 332, 645 | | 360, 645 | 409, 645 | | | 405, 646 | 292, 890 | 522, 645 |
| a? | 380, 645 | 428, 890 | 332, 645 | 434, 890 | 338, 645 | 346, 890 | 376, 645 | 449, 890 | 310, 810 | 302, 810 | 432, 890 | 322, 890 | 529, 645 |
| a$\delta$ | -39, 2 | | 11, 0 | -216, -245 | -6, 0 | | -16, 0 | -40, -245 | | | -27, -244 | -30, 0 | -7, 0 |
| b | 174, -108 | 214, -108 | 163, -113 | 221, -107 | 158, -108 | 222, -111 | 197, -319 | 229, -111 | 136, -107 | 0, -319 | 223, -107 | 115, -107 | 382, -105 |
| b? | 220, -110 | 216, -110 | 171, -110 | 221, -110 | 177, -110 | 89, -319 | 171, -319 | 236, -110 | 115, -110 | 70, -281 | 220, -110 | 110, -110 | 368, -110 |
| b$\delta$ | -46, 2 | -2, 2 | -8, -3 | 0, 3 | -19, 2 | 133, 208 | 26, 0 | -7, -1 | 21, 3 | -70, -38 | 3, 3 | 5, 3 | 14, 5 |
| sb | **aȧạ** | **bb̈ḇ** | **cċc̣** | **dd̈ḍ** | **eėẹ** | **fff̣** | **gġg̣** | **hḧḥ** | **iii̇** | **jjj̈** | **kk̇ḳ** | **lll̇** | **mm̈ṃ** |
| patch | **aȧạ** | **bb̈ḇ** | **cċc̣** | **dd̈ḍ** | **eėẹ** | **fff̣** | **gġg̣** | **hḧḥ** | **iii̇** | **jjj̈** | **kk̇ḳ** | **lll̇** | **mm̈ṃ** |
| ideal | x, b | a, b-13 | x, b | a, b | x, b | a, b | x+33, d | a, b | a-48, b | a-48, d | a, b | a, b | x, b |
| a | | | | | | | 231, 645 | | 137, 795 | | | 142, 885 | 396, 645 |
| a? | 255, 645 | 236, 885 | 224, 645 | 274, 885 | 232, 645 | 226, 885 | 264, 676 | 275, 885 | 156, 837 | 78, 837 | 262, 885 | 148, 885 | 408, 645 |
| a$\delta$ | | | | | | | -32, -31 | | -18, -42 | | | -6, 0 | -12, 0 |
| b | | | | | | | 229, -319 | 277, -110 | 136, -107 | 109, -319 | | 135, -110 | 396, -105 |
| b? | 255, -110 | 236, -121 | 224, -110 | 274, -110 | 232, -110 | 226, -110 | 264, -319 | 275, -110 | 156, -110 | 78, -319 | 262, -110 | 148, -110 | 408, -110 |
| b$\delta$ | | | | | | | -34, 0 | 2, 0 | -20, 3 | 31, 0 | | -14, 0 | -12, 5 |
| si | ***aȧạ*** | ***bb̈ḇ*** | ***cċc̣*** | ***dd̈ḍ*** | ***eėẹ*** | ***fff̣*** | ***gġg̣*** | ***hḧḥ*** | ***iii̇*** | ***jjj̈*** | ***kk̇ḳ*** | ***lll̇*** | ***mm̈ṃ*** |
| patch | ***aȧạ*** | ***bb̈ḇ*** | ***cċc̣*** | ***dd̈ḍ*** | ***eėẹ*** | ***fff̣*** | ***gġg̣*** | ***hḧḥ*** | ***iii̇*** | ***jjj̈*** | ***kk̇ḳ*** | ***lll̇*** | ***mm̈ṃ*** |
| ideal | x, b | a, b | x, b | a, b | x, b | a, d | x, d | a, b | a-67, b | a-67, d+25 | a, b | a, b | x, b |
| a | 379, 645 | | 242, 645 | 218, 645 | 243, 645 | | 291, 645 | 469, 645 | 137, 795 | | 306, 784 | 174, 890 | 392, 645 |
| a? | 251, 645 | 290, 890 | 203, 645 | 309, 890 | 218, 645 | 216, 890 | 220, 645 | 331, 890 | 180, 823 | 192, 823 | 316, 890 | 194, 890 | 407, 645 |
| a$\delta$ | 128, 0 | | 39, 0 | -91, -245 | 25, 0 | | 71, 0 | 138, -245 | -43, -28 | | -10, -106 | -20, 0 | -15, 0 |
| b | 196, -110 | 234, -110 | 236, -110 | 221, -110 | 234, -110 | 132, -110 | 199, -319 | 272, -110 | 136, -110 | 67, -319 | 290, -214 | 135, -110 | 400, -110 |
| b? | 260, -110 | 248, -110 | 212, -110 | 268, -110 | 227, -110 | 132, -319 | 186, -319 | 290, -110 | 153, -110 | 127, -292 | 275, -110 | 153, -110 | 415, -110 |
| b$\delta$ | -64, 0 | -14, 0 | 24, 0 | -47, 0 | 7, 0 | 0, 209 | 13, 0 | -18, 0 | -17, 0 | -60, -27 | 15, -104 | -18, 0 | -15, 0 |

Table 4: Font metrics for a–z 2/2

| hex | 006E | 006F | 0070 | 0071 | 0072 | 0073 | 0074 | 0075 | 0076 | 0077 | 0078 | 0079 | 007A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | nn̈n̲ | oo̱o̲ | pp̈p̲ | qq̈q̲ | rr̈r̲ | ss̈s̲ | tẗt̲ | uu̱u̲ | vv̇v̲ | wẘw̲ | xx̌x̲ | yy̋y̲ | zz̀z̲ |
| patch | nn̈n̲ | oo̱o̲ | pp̈p̲ | qq̈q̲ | rr̈r̲ | ss̈s̲ | tẗt̲ | uu̱u̲ | vv̇v̲ | wẘw̲ | xx̌x̲ | yy̋y̲ | zz̀z̲ |
| ideal | x, b | x, b | x, d | x, d | x, b | x, b | a-116, b | x, b | x, b | x, b | x, b | x, d | x, b |
| a | 264, 646 | 238, 645 | 271, 646 | 238, 645 | 191, 646 | 206, 645 |  | 255, 645 | 256, 646 | 345, 646 | 256, 646 | 262, 645 | 220, 646 |
| a? | 276, 645 | 252, 645 | 242, 645 | 268, 645 | 190, 645 | 199, 645 | 163, 769 | 271, 645 | 249, 645 | 373, 645 | 245, 645 | 260, 645 | 220, 645 |
| a$\delta$ | -12, 1 | -14, 0 | 28, 1 | -30, 0 | 1, 1 | 7, 0 |  | -16, 0 | 7, 1 | -28, 1 | 11, 1 | 2, 0 | 0, 1 |
| b | 265, -110 | 243, -110 | 179, -319 | 348, -319 | 135, -110 | 196, -110 | 163, -110 | 239, -110 | 241, -110 | 352, -110 | 224, -110 | 242, -319 | 207, -110 |
| b? | 276, -110 | 252, -110 | 242, -319 | 268, -319 | 190, -110 | 199, -110 | 163, -110 | 271, -110 | 249, -110 | 373, -110 | 245, -110 | 260, -319 | 220, -110 |
| b$\delta$ | -10, 0 | -9, 0 | -64, 0 | 80, 0 | -55, 0 | -3, 0 | 0, 0 | -32, 0 | -8, 0 | -21, 0 | -21, 0 | -18, 0 | -14, 0 |
| it | *nn̈n̲* | *oo̱o̲* | *pp̈p̲* | *qq̈q̲* | *rr̈r̲* | *ss̈s̲* | *tẗt̲* | *uu̱u̲* | *vv̇v̲* | *wẘw̲* | *xx̌x̲* | *yy̋y̲* | *zz̀z̲* |
| patch | *nn̈n̲* | *oo̱o̲* | *pp̈p̲* | *qq̈q̲* | *rr̈r̲* | *ss̈s̲* | *tẗt̲* | *uu̱u̲* | *vv̇v̲* | *wẘw̲* | *xx̌x̲* | *yy̋y̲* | *zz̀z̲* |
| ideal | x, b | x, b | x+62, d | x+17, d | x, b | x, b | x+132, b | x, b | x, b | x, b | x, b | x, d | x+15, b |
| a | 394, 646 | 359, 645 | 411, 646 | 369, 645 | 312, 646 | 327, 645 |  | 386, 646 |  | 348, 646 | 366, 646 | 367, 644 | 360, 646 |
| a? | 396, 645 | 361, 645 | 394, 705 | 386, 660 | 316, 645 | 314, 645 | 318, 775 | 398, 645 | 373, 645 | 481, 645 | 375, 645 | 389, 645 | 358, 658 |
| a$\delta$ | -2, 1 | -2, 0 | 17, -59 | -17, -15 | -4, 1 | 13, 0 |  | -12, 1 |  | -133, 1 | -9, 1 | -22, -1 | 2, -12 |
| b | 235, -111 | 186, -107 | 248, -110 | 169, -110 | 113, -110 | 169, -111 | 133, -107 | 226, -111 |  | 355, -110 | 224, -110 | 368, -107 | 187, -111 |
| b? | 236, -110 | 200, -110 | 177, -319 | 178, -319 | 155, -110 | 153, -110 | 130, -110 | 237, -110 | 213, -110 | 321, -110 | 214, -110 | 184, -319 | 195, -110 |
| b$\delta$ | -1, -1 | -14, 3 | 71, 209 | -9, 209 | -42, 0 | 16, -1 | 3, 3 | -11, -1 |  | 34, 0 | 10, 0 | 184, 212 | -8, -1 |
| sb | **nn̈n̲** | **oo̱o̲** | **pp̈p̲** | **qq̈q̲** | **rr̈r̲** | **ss̈s̲** | **tẗt̲** | **uu̱u̲** | **vv̇v̲** | **wẘw̲** | **xx̌x̲** | **yy̋y̲** | **zz̀z̲** |
| patch | **nn̈n̲** | **oo̱o̲** | **pp̈p̲** | **qq̈q̲** | **rr̈r̲** | **ss̈s̲** | **tẗt̲** | **uu̱u̲** | **vv̇v̲** | **wẘw̲** | **xx̌x̲** | **yy̋y̲** | **zz̀z̲** |
| ideal | x, b | x, b | x, d | x, d | x, b | x, b | a-103, b | x, b | x, b | x, b | x, b | x, d | x, b |
| a | 264, 646 |  |  |  |  |  |  | 264, 646 |  |  |  |  |  |
| a? | 290, 645 | 260, 645 | 254, 645 | 273, 645 | 212, 645 | 201, 645 | 168, 782 | 276, 645 | 262, 645 | 384, 645 | 260, 645 | 264, 645 | 222, 645 |
| a$\delta$ | -26, 1 |  |  |  |  |  |  | -12, 1 |  |  |  |  |  |
| b | 265, -111 |  |  |  |  |  | 163, -107 | 248, -111 |  |  |  |  |  |
| b? | 290, -110 | 260, -110 | 254, -319 | 273, -319 | 212, -110 | 201, -110 | 168, -110 | 276, -110 | 262, -110 | 384, -110 | 260, -110 | 264, -319 | 222, -110 |
| b$\delta$ | -24, -1 |  |  |  |  |  | -5, 3 | -28, -1 |  |  |  |  |  |
| si | ***nn̈n̲*** | ***oo̱o̲*** | ***pp̈p̲*** | ***qq̈q̲*** | ***rr̈r̲*** | ***ss̈s̲*** | ***tẗt̲*** | ***uu̱u̲*** | ***vv̇v̲*** | ***wẘw̲*** | ***xx̌x̲*** | ***yy̋y̲*** | ***zz̀z̲*** |
| patch | ***nn̈n̲*** | ***oo̱o̲*** | ***pp̈p̲*** | ***qq̈q̲*** | ***rr̈r̲*** | ***ss̈s̲*** | ***tẗt̲*** | ***uu̱u̲*** | ***vv̇v̲*** | ***wẘw̲*** | ***xx̌x̲*** | ***yy̋y̲*** | ***zz̀z̲*** |
| ideal | x, b | x, b | x+36, d | x+24, d | x, b | x, b | a-125, b | x, b | x, b | x, b | x, b | x, d | x+23, b |
| a | 264, 646 | 238, 645 | 271, 646 | 238, 645 | 191, 646 | 326, 645 |  | 255, 645 | 256, 646 | 345, 646 |  | 262, 645 | 220, 646 |
| a? | 284, 645 | 235, 645 | 257, 679 | 272, 667 | 203, 645 | 190, 645 | 189, 765 | 262, 645 | 220, 645 | 332, 645 | 240, 645 | 236, 645 | 214, 666 |
| a$\delta$ | -20, 1 | 3, 0 | 14, -33 | -34, -22 | -12, 1 | 136, 0 |  | -7, 0 | 36, 1 | 13, 1 |  | 26, 0 | 6, -20 |
| b | 265, -110 | 243, -110 | 179, -319 | 348, -319 | 135, -110 | 196, -110 | 133, -107 | 239, -110 | 241, -110 | 352, -110 |  | 242, -319 | 207, -110 |
| b? | 293, -110 | 243, -110 | 216, -319 | 233, -319 | 212, -110 | 198, -110 | 173, -110 | 270, -110 | 228, -110 | 341, -110 | 248, -110 | 202, -319 | 218, -110 |
| b$\delta$ | -28, 0 | 0, 0 | -37, 0 | 115, 0 | -77, 0 | -2, 0 | -40, 3 | -31, 0 | 13, 0 | 11, 0 |  | 40, 0 | -11, 0 |

Table 5: Font metrics for Latin and IPA 1/7

| hex | 00C6 | 00D0 | 00D8 | 00DE | 00DF | 00E6 | 00F0 | 00F8 | 00FE | 0110 | 0111 | 0126 | 0127 | 0131 | 0141 | 0142 | 014B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Lu | Lu | Lu | Ll | Ll | Ll | Ll | Ll | Lu | Ll | Lu | Ll | Ll | Lu | Ll | Ll |
| reg | ÆǼǢ | ÐḊḌ | ØǪǾ | ÞṖ�example | ßßß | æǽǣ | ððð | øǿø | þṗþ | ĐḊḌ | đḋḍ | ĦḢḤ | ħḣḥ | ıi̇į | ŁĿĻ | łŀļ | ŋṅŋ |
| patch | ÆǼǢ | ÐḊḌ | ØǪǾ | ÞṖ�example | ßßß | æǽǣ | ððð | øǿø | þṗþ | ĐḊḌ | đḋḍ | ĦḢḤ | ħḣḥ | ıi̇į | ŁĿĻ | łŀļ | ŋṅŋ |
| ideal | c, b | c, b | c, b-13 | c, b | a, b | x, b | a, b | x+25, b-28 | a, d | c, b | a, b | c, b | a, b | x, b | c, b | a, b | x, d+9 |
| a | 480, 850 | 341, 850 | 351, 850 | 281, 850 | | 350, 645 | | | 278, 646 | | | | | 132, 645 | | | 264, 646 |
| a? | 426, 850 | 333, 850 | 352, 850 | 250, 850 | 259, 885 | 344, 645 | 240, 885 | 250, 668 | 236, 885 | 336, 850 | 270, 885 | 362, 850 | 270, 885 | 142, 645 | 256, 850 | 133, 885 | 240, 645 |
| a$\delta$ | 54, 0 | 8, 0 | 0, 0 | 32, 0 | | 6, 0 | | | 42, -239 | | | | | -10, 0 | | | 24, 1 |
| b | 549, -107 | | 350, -110 | | | 327, -109 | | 243, -110 | | | | | | 139, -108 | | | 265, -111 |
| b? | 426, -110 | 333, -110 | 352, -121 | 250, -110 | 259, -110 | 344, -110 | 240, -110 | 250, -136 | 236, -319 | 336, -110 | 270, -110 | 362, -110 | 270, -110 | 142, -110 | 256, -110 | 133, -110 | 240, -308 |
| b$\delta$ | 122, 3 | | -2, 11 | | | -16, 1 | | -8, 26 | | | | | | -3, 2 | | | 25, 197 |
| it | *ÆǼǢ* | *ÐḊḌ* | *ØǪǾ* | *ÞṖṗ* | *ßßß* | *æǽǣ* | *ððð* | *øǿø* | *þṗþ* | *ĐḊḌ* | *đḋḍ* | *ĦḢḤ* | *ħḣḥ* | *ıi̇į* | *ŁĿĻ* | *łŀļ* | *ŋṅŋ* |
| patch | *ÆǼǢ* | *ÐḊḌ* | *ØǪǾ* | *ÞṖṗ* | *ßßß* | *æǽǣ* | *ððð* | *øǿø* | *þṗþ* | *ĐḊḌ* | *đḋḍ* | *ĦḢḤ* | *ħḣḥ* | *ıi̇į* | *ŁĿĻ* | *łŀļ* | *ŋṅŋ* |
| ideal | c, b | c, b | c+25, b-43 | c, b | a, d | x, b | a+6, b | x+48, b-48 | a, d-7 | c, b | a, b | c, b | a, b | x, b | c, b | a, b | x, d |
| a | 615, 850 | | | | | 480, 645 | | | 278, 646 | | | | | 267, 645 | | | 264, 646 |
| a? | 595, 850 | 514, 850 | 520, 873 | 431, 850 | 461, 890 | 481, 645 | 429, 896 | 370, 691 | 439, 890 | 514, 850 | 434, 890 | 517, 850 | 449, 890 | 275, 645 | 440, 850 | 322, 890 | 382, 645 |
| a$\delta$ | 20, 0 | | | | | -1, 0 | | | -161, -244 | | | | | -8, 0 | | | -118, 1 |
| b | 549, -107 | | 310, -113 | | | 327, -109 | | 186, -107 | | | | | | 139, -108 | | | 265, -111 |
| b? | 391, -110 | 310, -110 | 302, -151 | 227, -110 | 204, -319 | 320, -110 | 216, -110 | 190, -156 | 181, -324 | 310, -110 | 221, -110 | 313, -110 | 236, -110 | 115, -110 | 236, -110 | 110, -110 | 177, -319 |
| b$\delta$ | 158, 3 | | 8, 38 | | | 7, 1 | | -4, 49 | | | | | | 24, 2 | | | 88, 208 |
| sb | **ÆǼǢ** | **ÐḊḌ** | **ØǪǾ** | **ÞṖṗ** | **ßßß** | **æǽǣ** | **ððð** | **øǿø** | **þṗþ** | **ĐḊḌ** | **đḋḍ** | **ĦḢḤ** | **ħḣḥ** | **ıi̇į** | **ŁĿĻ** | **łŀļ** | **ŋṅŋ** |
| patch | **ÆǼǢ** | **ÐḊḌ** | **ØǪǾ** | **ÞṖṗ** | **ßßß** | **æǽǣ** | **ððð** | **øǿø** | **þṗþ** | **ĐḊḌ** | **đḋḍ** | **ĦḢḤ** | **ħḣḥ** | **ıi̇į** | **ŁĿĻ** | **łŀļ** | **ŋṅŋ** |
| ideal | c, b | c, b | c+25, b-45 | c, b | a, b | x, b | a+19, b | x+41, b-34 | a, d | c, b | a, b | c, b | a, b | x, b | c, b | a, b | x, d+36 |
| a | 510, 850 | | | | | 355, 645 | | | | | | | | 151, 645 | | | |
| a? | 446, 805 | 349, 805 | 366, 828 | 287, 805 | 284, 885 | 362, 645 | 256, 904 | 260, 684 | 260, 885 | 349, 805 | 274, 885 | 362, 805 | 275, 885 | 156, 645 | 258, 805 | 142, 885 | 248, 645 |
| a$\delta$ | 64, 45 | | | | | -8, 0 | | | | | | | | -4, 0 | | | |
| b | | | | | | | | | | | | | | 135, -108 | | | |
| b? | 446, -110 | 349, -110 | 366, -153 | 287, -110 | 284, -110 | 362, -110 | 256, -110 | 260, -142 | 260, -319 | 349, -110 | 274, -110 | 362, -110 | 275, -110 | 156, -110 | 258, -110 | 142, -110 | 248, -281 |
| b$\delta$ | | | | | | | | | | | | | | -20, 2 | | | |
| si | ***ÆǼǢ*** | ***ÐḊḌ*** | ***ØǪǾ*** | ***ÞṖṗ*** | ***ßßß*** | ***æǽǣ*** | ***ððð*** | ***øǿø*** | ***þṗþ*** | ***ĐḊḌ*** | ***đḋḍ*** | ***ĦḢḤ*** | ***ħḣḥ*** | ***ıi̇į*** | ***ŁĿĻ*** | ***łŀļ*** | ***ŋṅŋ*** |
| patch | ***ÆǼǢ*** | ***ÐḊḌ*** | ***ØǪǾ*** | ***ÞṖṗ*** | ***ßßß*** | ***æǽǣ*** | ***ððð*** | ***øǿø*** | ***þṗþ*** | ***ĐḊḌ*** | ***đḋḍ*** | ***ĦḢḤ*** | ***ħḣḥ*** | ***ıi̇į*** | ***ŁĿĻ*** | ***łŀļ*** | ***ŋṅŋ*** |
| ideal | c, b | c, b | c+25, b-40 | c, b | a, d | x, b | a+19, b | x+22, b-30 | a, d | c, b | a, b | c, b | a, b | x, b | c, b | a, b | x, d+16 |
| a | 500, 850 | 341, 850 | 351, 850 | 281, 850 | | 350, 645 | | | 278, 646 | | | | | 132, 645 | | | 264, 646 |
| a? | 459, 850 | 367, 850 | 395, 873 | 303, 850 | 325, 890 | 339, 645 | 291, 909 | 239, 665 | 300, 890 | 367, 850 | 309, 890 | 396, 850 | 331, 890 | 144, 645 | 300, 850 | 190, 890 | 253, 645 |
| a$\delta$ | 41, 0 | -26, 0 | -44, -23 | -22, 0 | | 11, 0 | | | -22, -244 | | | | | -12, 0 | | | 11, 1 |
| b | 549, -107 | | 350, -110 | | | 327, -109 | | 243, -110 | | | | | | 139, -108 | | | 265, -111 |
| b? | 426, -110 | 334, -110 | 349, -148 | 270, -110 | 241, -319 | 347, -110 | 246, -110 | 237, -138 | 216, -319 | 334, -110 | 268, -110 | 363, -110 | 290, -110 | 153, -110 | 267, -110 | 148, -110 | 222, -301 |
| b$\delta$ | 123, 3 | | 1, 38 | | | -20, 1 | | 6, 28 | | | | | | -14, 2 | | | 43, 190 |

Table 6: Font metrics for Latin and IPA 2/7

| hex | 0152 | 0153 | 017F | 0181 | 0182 | 0186 | 0187 | 0189 | 018A | 018B | 018E | 018F | 0190 | 0191 | 0192 | 0193 | 0194 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Ll | Ll | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Ll | Lu | Lu |
| reg | Œ Ė Ę | œ ė ę | ſ f̈ | Ɓ Ḃ Ḅ | Ƃ Ƃ̇ Ƃ̣ | Ɔ Ɔ̇ Ọ | Ƈ Ƈ̇ Ƈ̣ | Ɖ Ɖ̇ Ɖ̣ | Ɗ Ɗ̇ Ɗ̣ | ƌ ƌ̇ ƌ̣ | Ǝ Ė Ę | Ə ė ę | Ɛ ė ę | Ƒ Ḟ Ƒ̣ | f f̈ f̣ | Ɠ Ġ Ɠ̣ | Ɣ Ɣ̇ Ɣ̣ |
| patch | Œ Ė Ę | œ ė ę | ſ f̈ | Ɓ Ḃ Ḅ | Ƃ Ƃ̇ Ƃ̣ | Ɔ Ɔ̇ Ọ | Ƈ Ƈ̇ Ƈ̣ | Ɖ Ɖ̇ Ɖ̣ | Ɗ Ɗ̇ Ɗ̣ | ƌ ƌ̇ ƌ̣ | Ǝ Ė Ę | Ə ė ę | Ɛ ė ę | Ƒ Ḟ Ƒ̣ | f f̈ f̣ | Ɠ Ġ Ɠ̣ | Ɣ Ɣ̇ Ɣ̣ |
| ideal | c, b | x, b | a, b | c, b | c, b | c, b | a+45, b | c, b | c, b | c, b | c, b | c, b | c, b | c, d+60 | a, d | a+45, b | c, d |
| a | 460, 850 | 405, 645 | | 327, 805 | | | | | 306, 805 | | | | | | | | 331, 805 |
| a? | 436, 850 | 388, 645 | 190, 885 | 303, 850 | 274, 850 | 322, 850 | 369, 930 | 333, 850 | 368, 850 | 292, 850 | 280, 850 | 326, 850 | 256, 850 | 183, 850 | 201, 885 | 372, 930 | 308, 850 |
| a$\delta$ | 24, 0 | 18, 0 | | 24, -45 | | | | | -62, -45 | | | | | | | | 22, -45 |
| b | | | | 328, -108 | | | | | 369, -107 | | | | | | | | |
| b? | 436, -110 | 388, -110 | 190, -110 | 303, -110 | 274, -110 | 322, -110 | 369, -110 | 333, -110 | 368, -110 | 292, -110 | 280, -110 | 326, -110 | 256, -110 | 183, -257 | 201, -319 | 372, -110 | 308, -319 |
| b$\delta$ | | | | 25, 2 | | | | | 0, 3 | | | | | | | | |
| it | *Œ Ė Ę* | *œ ė ę* | *ſ f̈* | *Ɓ Ḃ Ḅ* | *Ƃ Ƃ̇ Ƃ̣* | *Ɔ Ɔ̇ Ọ* | *Ƈ Ƈ̇ Ƈ̣* | *Ɖ Ɖ̇ Ɖ̣* | *Ɗ Ɗ̇ Ɗ̣* | *ƌ ƌ̇ ƌ̣* | *Ǝ Ė Ę* | *Ə ė ę* | *Ɛ ė ę* | *Ƒ Ḟ Ƒ̣* | *f f̈ f̣* | *Ɠ Ġ Ɠ̣* | *Ɣ Ɣ̇ Ɣ̣* |
| patch | *Œ Ė Ę* | *œ ė ę* | *ſ f̈* | *Ɓ Ḃ Ḅ* | *Ƃ Ƃ̇ Ƃ̣* | *Ɔ Ɔ̇ Ọ* | *Ƈ Ƈ̇ Ƈ̣* | *Ɖ Ɖ̇ Ɖ̣* | *Ɗ Ɗ̇ Ɗ̣* | *ƌ ƌ̇ ƌ̣* | *Ǝ Ė Ę* | *Ə ė ę* | *Ɛ ė ę* | *Ƒ Ḟ Ƒ̣* | *f f̈ f̣* | *Ɠ Ġ Ɠ̣* | *Ɣ Ɣ̇ Ɣ̣* |
| ideal | c, b | x, b | a, d | c, b | c, b | c+14, b | a+45, b | c, b | c, b | c, b | c, b | c, b | c, b | c, d+60 | a, d+60 | a+45, b | c, d+20 |
| a | 575, 850 | 480, 645 | | 327, 805 | | | | | 306, 805 | | | | | | | | 331, 805 |
| a? | 587, 850 | 490, 645 | 339, 890 | 488, 850 | 465, 850 | 506, 862 | 521, 935 | 534, 850 | 553, 850 | 450, 850 | 465, 850 | 515, 850 | 439, 850 | 419, 850 | 414, 890 | 559, 935 | 462, 850 |
| a$\delta$ | -12, 0 | -10, 0 | | -161, -45 | | | | | -247, -45 | | | | | | | | -131, -45 |
| b | | | | 328, -108 | | | | | 369, -107 | | | | | | | | |
| b? | 383, -110 | 329, -110 | 82, -319 | 284, -110 | 261, -110 | 299, -110 | 299, -110 | 330, -110 | 349, -110 | 246, -110 | 261, -110 | 311, -110 | 235, -110 | 184, -257 | 170, -257 | 337, -110 | 218, -297 |
| b$\delta$ | | | | 44, 2 | | | | | 20, 3 | | | | | | | | |
| sb | **Œ Ė Ę** | **œ ė ę** | **ſ f̈** | **Ɓ Ḃ Ḅ** | **Ƃ Ƃ̇ Ƃ̣** | **Ɔ Ɔ̇ Ọ** | **Ƈ Ƈ̇ Ƈ̣** | **Ɖ Ɖ̇ Ɖ̣** | **Ɗ Ɗ̇ Ɗ̣** | **ƌ ƌ̇ ƌ̣** | **Ǝ Ė Ę** | **Ə ė ę** | **Ɛ ė ę** | **Ƒ Ḟ Ƒ̣** | **f f̈ f̣** | **Ɠ Ġ Ɠ̣** | **Ɣ Ɣ̇ Ɣ̣** |
| patch | **Œ Ė Ę** | **œ ė ę** | **ſ f̈** | **Ɓ Ḃ Ḅ** | **Ƃ Ƃ̇ Ƃ̣** | **Ɔ Ɔ̇ Ọ** | **Ƈ Ƈ̇ Ƈ̣** | **Ɖ Ɖ̇ Ɖ̣** | **Ɗ Ɗ̇ Ɗ̣** | **ƌ ƌ̇ ƌ̣** | **Ǝ Ė Ę** | **Ə ė ę** | **Ɛ ė ę** | **Ƒ Ḟ Ƒ̣** | **f f̈ f̣** | **Ɠ Ġ Ɠ̣** | **Ɣ Ɣ̇ Ɣ̣** |
| ideal | c, b | x, b | a, b | c, b | c, b | c, b | a+45, b | c, b | c, b | c, b | c, b | c, b | c, b | c, d+36 | a, d+62 | a+45, b | c, d+22 |
| a | 450, 850 | 430, 645 | | | | 275, 833 | | | | | | | | | | | |
| a? | 432, 805 | 402, 645 | 217, 885 | 316, 805 | 268, 805 | 314, 805 | 368, 930 | 349, 805 | 381, 805 | 292, 805 | 302, 805 | 353, 805 | 269, 805 | 236, 805 | 221, 885 | 374, 930 | 294, 805 |
| a$\delta$ | 18, 45 | 28, 0 | | | | -38, 28 | | | | | | | | | | | |
| b | | | | | | | | | | | | | | | | | |
| b? | 432, -110 | 402, -110 | 217, -110 | 316, -110 | 268, -110 | 314, -110 | 368, -110 | 349, -110 | 381, -110 | 292, -110 | 302, -110 | 353, -110 | 269, -110 | 236, -281 | 221, -255 | 374, -110 | 294, -295 |
| b$\delta$ | | | | | | | | | | | | | | | | | |
| si | ***Œ Ė Ę*** | ***œ ė ę*** | ***ſ f̈*** | ***Ɓ Ḃ Ḅ*** | ***Ƃ Ƃ̇ Ƃ̣*** | ***Ɔ Ɔ̇ Ọ*** | ***Ƈ Ƈ̇ Ƈ̣*** | ***Ɖ Ɖ̇ Ɖ̣*** | ***Ɗ Ɗ̇ Ɗ̣*** | ***ƌ ƌ̇ ƌ̣*** | ***Ǝ Ė Ę*** | ***Ə ė ę*** | ***Ɛ ė ę*** | ***Ƒ Ḟ Ƒ̣*** | ***f f̈ f̣*** | ***Ɠ Ġ Ɠ̣*** | ***Ɣ Ɣ̇ Ɣ̣*** |
| patch | ***Œ Ė Ę*** | ***œ ė ę*** | ***ſ f̈*** | ***Ɓ Ḃ Ḅ*** | ***Ƃ Ƃ̇ Ƃ̣*** | ***Ɔ Ɔ̇ Ọ*** | ***Ƈ Ƈ̇ Ƈ̣*** | ***Ɖ Ɖ̇ Ɖ̣*** | ***Ɗ Ɗ̇ Ɗ̣*** | ***ƌ ƌ̇ ƌ̣*** | ***Ǝ Ė Ę*** | ***Ə ė ę*** | ***Ɛ ė ę*** | ***Ƒ Ḟ Ƒ̣*** | ***f f̈ f̣*** | ***Ɠ Ġ Ɠ̣*** | ***Ɣ Ɣ̇ Ɣ̣*** |
| ideal | c, b | x, b | a+13, d-9 | c, b | c, b | c, b | a+45, b | c, b | c, b | c, b | c, b | c, b | c, b | c, d+36 | a, d+62 | a+45, b | c, d+22 |
| a | 480, 850 | 415, 645 | | 327, 805 | | | | | 306, 805 | | | | | | | | 331, 805 |
| a? | 450, 850 | 384, 645 | 209, 903 | 359, 850 | 303, 850 | 357, 850 | 376, 935 | 388, 850 | 424, 850 | 304, 850 | 320, 850 | 369, 850 | 293, 850 | 288, 850 | 269, 890 | 413, 935 | 316, 850 |
| a$\delta$ | 30, 0 | 31, 0 | | -32, -45 | | | | | -118, -45 | | | | | | | | 15, -45 |
| b | | | | 328, -108 | | | | | 369, -107 | | | | | | | | |
| b? | 417, -110 | 393, -110 | 121, -326 | 326, -110 | 270, -110 | 324, -110 | 326, -110 | 355, -110 | 391, -110 | 271, -110 | 287, -110 | 336, -110 | 260, -110 | 220, -281 | 198, -255 | 362, -110 | 245, -295 |
| b$\delta$ | | | | 2, 2 | | | | | -22, 3 | | | | | | | | |

Table 7: Font metrics for Latin and IPA 3/7

| hex | 0196 | 0197 | 0198 | 019C | 019D | 019F | 01B7 | 0237 | 0250 | 0251 | 0252 | 0253 | 0254 | 0255 | 0256 | 0257 | 0258 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | lll | ɨɨ | ĸĸĸ | ɯɯɯ | ɲɲɲ | ɵɵɵ | ʒʒʒ | jjj | ɐɐɐ | ɑɑɑ | ɒɒɒ | ɓɓɓ | ɔɔɔ | ɕɕɕ | ɖɖɖ | ɗɗɗ | ɘɘɘ |
| patch | lll | ɨɨ | ĸĸĸ | ɯɯɯ | ɲɲɲ | ɵɵɵ | ʒʒʒ | jjj | ɐɐɐ | ɑɑɑ | ɒɒɒ | ɓɓɓ | ɔɔɔ | ɕɕɕ | ɖɖɖ | ɗɗɗ | ɘɘɘ |
| ideal | a, b | c, b | c, b | c, b | c, d+45 | c, b | c+22, d+21 | x, d+9 | x, b | x, b | x, b | a, b | x, b | x, b-106 | a, d+44 | a, b | x, b |
| a | | | | 431, 801 | | | | 140, 645 | 228, 645 | 228, 645 | 264, 645 | 480, 630 | 216, 645 | 209, 645 | 218, 645 | 218, 645 | 204, 645 |
| a? | 152, 885 | 148, 850 | 339, 850 | 434, 850 | 324, 850 | 352, 850 | 303, 870 | 74, 645 | 216, 645 | 272, 645 | 247, 645 | 265, 885 | 216, 645 | 222, 645 | 314, 885 | 322, 885 | 220, 645 |
| a$\delta$ | | | | -2, -49 | | | | 66, 0 | 12, 0 | -44, 0 | 17, 0 | 215, -255 | 0, 0 | -13, 0 | -96, -240 | -104, -240 | -16, 0 |
| b | | | | 450, -108 | | 350, -110 | 272, -319 | | 229, -114 | 212, -114 | 259, -114 | 212, -114 | 177, -114 | 212, -114 | 212, -114 | 212, -114 | 212, -114 |
| b? | 152, -110 | 148, -110 | 339, -110 | 434, -110 | 324, -272 | 352, -110 | 303, -296 | 74, -308 | 216, -110 | 272, -110 | 247, -110 | 265, -110 | 216, -110 | 222, -214 | 314, -273 | 322, -110 | 220, -110 |
| b$\delta$ | | | | 16, 2 | | -2, 0 | -31, -23 | | 12, -4 | -60, -4 | 12, -4 | -53, -4 | -38, -4 | -10, 100 | -102, 159 | -110, -4 | -8, -4 |
| it | *lll* | *ɨɨ* | *ĸĸĸ* | *ɯɯɯ* | *ɲɲɲ* | *ɵɵɵ* | *ʒʒʒ* | *jjj* | *ɐɐɐ* | *ɑɑɑ* | *ɒɒɒ* | *ɓɓɓ* | *ɔɔɔ* | *ɕɕɕ* | *ɖɖɖ* | *ɗɗɗ* | *ɘɘɘ* |
| patch | *lll* | *ɨɨ* | *ĸĸĸ* | *ɯɯɯ* | *ɲɲɲ* | *ɵɵɵ* | *ʒʒʒ* | *jjj* | *ɐɐɐ* | *ɑɑɑ* | *ɒɒɒ* | *ɓɓɓ* | *ɔɔɔ* | *ɕɕɕ* | *ɖɖɖ* | *ɗɗɗ* | *ɘɘɘ* |
| ideal | a, b | c, b | c, b | c, b | c, d+45 | c, b | c+22, d+21 | x, d+36 | x+16, b | x, b | x, b | a+6, b | x, b | x, b-100 | a+9, d+44 | a, b | x, b |
| a | | | | 431, 801 | | | | 140, 645 | | | | | 322, 654 | | 218, 645 | 218, 645 | 204, 645 |
| a? | 322, 890 | 321, 850 | 499, 850 | 609, 850 | 528, 850 | 515, 850 | 489, 870 | 267, 645 | 368, 659 | 380, 645 | 391, 645 | 431, 896 | 332, 645 | 352, 645 | 465, 899 | 464, 890 | 356, 645 |
| a$\delta$ | | | | -178, -49 | | | | -127, 0 | | | | | -10, 9 | | -247, -254 | -246, -245 | -152, 0 |
| b | | | | 450, -108 | | | | | 229, -114 | 212, -114 | 243, -114 | 212, -114 | 172, -114 | 212, -114 | 212, -114 | 212, -114 | 212, -114 |
| b? | 110, -110 | 117, -110 | 295, -110 | 405, -110 | 290, -272 | 311, -110 | 241, -296 | 70, -281 | 205, -110 | 220, -110 | 231, -110 | 218, -110 | 171, -110 | 171, -208 | 215, -273 | 251, -110 | 195, -110 |
| b$\delta$ | | | | 45, 2 | | | | | 24, -4 | -8, -4 | 12, -4 | -6, -4 | 1, -4 | 41, 94 | -3, 159 | -39, -4 | 17, -4 |
| sb | **lll** | **ɨɨ** | **ĸĸĸ** | **ɯɯɯ** | **ɲɲɲ** | **ɵɵɵ** | **ʒʒʒ** | **jjj** | **ɐɐɐ** | **ɑɑɑ** | **ɒɒɒ** | **ɓɓɓ** | **ɔɔɔ** | **ɕɕɕ** | **ɖɖɖ** | **ɗɗɗ** | **ɘɘɘ** |
| patch | **lll** | **ɨɨ** | **ĸĸĸ** | **ɯɯɯ** | **ɲɲɲ** | **ɵɵɵ** | **ʒʒʒ** | **jjj** | **ɐɐɐ** | **ɑɑɑ** | **ɒɒɒ** | **ɓɓɓ** | **ɔɔɔ** | **ɕɕɕ** | **ɖɖɖ** | **ɗɗɗ** | **ɘɘɘ** |
| ideal | a, b | c, b | c, b | c, b | c, d+47 | c, b | a-24, d+24 | x, d | x, b-14 | x, b | x, b | a, b | x, b-14 | x, b-101 | a, d+46 | a, b | x, b |
| a | | | | | | | | 163, 645 | | | | | 199, 645 | | | | |
| a? | 170, 885 | 158, 805 | 338, 805 | 444, 805 | 336, 805 | 366, 805 | 316, 861 | 78, 645 | 207, 645 | 276, 645 | 251, 645 | 268, 885 | 199, 645 | 232, 645 | 322, 885 | 324, 885 | 230, 645 |
| a$\delta$ | | | | | | | | 85, 0 | | | | | 0, 0 | | | | |
| b | | | | | | | 272, -319 | | 228, -114 | | | | 184, -114 | | | | |
| b? | 170, -110 | 158, -110 | 338, -110 | 444, -110 | 336, -270 | 366, -110 | 316, -293 | 78, -319 | 207, -122 | 276, -110 | 251, -110 | 268, -110 | 199, -122 | 232, -209 | 322, -271 | 324, -110 | 230, -110 |
| b$\delta$ | | | | | | | -44, -26 | | 21, 8 | | | | -15, 8 | | | | |
| si | ***lll*** | ***ɨɨ*** | ***ĸĸĸ*** | ***ɯɯɯ*** | ***ɲɲɲ*** | ***ɵɵɵ*** | ***ʒʒʒ*** | ***jjj*** | ***ɐɐɐ*** | ***ɑɑɑ*** | ***ɒɒɒ*** | ***ɓɓɓ*** | ***ɔɔɔ*** | ***ɕɕɕ*** | ***ɖɖɖ*** | ***ɗɗɗ*** | ***ɘɘɘ*** |
| patch | ***lll*** | ***ɨɨ*** | ***ĸĸĸ*** | ***ɯɯɯ*** | ***ɲɲɲ*** | ***ɵɵɵ*** | ***ʒʒʒ*** | ***jjj*** | ***ɐɐɐ*** | ***ɑɑɑ*** | ***ɒɒɒ*** | ***ɓɓɓ*** | ***ɔɔɔ*** | ***ɕɕɕ*** | ***ɖɖɖ*** | ***ɗɗɗ*** | ***ɘɘɘ*** |
| ideal | a, b | c, b | c, b | c, b | c, d+47 | c, b | a-25, d+24 | x, d+25 | x, b | x, b | x, b | a, b | x, b | x, b-101 | a, d+46 | a, b | x, b |
| a | | | | 431, 801 | | | | 140, 645 | | 228, 645 | 264, 645 | | 160, 645 | | 218, 645 | 218, 645 | 204, 645 |
| a? | 194, 890 | 181, 850 | 353, 850 | 463, 850 | 383, 850 | 387, 850 | 332, 865 | 121, 645 | 221, 645 | 247, 645 | 247, 645 | 284, 890 | 199, 645 | 204, 645 | 302, 890 | 318, 890 | 207, 645 |
| a$\delta$ | | | | -32, -49 | | | | 19, 0 | | -19, 0 | 17, 0 | | -39, 0 | | -84, -245 | -100, -245 | -3, 0 |
| b | | | | 450, -108 | | | 236, -319 | | 229, -114 | 212, -114 | 259, -114 | 212, -114 | 177, -114 | 212, -114 | 212, -114 | 212, -114 | 212, -114 |
| b? | 153, -110 | 148, -110 | 320, -110 | 430, -110 | 317, -270 | 354, -110 | 258, -293 | 93, -292 | 230, -110 | 256, -110 | 256, -110 | 243, -110 | 207, -110 | 192, -209 | 227, -271 | 276, -110 | 216, -110 |
| b$\delta$ | | | | 20, 2 | | | -22, -26 | | -1, -4 | -44, -4 | 3, -4 | -31, -4 | -30, -4 | 20, 95 | -15, 157 | -64, -4 | -4, -4 |

Table 8: Font metrics for Latin and IPA 4/7

| hex | 0259 | 025A | 025B | 025C | 025D | 025E | 025F | 0260 | 0261 | 0262 | 0263 | 0264 | 0265 | 0266 | 0267 | 0268 | 0269 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | ə ɚ ɛ | ɝ ɚ ɜ | ɛ ɛ ɜ | ɜ ɜ ɜ | ɝ ɝ ɝ | ɵ ɵ ɞ | ɟ ɟ ɟ | ɠ ɠ ɠ | ɡ ɡ ɡ | ɢ ɢ ɢ | ɣ ɣ ɣ | ɤ ɤ ɤ | ɥ ɥ ɥ | ɦ ɦ ɦ | ɧ ɧ ɧ | ɨ ɨ ɨ | ᵻ ᵻ ᵻ |
| patch | ə ɚ ɛ | ɝ ɚ ɜ | ɛ ɛ ɜ | ɜ ɜ ɜ | ɝ ɝ ɝ | ɵ ɵ ɞ | ɟ ɟ ɟ | ɠ ɠ ɠ | ɡ ɡ ɡ | ɢ ɢ ɢ | ɣ ɣ ɣ | ɤ ɤ ɤ | ɥ ɥ ɥ | ɦ ɦ ɦ | ɧ ɧ ɧ | ɨ ɨ ɨ | ᵻ ᵻ ᵻ |
| ideal | x, b | x+45, b | x, b | x, b | a-90, b | x, b | x, d+26 | a, d | x, d | x, b | x, d+30 | x, b | x, d | a, b | a, d+11 | a-51, b | x, b |
| a | 195, 645 | 229, 645 | 209, 645 | 194, 645 | 209, 645 | 239, 645 | 169, 645 | 209, 645 | 239, 645 | 265, 645 | 259, 645 | 248, 645 | 268, 645 | 299, 565 | 299, 565 | 155, 795 | 123, 645 |
| a? | 232, 645 | 306, 688 | 202, 645 | 196, 645 | 234, 795 | 242, 645 | 124, 645 | 320, 885 | 228, 645 | 259, 645 | 258, 645 | 221, 645 | 272, 645 | 262, 885 | 228, 885 | 150, 834 | 170, 645 |
| a$\delta$ | -37, 0 | -78, -43 | 8, 0 | -2, 0 | -26, -150 | -2, 0 | 44, 0 | -112, -240 | 11, 0 | 6, 0 | 1, 0 | 27, 0 | -4, 0 | 36, -320 | 72, -320 | 6, -39 | -47, 0 |
| b | 186, -114 | 212, -114 | 212, -114 | 184, -114 | 212, -114 | 256, -114 | 196, -110 | 236, -110 | 226, -320 | 248, -110 | 251, -277 | 216, -114 | 219, -129 | 240, -114 | 240, -114 | 134, -114 | 153, -114 |
| b? | 232, -110 | 306, -110 | 202, -110 | 196, -110 | 234, -110 | 242, -110 | 124, -291 | 320, -319 | 228, -319 | 259, -110 | 258, -287 | 221, -110 | 272, -319 | 262, -110 | 228, -306 | 150, -110 | 170, -110 |
| b$\delta$ | -46, -4 | -94, -4 | 10, -4 | -12, -4 | -22, -4 | 14, -4 | 72, 181 | -84, 209 | -2, -1 | -11, 0 | -7, 10 | -5, -4 | -52, 190 | -22, -4 | 12, 192 | -16, -4 | -17, -4 |
| it | *ə ɚ ɛ* | *ɝ ɚ ɜ* | *ɛ ɛ ɜ* | *ɜ ɜ ɜ* | *ɝ ɝ ɝ* | *ɵ ɵ ɞ* | *ɟ ɟ ɟ* | *ɠ ɠ ɠ* | *ɡ ɡ ɡ* | *ɢ ɢ ɢ* | *ɣ ɣ ɣ* | *ɤ ɤ ɤ* | *ɥ ɥ ɥ* | *ɦ ɦ ɦ* | *ɧ ɧ ɧ* | *ɨ ɨ ɨ* | *ᵻ ᵻ ᵻ* |
| patch | *ə ɚ ɛ* | *ɝ ɚ ɜ* | *ɛ ɛ ɜ* | *ɜ ɜ ɜ* | *ɝ ɝ ɝ* | *ɵ ɵ ɞ* | *ɟ ɟ ɟ* | *ɠ ɠ ɠ* | *ɡ ɡ ɡ* | *ɢ ɢ ɢ* | *ɣ ɣ ɣ* | *ɤ ɤ ɤ* | *ɥ ɥ ɥ* | *ɦ ɦ ɦ* | *ɧ ɧ ɧ* | *ɨ ɨ ɨ* | *ᵻ ᵻ ᵻ* |
| ideal | x, b | x+44, b | x, b | x, b | x+132, b | x+17, b-25 | x, d | a+6, d+31 | x+17, d+31 | x, b | x, d+45 | x, b+2 | x, d | a, b | a, d | a-80, b | x, b |
| a | | | 322, 654 | | | | | | | 265, 645 | | | 258, 645 | 299, 565 | | | 123, 645 |
| a? | 356, 645 | 418, 687 | 338, 645 | 331, 645 | 413, 775 | 378, 660 | 305, 645 | 431, 896 | 381, 660 | 388, 645 | 387, 645 | 338, 645 | 396, 645 | 449, 890 | 449, 890 | 316, 810 | 266, 645 |
| a$\delta$ | | | -16, 9 | | | | | | | -123, 0 | | | -138, 0 | -150, -325 | | | -143, 0 |
| b | 212, -114 | 212, -114 | 212, -114 | 184, -114 | 212, -114 | 256, -114 | | | | 248, -114 | | 177, -114 | 219, -129 | 240, -114 | | 134, -114 | 153, -114 |
| b? | 195, -110 | 248, -110 | 178, -110 | 170, -110 | 225, -110 | 209, -133 | 100, -319 | 180, -286 | 180, -286 | 227, -110 | 192, -272 | 178, -106 | 191, -319 | 236, -110 | 192, -319 | 121, -110 | 106, -110 |
| b$\delta$ | 17, -4 | -36, -4 | 34, -4 | 14, -4 | -13, -4 | 47, 19 | | | | 21, -4 | | -1, -8 | 28, 190 | 4, -4 | | 13, -4 | 47, -4 |
| sb | **ə ɚ ɛ** | **ɝ ɚ ɜ** | **ɛ ɛ ɜ** | **ɜ ɜ ɜ** | **ɝ ɝ ɝ** | **ɵ ɵ ɞ** | **ɟ ɟ ɟ** | **ɠ ɠ ɠ** | **ɡ ɡ ɡ** | **ɢ ɢ ɢ** | **ɣ ɣ ɣ** | **ɤ ɤ ɤ** | **ɥ ɥ ɥ** | **ɦ ɦ ɦ** | **ɧ ɧ ɧ** | **ɨ ɨ ɨ** | **ᵻ ᵻ ᵻ** |
| patch | **ə ɚ ɛ** | **ɝ ɚ ɜ** | **ɛ ɛ ɜ** | **ɜ ɜ ɜ** | **ɝ ɝ ɝ** | **ɵ ɵ ɞ** | **ɟ ɟ ɟ** | **ɠ ɠ ɠ** | **ɡ ɡ ɡ** | **ɢ ɢ ɢ** | **ɣ ɣ ɣ** | **ɤ ɤ ɤ** | **ɥ ɥ ɥ** | **ɦ ɦ ɦ** | **ɧ ɧ ɧ** | **ɨ ɨ ɨ** | **ᵻ ᵻ ᵻ** |
| ideal | x, b | x+50, b | x, b | x, b | a-128, b | x, b | x, d | a+6, d+33 | x, d | x+23, b | x, d+32 | x, b | x, d | a, b | a, d | a-67, b | x, b |
| a | | | 229, 645 | | | | | | | | | | | | | | |
| a? | 228, 645 | 319, 693 | 212, 645 | 216, 645 | 276, 757 | 249, 645 | 144, 645 | 336, 891 | 235, 645 | 266, 666 | 272, 645 | 202, 645 | 278, 645 | 274, 885 | 243, 885 | 152, 818 | 164, 645 |
| a$\delta$ | | | 16, 0 | | | | | | | | | | | | | | |
| b | 212, -114 | | | | | | | | | | | | | | | 134, -114 | |
| b? | 228, -110 | 319, -110 | 212, -110 | 216, -110 | 276, -110 | 249, -110 | 144, -319 | 336, -284 | 235, -319 | 266, -110 | 272, -285 | 202, -110 | 278, -319 | 274, -110 | 243, -319 | 152, -110 | 164, -110 |
| b$\delta$ | -16, -4 | | | | | | | | | | | | | | | -18, -4 | |
| si | ***ə ɚ ɛ*** | ***ɝ ɚ ɜ*** | ***ɛ ɛ ɜ*** | ***ɜ ɜ ɜ*** | ***ɝ ɝ ɝ*** | ***ɵ ɵ ɞ*** | ***ɟ ɟ ɟ*** | ***ɠ ɠ ɠ*** | ***ɡ ɡ ɡ*** | ***ɢ ɢ ɢ*** | ***ɣ ɣ ɣ*** | ***ɤ ɤ ɤ*** | ***ɥ ɥ ɥ*** | ***ɦ ɦ ɦ*** | ***ɧ ɧ ɧ*** | ***ɨ ɨ ɨ*** | ***ᵻ ᵻ ᵻ*** |
| patch | ***ə ɚ ɛ*** | ***ɝ ɚ ɜ*** | ***ɛ ɛ ɜ*** | ***ɜ ɜ ɜ*** | ***ɝ ɝ ɝ*** | ***ɵ ɵ ɞ*** | ***ɟ ɟ ɟ*** | ***ɠ ɠ ɠ*** | ***ɡ ɡ ɡ*** | ***ɢ ɢ ɢ*** | ***ɣ ɣ ɣ*** | ***ɤ ɤ ɤ*** | ***ɥ ɥ ɥ*** | ***ɦ ɦ ɦ*** | ***ɧ ɧ ɧ*** | ***ɨ ɨ ɨ*** | ***ᵻ ᵻ ᵻ*** |
| ideal | x, b | x+50, b | x, b | x, b | a-128, b | x, b | x, d | a, d+33 | x, d | x+23, b | x, d+32 | x, b | x, d | a, b | a, d | a-67, b | x, b |
| a | | | 209, 645 | 194, 645 | | | | | | 265, 645 | | | 258, 645 | 299, 565 | 299, 565 | 137, 795 | 123, 645 |
| a? | 218, 645 | 299, 693 | 194, 645 | 194, 645 | 261, 762 | 232, 645 | 141, 645 | 309, 890 | 238, 645 | 248, 666 | 255, 645 | 185, 645 | 258, 645 | 302, 890 | 301, 890 | 180, 823 | 138, 645 |
| a$\delta$ | | | 15, 0 | 0, 0 | | | | | | 17, -21 | | | 0, 0 | -3, -325 | -2, -325 | -43, -28 | -15, 0 |
| b | | 212, -114 | 212, -114 | 184, -114 | 212, -114 | 256, -114 | | | | 248, -110 | | 177, -114 | 219, -129 | 240, -114 | 240, -114 | 136, -110 | 153, -114 |
| b? | 227, -110 | 298, -110 | 203, -110 | 203, -110 | 246, -110 | 241, -110 | 107, -319 | 232, -284 | 204, -319 | 252, -110 | 228, -285 | 194, -110 | 224, -319 | 260, -110 | 217, -319 | 153, -110 | 147, -110 |
| b$\delta$ | | -86, -4 | 9, -4 | -19, -4 | -34, -4 | 15, -4 | | | | -4, 0 | | -17, -4 | -5, 190 | -20, -4 | 23, 205 | -17, 0 | 6, -4 |

Table 9: Font metrics for Latin and IPA 5/7

| hex | 026A | 026B | 026C | 026D | 026E | 026F | 0270 | 0271 | 0272 | 0273 | 0274 | 0275 | 0277 | 0278 | 0279 | 027A | 027B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | ɪi̱ | ɬi̱ | ɬɬi̱ | ꞎ | ɮʑʒ | ɯu̇ɯ | ɰu̇ɰ | ɱɲ̇ɱ | ɲɲ̇ɲ | ɳɳ̇ɳ | ɴɴ̇ɴ | ɵɘ̟ɵ | ѡѡ̇ѡ | ɸɸɸ | ɹɾɾ | ɻɻ̇ɻ | ɽɽ̇ɽ |
| patch | ɪi̱ | ɬi̱ | ɬɬi̱ | ꞎ | ɮʑʒ | ɯu̇ɯ | ɰu̇ɰ | ɱɲ̇ɱ | ɲɲ̇ɲ | ɳɳ̇ɳ | ɴɴ̇ɴ | ɵɘ̟ɵ | ѡѡ̇ѡ | ɸɸɸ | ɹɾɾ | ɻɻ̇ɻ | ɽɽ̇ɽ |
| ideal | x, b | a, b | a, b | a, d+44 | a, d | x, b-13 | x, d | x, d+11 | x, d+11 | x, d+11 | x, b | x, b | x, b | a, d | x, b-13 | a-51, b-13 | x, d+58 |
| a | 133, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 407, 645 | 209, 645 | 383, 645 | 264, 646 | 263, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | |
| a? | 136, 645 | 144, 885 | 182, 885 | 182, 885 | 250, 885 | 390, 645 | 379, 645 | 366, 645 | 234, 645 | 296, 645 | 278, 645 | 252, 645 | 357, 645 | 286, 885 | 178, 645 | 180, 834 | 226, 645 |
| a$\delta$ | -3, 0 | 65, -240 | 27, -240 | 27, -240 | -42, -240 | 16, 0 | -170, 0 | 18, 0 | 30, 1 | -32, 0 | -70, 0 | -43, 0 | -148, 0 | -76, -240 | 32, 0 | 30, -189 | |
| b | 139, -114 | 146, -114 | 218, -120 | 196, -110 | 196, -110 | 410, -114 | 294, -114 | 383, -105 | 265, -111 | 262, -112 | 247, -114 | 196, -110 | 341, -114 | 196, -110 | 190, -114 | 203, -114 | 196, -110 |
| b? | 136, -110 | 144, -110 | 182, -110 | 182, -273 | 250, -319 | 390, -121 | 379, -319 | 366, -306 | 234, -306 | 296, -306 | 278, -110 | 252, -110 | 357, -110 | 286, -319 | 178, -121 | 180, -121 | 226, -259 |
| b$\delta$ | 3, -4 | 2, -4 | 36, -10 | 14, 163 | -54, 209 | 20, 7 | -85, 205 | 18, 201 | 30, 195 | -34, 194 | -32, -4 | -56, 0 | -16, -4 | -90, 209 | 12, 7 | 24, 7 | -30, 149 |
| it | *ɪi̱* | *ɬi̱* | *ɬɬi̱* | *ꞎ* | *ɮʑʒ* | *ɯu̇ɯ* | *ɰu̇ɰ* | *ɱɲ̇ɱ* | *ɲɲ̇ɲ* | *ɳɳ̇ɳ* | *ɴɴ̇ɴ* | *ɵɘ̟ɵ* | *ѡѡ̇ѡ* | *ɸɸɸ* | *ɹɾɾ* | *ɻɻ̇ɻ* | *ɽɽ̇ɽ* |
| patch | *ɪi̱* | *ɬi̱* | *ɬɬi̱* | *ꞎ* | *ɮʑʒ* | *ɯu̇ɯ* | *ɰu̇ɰ* | *ɱɲ̇ɱ* | *ɲɲ̇ɲ* | *ɳɳ̇ɳ* | *ɴɴ̇ɴ* | *ɵɘ̟ɵ* | *ѡѡ̇ѡ* | *ɸɸɸ* | *ɹɾɾ* | *ɻɻ̇ɻ* | *ɽɽ̇ɽ* |
| ideal | x, b | a, b | a, b | a, d+30 | a, d+8 | x, b-15 | x, d | x+21, d | x+19, d | x, d | x, b | x+14, b | x, b-18 | a, d+32 | x, b | a-71, b-19 | x, d+58 |
| a | 128, 645 | | | | | 407, 645 | | 383, 645 | 264, 646 | 263, 645 | | | | | | | |
| a? | 281, 645 | 333, 890 | 368, 890 | 341, 890 | 447, 890 | 517, 645 | 515, 645 | 519, 664 | 400, 662 | 367, 645 | 418, 645 | 381, 657 | 494, 645 | 481, 890 | 315, 645 | 362, 819 | 352, 645 |
| a$\delta$ | -153, 0 | | | | | -110, 0 | | -136, -19 | -136, -16 | -104, 0 | | | | | | | |
| b | 139, -114 | 146, -114 | 218, -120 | | | 410, -114 | 294, -114 | 383, -105 | 265, -111 | 262, -112 | 247, -114 | | 341, -114 | | 190, -114 | 203, -114 | |
| b? | 120, -110 | 120, -110 | 156, -110 | 91, -287 | 192, -309 | 354, -123 | 310, -319 | 310, -319 | 191, -319 | 162, -319 | 258, -110 | 218, -110 | 330, -126 | 231, -285 | 155, -110 | 161, -127 | 160, -259 |
| b$\delta$ | 19, -4 | 26, -4 | 62, -10 | | | 56, 9 | -16, 205 | 73, 214 | 74, 208 | 100, 207 | -11, -4 | | 11, 12 | | 35, -4 | 42, 13 | |
| sb | **ɪi̱** | **ɬi̱** | **ɬɬi̱** | **ꞎ** | **ɮʑʒ** | **ɯu̇ɯ** | **ɰu̇ɰ** | **ɱɲ̇ɱ** | **ɲɲ̇ɲ** | **ɳɳ̇ɳ** | **ɴɴ̇ɴ** | **ɵɘ̟ɵ** | **ѡѡ̇ѡ** | **ɸɸɸ** | **ɹɾɾ** | **ɻɻ̇ɻ** | **ɽɽ̇ɽ** |
| patch | **ɪi̱** | **ɬi̱** | **ɬɬi̱** | **ꞎ** | **ɮʑʒ** | **ɯu̇ɯ** | **ɰu̇ɰ** | **ɱɲ̇ɱ** | **ɲɲ̇ɲ** | **ɳɳ̇ɳ** | **ɴɴ̇ɴ** | **ɵɘ̟ɵ** | **ѡѡ̇ѡ** | **ɸɸɸ** | **ɹɾɾ** | **ɻɻ̇ɻ** | **ɽɽ̇ɽ** |
| ideal | x, b | a, b | a, b | a, d | a, d | x-2, b-17 | x, d | x, d | x, d | x, d | x, b | x, b | x, b | a, d+34 | x-2, b-17 | a-69, b | x, d+60 |
| a | 146, 645 | | | | | 377, 645 | | | | | | | | | | | |
| a? | 152, 645 | 156, 885 | 190, 885 | 194, 885 | 274, 885 | 354, 641 | 388, 645 | 363, 645 | 246, 645 | 306, 645 | 290, 645 | 260, 645 | 369, 645 | 297, 885 | 144, 641 | 205, 816 | 238, 645 |
| a$\delta$ | -6, 0 | | | | | 23, 4 | | | | | | | | | | | |
| b | 156, -114 | 146, -114 | 218, -120 | | | 380, -114 | | | | | | 229, -114 | | | 190, -114 | | |
| b? | 152, -110 | 156, -110 | 190, -110 | 194, -319 | 274, -319 | 354, -125 | 388, -319 | 363, -319 | 246, -319 | 306, -319 | 290, -110 | 260, -110 | 369, -110 | 297, -283 | 144, -125 | 205, -110 | 238, -257 |
| b$\delta$ | 4, -4 | -10, -4 | 28, -10 | | | 26, 11 | | | | | | -30, -4 | | | 46, 11 | | |
| si | ***ɪi̱*** | ***ɬi̱*** | ***ɬɬi̱*** | ***ꞎ*** | ***ɮʑʒ*** | ***ɯu̇ɯ*** | ***ɰu̇ɰ*** | ***ɱɲ̇ɱ*** | ***ɲɲ̇ɲ*** | ***ɳɳ̇ɳ*** | ***ɴɴ̇ɴ*** | ***ɵɘ̟ɵ*** | ***ѡѡ̇ѡ*** | ***ɸɸɸ*** | ***ɹɾɾ*** | ***ɻɻ̇ɻ*** | ***ɽɽ̇ɽ*** |
| patch | ***ɪi̱*** | ***ɬi̱*** | ***ɬɬi̱*** | ***ꞎ*** | ***ɮʑʒ*** | ***ɯu̇ɯ*** | ***ɰu̇ɰ*** | ***ɱɲ̇ɱ*** | ***ɲɲ̇ɲ*** | ***ɳɳ̇ɳ*** | ***ɴɴ̇ɴ*** | ***ɵɘ̟ɵ*** | ***ѡѡ̇ѡ*** | ***ɸɸɸ*** | ***ɹɾɾ*** | ***ɻɻ̇ɻ*** | ***ɽɽ̇ɽ*** |
| ideal | x, b | a, b | a, b | a, d | a, d | x-3, b | x, d | x, d | x, d | x, d | x, b | x, b | x, b | a, d+34 | x-3, b | a-69, b | x, d+60 |
| a | 128, 645 | | | | | 407, 645 | | 383, 645 | 264, 646 | 263, 645 | | | | | | | |
| a? | 137, 645 | 187, 890 | 223, 890 | 187, 890 | 301, 890 | 372, 640 | 373, 645 | 373, 645 | 254, 645 | 254, 645 | 275, 645 | 229, 645 | 350, 645 | 328, 890 | 171, 640 | 217, 821 | 194, 645 |
| a$\delta$ | -9, 0 | | | | | 35, 5 | | 10, 0 | 10, 1 | 9, 0 | | | | | | | |
| b | 139, -114 | 146, -114 | 218, -120 | | | 410, -114 | 294, -114 | 383, -105 | 265, -111 | 262, -112 | 247, -114 | | 341, -114 | | 190, -114 | 203, -114 | |
| b? | 145, -110 | 145, -110 | 181, -110 | 103, -319 | 217, -319 | 382, -110 | 339, -319 | 339, -319 | 220, -319 | 220, -319 | 283, -110 | 238, -110 | 359, -110 | 251, -283 | 180, -110 | 190, -110 | 172, -257 |
| b$\delta$ | -6, -4 | 1, -4 | 37, -10 | | | 28, -4 | -45, 205 | 44, 214 | 45, 208 | 42, 207 | -36, -4 | | -18, -4 | | 10, -4 | 13, -4 | |

Table 10: Font metrics for Latin and IPA 6/7

| hex | 027C | 027D | 027E | 027F | 0280 | 0281 | 0282 | 0283 | 0284 | 0285 | 0286 | 0287 | 0288 | 0289 | 028A | 028B | 028C |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | ɼ | ɽ | ɾ | ɿ | ʀ | ʁ | ʂ | ʃ | ʄ | ʅ | ʆ | ʇ | ʈ | ʉ | ʊ | ʋ | ʌ |
| patch | ɼ | ɽ | ɾ | ɿ | ʀ | ʁ | ʂ | ʃ | ʄ | ʅ | ʆ | ʇ | ʈ | ʉ | ʊ | ʋ | ʌ |
| ideal | x, d | x, d | x, b | x, b | x, b | x, b | x, d+56 | a, d+18 | a, d+18 | x+121, d+18 | a, d+7 | x, d+79 | a-114, d+11 | x, b | x, b | x, b | x, b |
| a | 222, 703 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 259, 645 | 209, 645 |
| a? | 186, 645 | 186, 645 | 174, 645 | 170, 645 | 238, 645 | 238, 645 | 199, 645 | 140, 885 | 140, 885 | 170, 764 | 150, 885 | 150, 645 | 158, 771 | 271, 645 | 254, 645 | 235, 645 | 244, 645 |
| aδ | 36, 58 | 22, 0 | 34, 0 | 38, 0 | -30, 0 | -30, 0 | 10, 0 | 68, -240 | 68, -240 | 38, -119 | 60, -240 | 59, 0 | 52, -126 | -62, 0 | -44, 0 | 24, 0 | -35, 0 |
| b | 127, -305 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 258, -114 | 251, -114 | 196, -110 |
| b? | 186, -319 | 186, -319 | 174, -110 | 170, -110 | 238, -110 | 238, -110 | 199, -261 | 140, -299 | 140, -299 | 170, -299 | 150, -310 | 150, -238 | 158, -306 | 271, -110 | 254, -110 | 235, -110 | 244, -110 |
| bδ | -60, 14 | 10, 209 | 22, 0 | 26, 0 | -42, 0 | -42, 0 | -3, 151 | 56, 189 | 56, 189 | 26, 189 | 46, 200 | 46, 128 | 38, 196 | -75, 0 | 4, -4 | 16, -4 | -48, 0 |
| it | *ɼ* | *ɽ* | *ɾ* | *ɿ* | *ʀ* | *ʁ* | *ʂ* | *ʃ* | *ʄ* | *ʅ* | *ʆ* | *ʇ* | *ʈ* | *ʉ* | *ʊ* | *ʋ* | *ʌ* |
| patch | *ɼ* | *ɽ* | *ɾ* | *ɿ* | *ʀ* | *ʁ* | *ʂ* | *ʃ* | *ʄ* | *ʅ* | *ʆ* | *ʇ* | *ʈ* | *ʉ* | *ʊ* | *ʋ* | *ʌ* |
| ideal | x+19, d+34 | x+17, d+38 | x, b | x, b | x, b | x, b | x+15, d+52 | a+10, d+18 | a+10, d+18 | a+10, d+18 | a+10, d+34 | x, d+78 | a-114, d+6 | x, b | x, b-13 | x, b-16 | x, b |
| a | 222, 703 | | | | | | | | | | | | | | | 259, 645 | |
| a? | 329, 662 | 323, 660 | 311, 645 | 307, 645 | 358, 645 | 349, 645 | 317, 658 | 343, 900 | 343, 900 | 343, 900 | 343, 900 | 288, 645 | 314, 776 | 398, 645 | 392, 645 | 382, 645 | 383, 645 |
| aδ | -107, 41 | | | | | | | | | | | | | | | -123, 0 | |
| b | 127, -305 | | | | | | | | | | | | | | 258, -114 | 251, -114 | |
| b? | 128, -283 | 124, -279 | 151, -110 | 147, -110 | 197, -110 | 189, -110 | 121, -265 | 88, -299 | 88, -299 | 88, -299 | 92, -283 | 100, -239 | 83, -311 | 238, -110 | 229, -121 | 218, -124 | 222, -110 |
| bδ | -1, -22 | | | | | | | | | | | | | | 29, 7 | 33, 10 | |
| sb | **ɼ** | **ɽ** | **ɾ** | **ɿ** | **ʀ** | **ʁ** | **ʂ** | **ʃ** | **ʄ** | **ʅ** | **ʆ** | **ʇ** | **ʈ** | **ʉ** | **ʊ** | **ʋ** | **ʌ** |
| patch | **ɼ** | **ɽ** | **ɾ** | **ɿ** | **ʀ** | **ʁ** | **ʂ** | **ʃ** | **ʄ** | **ʅ** | **ʆ** | **ʇ** | **ʈ** | **ʉ** | **ʊ** | **ʋ** | **ʌ** |
| ideal | x, d+36 | x, d | x, b | x, b | x, b | x, b | x, d+53 | a, d+21 | a, d+21 | a, d+21 | a, d | x, d+67 | a-103, d | x, b | x, b | x, b | x, b |
| a | | | | | | | | | | | | | | | 266, 645 | | |
| a? | 206, 645 | 198, 645 | 186, 645 | 182, 645 | 236, 645 | 220, 645 | 201, 645 | 152, 885 | 152, 885 | 152, 885 | 162, 885 | 157, 645 | 182, 782 | 274, 645 | 267, 645 | 256, 645 | 231, 645 |
| aδ | | | | | | | | | | | | | | | -1, 0 | | |
| b | | | | | | | | | | | | | | | | | |
| b? | 206, -281 | 198, -319 | 186, -110 | 182, -110 | 236, -110 | 220, -110 | 201, -264 | 152, -296 | 152, -296 | 152, -296 | 162, -319 | 157, -250 | 182, -319 | 274, -110 | 267, -110 | 256, -110 | 231, -110 |
| bδ | | | | | | | | | | | | | | | | | |
| si | ***ɼ*** | ***ɽ*** | ***ɾ*** | ***ɿ*** | ***ʀ*** | ***ʁ*** | ***ʂ*** | ***ʃ*** | ***ʄ*** | ***ʅ*** | ***ʆ*** | ***ʇ*** | ***ʈ*** | ***ʉ*** | ***ʊ*** | ***ʋ*** | ***ʌ*** |
| patch | ***ɼ*** | ***ɽ*** | ***ɾ*** | ***ɿ*** | ***ʀ*** | ***ʁ*** | ***ʂ*** | ***ʃ*** | ***ʄ*** | ***ʅ*** | ***ʆ*** | ***ʇ*** | ***ʈ*** | ***ʉ*** | ***ʊ*** | ***ʋ*** | ***ʌ*** |
| ideal | x, d+36 | x, d | x, b | x, b | x-1, b | x, b | x, d+52 | a, d+21 | a, d+21 | a, d+21 | a, d | x, d+67 | a-103, d | x, b | x-1, b | x, b | x, b |
| a | 222, 703 | | | | | | | | | | | | | | | 259, 645 | |
| a? | 181, 645 | 177, 645 | 167, 645 | 167, 645 | 214, 642 | 214, 645 | 178, 645 | 195, 890 | 195, 890 | 195, 890 | 189, 890 | 149, 645 | 172, 787 | 255, 645 | 248, 642 | 241, 645 | 239, 645 |
| aδ | 41, 58 | | | | | | | | | | | | | | | 18, 0 | |
| b | 127, -305 | | | | | | | | | | | | | | 258, -114 | 251, -114 | |
| b? | 155, -281 | 143, -319 | 176, -110 | 176, -110 | 223, -110 | 223, -110 | 155, -265 | 116, -296 | 116, -296 | 116, -296 | 105, -319 | 129, -250 | 109, -319 | 264, -110 | 257, -110 | 249, -110 | 247, -110 |
| bδ | -28, -24 | | | | | | | | | | | | | | 1, -4 | 2, -4 | |

Table 11: Font metrics for Latin and IPA 7/7

| hex | 028D | 028E | 028F | 0290 | 0291 | 0292 | 0293 | 0294 | 0295 | 0296 | 0297 | 0299 | 029B | 029C | 029D | 029F |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Lu | Lu | Lu | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | ɱ̇ɱ̇ | ʎ̊ʎ̊ | ʏ̇ʏ̇ | ʐ̈ʐ̈ | ʑ̇ʑ̇ | ʒ̈ʒ̈ | ʒ̇ʒ̇ | ??? | ʏʏʏ | ꞎꞎꞎ | ⅽ̇ⅽ | ʙ̈ʙ̈ | ɢ̈ɢ̈ | ʜ̇ʜ̇ | j̈j̈ | ʟ̇ʟ̇ |
| patch | ɱ̇ɱ̇ | ʎ̊ʎ̊ | ʏ̇ʏ̇ | ʐ̈ʐ̈ | ʑ̇ʑ̇ | ʒ̈ʒ̈ | ʒ̇ʒ̇ | ??? | ʏʏʏ | ꞎꞎꞎ | ⅽ̇ⅽ | ʙ̈ʙ̈ | ɢ̈ɢ̈ | ʜ̇ʜ̇ | j̈j̈ | ʟ̇ʟ̇ |
| ideal | x, b | a-37, b | x, b | x, d+47 | x+19, d+110 | x, d | x, d | c, b | c, b | c, b | x, d+11 | x, b | a-121, b | x, b | a-51, d+55 | x, b |
| a | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 218, 646 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 |
| a? | 367, 645 | 248, 848 | 236, 645 | 262, 645 | 218, 662 | 216, 645 | 220, 645 | 211, 850 | 221, 850 | 208, 850 | 221, 645 | 232, 645 | 294, 764 | 293, 645 | 126, 834 | 207, 645 |
| a$\delta$ | -158, 0 | -39, -203 | -27, 0 | -54, 0 | -8, -17 | 2, 1 | -11, 0 | -2, -205 | -12, -205 | 1, -205 | -12, 0 | -24, 0 | -85, -119 | -84, 0 | 83, -189 | 2, 0 |
| b | 196, -110 | 196, -110 | 239, -107 | 196, -110 | 196, -110 | 185, -319 | 196, -110 | 202, -107 | 230, -107 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 305, -110 | 196, -110 | 196, -110 |
| b? | 367, -110 | 248, -110 | 236, -110 | 262, -270 | 218, -207 | 216, -319 | 220, -319 | 211, -110 | 221, -110 | 208, -110 | 221, -306 | 232, -110 | 294, -110 | 293, -110 | 126, -262 | 207, -110 |
| b$\delta$ | -171, 0 | -52, 0 | 3, 3 | -66, 160 | -22, 97 | -32, 0 | -24, 209 | -9, 3 | 9, 3 | -12, 0 | -25, 196 | -36, 0 | -98, 0 | 12, 0 | 70, 152 | -11, 0 |
| it | ɱ̇ɱ̇ | ʎ̊ʎ̊ | ʏ̇ʏ̇ | ʐ̈ʐ̈ | ʑ̇ʑ̇ | ʒ̈ʒ̈ | ʒ̇ʒ̇ | ??? | ʏʏʏ | ꞎꞎꞎ | ⅽ̇ⅽ | ʙ̈ʙ̈ | ɢ̈ɢ̈ | ʜ̇ʜ̇ | j̈j̈ | ʟ̇ʟ̇ |
| patch | ɱ̇ɱ̇ | ʎ̊ʎ̊ | ʏ̇ʏ̇ | ʐ̈ʐ̈ | ʑ̇ʑ̇ | ʒ̈ʒ̈ | ʒ̇ʒ̇ | ??? | ʏʏʏ | ꞎꞎꞎ | ⅽ̇ⅽ | ʙ̈ʙ̈ | ɢ̈ɢ̈ | ʜ̇ʜ̇ | j̈j̈ | ʟ̇ʟ̇ |
| ideal | x, b | a-52, b | x, b | x+19, d+41 | x+19, d+110 | x+15, d | x, d-80 | a-26, b | a-15, b | c, b-23 | x, d | x, b | a-94, b | x, b | a-80, d+63 | x, b |
| a | | | | | | 328, 646 | | | | | | | | | | |
| a? | 506, 645 | 433, 838 | 372, 645 | 346, 662 | 355, 662 | 353, 658 | 355, 645 | 376, 864 | 378, 875 | 373, 850 | 411, 645 | 352, 645 | 410, 796 | 416, 645 | 303, 810 | 355, 645 |
| a$\delta$ | | | | | | -25, -12 | | | | | | | | | | |
| b | | | 239, -107 | | | 130, -319 | | 202, -107 | 230, -107 | | | | | 273, -110 | | |
| b? | 346, -110 | 231, -110 | 212, -110 | 147, -276 | 171, -207 | 145, -319 | 133, -397 | 169, -110 | 169, -110 | 164, -131 | 206, -319 | 192, -110 | 217, -110 | 256, -110 | 77, -254 | 194, -110 |
| b$\delta$ | | | 27, 3 | | | -15, 0 | | 33, 3 | 61, 3 | | | | | 17, 0 | | |
| sb | ɱ̇ɱ̇ | ʎ̊ʎ̊ | ʏ̇ʏ̇ | ʐ̈ʐ̈ | ʑ̇ʑ̇ | ʒ̈ʒ̈ | ʒ̇ʒ̇ | ??? | ʏʏʏ | ꞎꞎꞎ | ⅽ̇ⅽ | ʙ̈ʙ̈ | ɢ̈ɢ̈ | ʜ̇ʜ̇ | j̈j̈ | ʟ̇ʟ̇ |
| patch | ɱ̇ɱ̇ | ʎ̊ʎ̊ | ʏ̇ʏ̇ | ʐ̈ʐ̈ | ʑ̇ʑ̇ | ʒ̈ʒ̈ | ʒ̇ʒ̇ | ??? | ʏʏʏ | ꞎꞎꞎ | ⅽ̇ⅽ | ʙ̈ʙ̈ | ɢ̈ɢ̈ | ʜ̇ʜ̇ | j̈j̈ | ʟ̇ʟ̇ |
| ideal | x, b | a-36, b | x, b | x, d+41 | x+23, d+107 | x+20, d | x+15, d-96 | c, b | c, b | c, b | x, d | x, b | a-121, b | x, b | a-64, d+65 | x, b |
| a | | | | | | 234, 645 | | | | | | | | | | |
| a? | 356, 645 | 243, 849 | 244, 645 | 254, 645 | 230, 666 | 227, 663 | 236, 658 | 206, 805 | 206, 805 | 206, 805 | 273, 645 | 246, 645 | 290, 764 | 316, 645 | 132, 821 | 219, 645 |
| a$\delta$ | | | | | | 7, -18 | | | | | | | | | | |
| b | | | | | | 193, -319 | | | | | | | | 351, -110 | | |
| b? | 356, -110 | 243, -110 | 244, -110 | 254, -276 | 230, -210 | 227, -319 | 236, -413 | 206, -110 | 206, -110 | 206, -110 | 273, -319 | 246, -110 | 290, -110 | 316, -110 | 132, -252 | 219, -110 |
| b$\delta$ | | | | | | -34, 0 | | | | | | | | 35, 0 | | |
| si | ɱ̇ɱ̇ | ʎ̊ʎ̊ | ʏ̇ʏ̇ | ʐ̈ʐ̈ | ʑ̇ʑ̇ | ʒ̈ʒ̈ | ʒ̇ʒ̇ | ??? | ʏʏʏ | ꞎꞎꞎ | ⅽ̇ⅽ | ʙ̈ʙ̈ | ɢ̈ɢ̈ | ʜ̇ʜ̇ | j̈j̈ | ʟ̇ʟ̇ |
| patch | ɱ̇ɱ̇ | ʎ̊ʎ̊ | ʏ̇ʏ̇ | ʐ̈ʐ̈ | ʑ̇ʑ̇ | ʒ̈ʒ̈ | ʒ̇ʒ̇ | ??? | ʏʏʏ | ꞎꞎꞎ | ⅽ̇ⅽ | ʙ̈ʙ̈ | ɢ̈ɢ̈ | ʜ̇ʜ̇ | j̈j̈ | ʟ̇ʟ̇ |
| ideal | x, b | a-56, b | x, b | x+23, d+38 | x+23, d+107 | x+24, d | x+15, d-96 | c, b | c, b | c, b | x, d+15 | x, b | a-94, b | x+28, b | a-78, d+65 | x, b |
| a | | | | | | 218, 646 | | | | | | | | | | |
| a? | 363, 645 | 286, 834 | 228, 645 | 206, 666 | 212, 666 | 211, 667 | 213, 658 | 227, 850 | 227, 850 | 227, 850 | 268, 645 | 213, 645 | 278, 796 | 342, 671 | 158, 812 | 211, 645 |
| a$\delta$ | | | | | | 7, -21 | | | | | | | | | | |
| b | | | 239, -107 | | | 141, -319 | | 202, -107 | 230, -107 | | | | | 327, -110 | | |
| b? | 371, -110 | 256, -110 | 237, -110 | 176, -279 | 196, -210 | 172, -319 | 157, -413 | 194, -110 | 194, -110 | 194, -110 | 237, -302 | 221, -110 | 256, -110 | 345, -110 | 103, -252 | 219, -110 |
| b$\delta$ | | | 2, 3 | | | -31, 0 | | 8, 3 | 36, 3 | | | | | -18, 0 | | |

Table 12: Font metrics for above marks 1/2

| hex | 0300 | 0301 | 0302 | 0303 | 0304 | 0306 | 0307 | 0308 | 0309 | 030A | 030B | 030C | 030D | 030F | 0310 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn |
| reg | ǹH̀ | ńH́ | n̂Ĥ | ñH̃ | n̄H̄ | n̆H̆ | ṅḢ | n̈Ḧ | n̊H̊ | n̊H̊ | ń̋H̋ | n̆H̆ | ṅḢ | n̏H̏ | n̊H̊ |
| patch | ǹH̀ | ńH́ | n̂Ĥ | ñH̃ | n̄H̄ | n̆H̆ | ṅḢ | n̈Ḧ | n̊H̊ | n̊H̊ | ń̋H̋ | n̆H̆ | ṅḢ | n̏H̏ | n̊H̊ |
| m | -142, 683 | -191, 682 | -139, 697 | -149, 697 | -209, 644 | -161, 681 | -180, 667 | -210, 645 | -160, 765 | -184, 708 | -219, 714 | -205, 676 | -155, 688 | -140, 758 | -251, 675 |
| m? | -159, 690 | -160, 691 | -140, 697 | -149, 711 | -212, 678 | -161, 672 | -180, 667 | -210, 669 | -161, 755 | -182, 711 | -170, 700 | -204, 676 | -154, 691 | -168, 724 | -251, 672 |
| m$\delta$ | 17, -7 | -31, -9 | 1, 0 | 0, -14 | 3, -34 | 0, 9 | 0, 0 | 0, -24 | 1, 10 | -2, -3 | -49, 14 | -1, 0 | -1, -3 | 28, 34 | 0, 3 |
| it | *ǹH* | *ńH* | *n̂H* | *ñH̃* | *n̄H̄* | *n̆H* | *ṅḢ* | *n̈Ḧ* | *n̊H̊* | *n̊H̊* | *ń̋H̋* | *n̆H̆* | *ṅḢ* | *n̏H̏* | *n̊H̊* |
| patch | *ǹH̀* | *ńH́* | *n̂Ĥ* | *ñH̃* | *n̄H̄* | *n̆H̆* | *ṅḢ* | *n̈Ḧ* | *n̊H̊* | *n̊H̊* | *ń̋H̋* | *n̆H̆* | *ṅḢ* | *n̏H̏* | *n̊H̊* |
| m | -3, 683 | -52, 682 | 3, 697 | -17, 697 | -78, 645 | -22, 681 | -40, 689 | -79, 645 | -4, 765 | -40, 708 | -74, 714 | -67, 676 | -15, 688 | 14, 758 | -114, 675 |
| m? | -18, 689 | -18, 690 | 3, 696 | -3, 716 | -71, 680 | -24, 671 | -35, 698 | -74, 669 | -8, 754 | -37, 712 | -26, 699 | -66, 675 | -13, 690 | -19, 723 | -114, 671 |
| m$\delta$ | 15, -6 | -34, -8 | 0, 1 | -14, -19 | -7, -35 | 2, 10 | -5, -9 | -5, -24 | 4, 11 | -3, -4 | -48, 15 | -1, 1 | -2, -2 | 33, 35 | 0, 4 |
| sb | **ǹH̀** | **ńH́** | **n̂Ĥ** | **ñH̃** | **n̄H̄** | **n̆H̆** | **ṅḢ** | **n̈Ḧ** | **n̊H̊** | **n̊H̊** | **ń̋H̋** | **n̆H̆** | **ṅḢ** | **n̏H̏** | **n̊H̊** |
| patch | **ǹH̀** | **ńH́** | **n̂Ĥ** | **ñH̃** | **n̄H̄** | **n̆H̆** | **ṅḢ** | **n̈Ḧ** | **n̊H̊** | **n̊H̊** | **ń̋H̋** | **n̆H̆** | **ṅḢ** | **n̏H̏** | **n̊H̊** |
| m | -127, 713 | -180, 713 | -127, 700 | -133, 679 | -205, 625 | -149, 684 | -169, 603 (1) | -198, 557 (1) | -162, 682 (1) | -170, 646 (1) | -197, 733 | -193, 676 | -137, 719 | -127, 758 | -239, 675 |
| m? | -144, 702 | -144, 699 | -128, 704 | -136, 719 | -200, 660 | -149, 680 | -168, 603 | -186, 580 | -148, 672 | -170, 630 | -158, 714 | -192, 686 | -137, 696 | -152, 737 | -235, 680 |
| m$\delta$ | 17, 11 | -36, 14 | 1, -4 | 3, -40 | -5, -35 | 0, 4 | -1, 0 | -12, -23 | -14, 10 | 0, 16 | -39, 19 | -1, -10 | 0, 23 | 25, 21 | -4, -5 |
| si | ***ǹH̀*** | ***ńH́*** | ***n̂Ĥ*** | ***ñH̃*** | ***n̄H̄*** | ***n̆H̆*** | ***ṅḢ*** | ***n̈Ḧ*** | ***n̊H̊*** | ***n̊H̊*** | ***ń̋H̋*** | ***n̆H̆*** | ***ṅḢ*** | ***n̏H̏*** | ***n̊H̊*** |
| patch | ***ǹH̀*** | ***ńH́*** | ***n̂Ĥ*** | ***ñH̃*** | ***n̄H̄*** | ***n̆H̆*** | ***ṅḢ*** | ***n̈Ḧ*** | ***n̊H̊*** | ***n̊H̊*** | ***ń̋H̋*** | ***n̆H̆*** | ***ṅḢ*** | ***n̏H̏*** | ***n̊H̊*** |
| m | -142, 683 | -191, 682 | -139, 697 | -134, 700 | -209, 645 | -161, 681 | -180, 667 | -210, 645 | -160, 765 | -184, 708 | -219, 714 | -205, 676 | -155, 688 | -140, 758 | -251, 675 |
| m? | -154, 694 | -152, 694 | -136, 697 | -141, 717 | -217, 651 | -162, 673 | -177, 692 | -243, 671 | -145, 759 | -174, 717 | -166, 704 | -205, 676 | -147, 686 | -156, 728 | -248, 673 |
| m$\delta$ | 12, -11 | -39, -12 | -3, 0 | 7, -17 | 8, -6 | 1, 8 | -3, -25 | 33, -26 | -15, 6 | -10, -9 | -53, 10 | 0, 0 | -8, 2 | 16, 30 | -3, 2 |

Table 13: Font metrics for above marks 2/2

| hex | 0311 | 0312 | 033D | 033E | 033F | 030E | 0346 | 034A | 034B | 034C | 0350 | 0351 | 0352 | 0357 | 035B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn |
| reg | n̂H̃ | ṅḢ̃ | n̆H̃ | ṅḢ̃ | n̄H̄ | n̋H̃ | n̂H̃ | n̆H̃ | n̊H̃ | n̄H̃ | n̆H̃ | ṅḢ̃ | n̂H̃ | ṅḢ̃ | n̆H̃ |
| patch | n̂H̃ | ṅḢ̃ | n̆H̃ | ṅḢ̃ | n̄H̄ | n̋H̃ | n̂H̃ | n̆H̃ | n̊H̃ | n̄H̃ | n̆H̃ | ṅḢ̃ | n̂H̃ | ṅḢ̃ | n̆H̃ |
| m | -205, 645 | -110, 605 | -265, 605 | -265, 605 | -235, 611 | -245, 719 | -251, 621 | -215, 645 | -205, 645 | -235, 645 | -201, 645 | -171, 645 | -251, 645 | -171, 645 | -164, 645 |
| m? | -208, 641 | -102, 611 | -260, 582 | -247, 584 | -240, 622 | -244, 682 | -255, 624 | -217, 637 | -217, 611 | -238, 647 | -183, 643 | -156, 652 | -251, 652 | -141, 662 | -174, 632 |
| m$\delta$ | 3, 4 | -8, -6 | -5, 23 | -18, 21 | 5, -11 | -1, 37 | 4, -3 | 2, 8 | 12, 34 | 3, -2 | -18, 2 | -15, -7 | 0, -7 | -30, -17 | 10, 13 |
| it | *n̂H̃* | *ṅḢ̃* | *n̆H̃* | *ṅḢ̃* | *n̄H̄* | *n̋H̃* | *n̂H̃* | *n̆H̃* | *n̊H̃* | *n̄H̃* | *n̆H̃* | *ṅḢ̃* | *n̂H̃* | *ṅḢ̃* | *n̆H̃* |
| patch | *n̂H̃* | *ṅḢ̃* | *n̆H̃* | *ṅḢ̃* | *n̄H̄* | *n̋H̃* | *n̂H̃* | *n̆H̃* | *n̊H̃* | *n̄H̃* | *n̆H̃* | *ṅḢ̃* | *n̂H̃* | *ṅḢ̃* | *n̆H̃* |
| m | -74, 645 | 13, 605 | -142, 605 | -142, 605 | -111, 611 | -99, 719 | -125, 621 | -84, 645 | -74, 645 | -104, 645 | -70, 645 | -40, 645 | -120, 645 | -40, 645 | -33, 645 |
| m? | -76, 640 | 24, 610 | -142, 581 | -128, 583 | -113, 621 | -105, 681 | -128, 623 | -87, 636 | -92, 610 | -107, 646 | -53, 642 | -23, 651 | -118, 651 | -6, 661 | -46, 631 |
| m$\delta$ | 2, 5 | -11, -5 | 0, 24 | -14, 22 | 2, -10 | 6, 38 | 3, -2 | 3, 9 | 18, 35 | 3, -1 | -17, 3 | -17, -6 | -2, -6 | -34, -16 | 13, 14 |
| sb | **n̂H̃** | **ṅḢ̃** | **n̆H̃** | **n̆H̃** | **n̄H̄** | **n̋H̃** | **n̂H̃** | **n̆H̃** | **n̊H̃** | **n̄H̃** | **n̆H̃** | **ṅḢ̃** | **n̂H̃** | **ṅḢ̃** | **n̆H̃** |
| patch | **n̂H̃** | **ṅḢ̃** | **n̆H̃** | **ṅḢ̃** | **n̄H̄** | **n̋H̃** | **n̂H̃** | **n̆H̃** | **n̊H̃** | **n̄H̃** | **n̆H̃** | **ṅḢ̃** | **n̂H̃** | **ṅḢ̃** | **n̆H̃** |
| m | -205, 645 | -97, 504 (1) | | | | -232, 719 | | | | | | 100, 547 (1) | | 100, 549 (1) | -180, 639 |
| m? | -220, 650 | -102, 528 | | | | -232, 696 | | | | | | 106, 547 | | 112, 547 | -184, 681 |
| m$\delta$ | 15, -5 | 5, -24 | | | | 0, 23 | | | | | | -6, 0 | | -12, 2 | 4, -42 |
| si | ***n̂H̃*** | ***ṅḢ̃*** | ***n̆H̃*** | ***n̆H̃*** | ***n̄H̄*** | ***n̋H̃*** | ***n̂H̃*** | ***n̆H̃*** | ***n̊H̃*** | ***n̄H̃*** | ***n̆H̃*** | ***ṅḢ̃*** | ***n̂H̃*** | ***ṅḢ̃*** | ***n̆H̃*** |
| patch | ***n̂H̃*** | ***ṅḢ̃*** | ***n̆H̃*** | ***ṅḢ̃*** | ***n̄H̄*** | ***n̋H̃*** | ***n̂H̃*** | ***n̆H̃*** | ***n̊H̃*** | ***n̄H̃*** | ***n̆H̃*** | ***ṅḢ̃*** | ***n̂H̃*** | ***ṅḢ̃*** | ***n̆H̃*** |
| m | -205, 645 | -110, 605 | | | | -245, 719 | | | | | | | | | |
| m? | -226, 644 | -116, 614 | | | | -242, 686 | | | | | | | | | |
| m$\delta$ | 21, 1 | 6, -9 | | | | -3, 33 | | | | | | | | | |

Table 14: Font metrics for below marks 1/2

| hex | 0316 | 0317 | 0318 | 0319 | 031C | 0320 | 0323 | 0324 | 0325 | 0326 | 0329 | 032A | 032B | 032C | 032D | 032E |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn |
| reg | n̖H | n̗H | n̘H | n̙H | n̜H | nH | n̩H | n̪H | n̫H | n̬H | ṋH | n̮H | n̯H | n̰H | ṉH | n̲H |
| patch | n̖H | n̗H | n̘H | n̙H | n̜H | nH | n̩H | n̪H | n̫H | n̬H | ṋH | n̮H | n̯H | n̰H | ṉH | n̲H |
| m | -219, -133 | -210, -132 | -224, -148 | -285, -147 | -216, -146 | -253, -148 | -216, -113 | -218, -116 | -218, -116 | -255, -82 | -257, -122 | -253, -113 | -251, -108 | -204, -116 | -195, -133 | -251, -110 |
| m? | -248, -124 | -224, -114 | -249, -102 | -249, -101 | -262, -98 | -252, -170 | -216, -152 | -217, -152 | -217, -78 | -245, -87 | -256, -106 | -255, -134 | -252, -119 | -204, -123 | -195, -128 | -252, -109 |
| m$\delta$ | 29, -9 | 14, -18 | 25, -46 | -36, -46 | 46, -48 | -1, 22 | 0, 39 | -1, 36 | -1, -38 | -10, 5 | -1, -16 | 2, 21 | 1, 11 | 0, 7 | 0, -5 | 1, -1 |
| it | *n̖H* | *n̗H* | *n̘H* | *n̙H* | *n̜H* | *nH* | *n̩H* | *n̪H* | *n̫H* | *n̬H* | *ṋH* | *n̮H* | *n̯H* | *n̰H* | *ṉH* | *n̲H* |
| patch | *n̖H* | *n̗H* | *n̘H* | *n̙H* | *n̜H* | *nH* | *n̩H* | *n̪H* | *n̫H* | *n̬H* | *ṋH* | *n̮H* | *n̯H* | *n̰H* | *ṉH* | *n̲H* |
| m | -246, -133 | -237, -132 | -254, -148 | -315, -147 | -246, -146 | -283, -148 | -239, -113 | -242, -116 | -242, -116 | -272, -82 | -282, -122 | -276, -113 | -273, -108 | -228, -116 | -222, -133 | -275, -110 |
| m? | -274, -126 | -248, -116 | -270, -104 | -270, -103 | -281, -100 | -287, -172 | -238, -111 | -239, -111 | -233, -82 | -263, -89 | -278, -108 | -283, -136 | -277, -121 | -229, -125 | -221, -130 | -269, -111 |
| m$\delta$ | 28, -7 | 11, -16 | 16, -44 | -45, -44 | 35, -46 | 4, 24 | -1, -2 | -3, -5 | -9, -34 | -9, 7 | -4, -14 | 7, 23 | 4, 13 | 1, 9 | -1, -3 | -6, 1 |
| sb | **n̖H** | **n̗H** | **n̘H** | **n̙H** | **n̜H** | **nH** | **n̩H** | **n̪H** | **n̫H** | **n̬H** | **ṋH** | **n̮H** | **n̯H** | **n̰H** | **ṉH** | **n̲H** |
| patch | **n̖H** | **n̗H** | **n̘H** | **n̙H** | **n̜H** | **nH** | **n̩H** | **n̪H** | **n̫H** | **n̬H** | **ṋH** | **n̮H** | **n̯H** | **n̰H** | **ṉH** | **n̲H** |
| m | -219, -133 | -210, -132 | -224, -148 | -285, -147 | -216, -146 | -253, -149 | -216, -113 | -218, -116 | -218, -116 | -255, -73 | -257, -124 | -253, -113 | -251, -108 | -208, -116 | -195, -133 | -239, -110 |
| m? | -248, -132 | -224, -122 | -249, -110 | -249, -110 | -262, -106 | -251, -179 | -216, -118 | -218, -113 | -216, -88 | -244, -86 | -256, -116 | -255, -142 | -252, -127 | -204, -131 | -195, -136 | -240, -105 |
| m$\delta$ | 29, -1 | 14, -10 | 25, -38 | -36, -37 | 46, -40 | -2, 30 | 0, 5 | 0, -3 | -2, -28 | -11, 13 | -1, -8 | 2, 29 | 1, 19 | -4, 15 | 0, 3 | 1, -5 |
| si | ***n̖H*** | ***n̗H*** | ***n̘H*** | ***n̙H*** | ***n̜H*** | ***nH*** | ***n̩H*** | ***n̪H*** | ***n̫H*** | ***n̬H*** | ***ṋH*** | ***n̮H*** | ***n̯H*** | ***n̰H*** | ***ṉH*** | ***n̲H*** |
| patch | ***n̖H*** | ***n̗H*** | ***n̘H*** | ***n̙H*** | ***n̜H*** | ***nH*** | ***n̩H*** | ***n̪H*** | ***n̫H*** | ***n̬H*** | ***ṋH*** | ***n̮H*** | ***n̯H*** | ***n̰H*** | ***ṉH*** | ***n̲H*** |
| m | -219, -133 | -210, -132 | -224, -148 | -285, -147 | -216, -146 | -253, -148 | -216, -113 | -218, -116 | -218, -116 | -255, -82 | -257, -122 | -253, -113 | -251, -108 | -204, -116 | -195, -133 | -251, -110 |
| m? | -251, -132 | -225, -122 | -246, -110 | -259, -110 | -260, -106 | -263, -179 | -214, -117 | -216, -113 | -211, -88 | -237, -86 | -256, -116 | -260, -142 | -253, -127 | -206, -127 | -199, -136 | -244, -113 |
| m$\delta$ | 32, -1 | 15, -10 | 22, -38 | -26, -37 | 44, -40 | 10, 31 | -2, 4 | -2, -3 | -7, -28 | -18, 4 | -1, -6 | 7, 29 | 2, 19 | 2, 11 | 4, 3 | -7, 3 |

Table 15: Font metrics for below marks 2/2

| hex | 032F | 031D | 031E | 031F | 0330 | 0331 | 0333 | 0339 | 033A | 033B | 033C | 0347 | 0348 | 0349 | 034D | 034E |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn |
| reg | n̯H̯ | n̝H̯ | n̞H̯ | n̟H̯ | n̠H̯ | n̡H̯ | n̢H̯ | ṇH̯ | n̤H̯ | n̥H̯ | n̦H̯ | ņH̯ | n̨H̯ | n̩H̯ | n̪H̯ | n̫H̯ |
| patch | n̯H̯ | n̝H̯ | n̞H̯ | n̟H̯ | n̠H̯ | n̡H̯ | n̢H̯ | ṇH̯ | n̤H̯ | n̥H̯ | n̦H̯ | ņH̯ | n̨H̯ | n̩H̯ | n̪H̯ | n̫H̯ |
| m | -250, -116 | -249, -143 | -247, -106 | -249, -145 | -213, -70 | -244, -110 | -244, -110 | -216, -146 | -253, -113 | -246, -113 | -251, -108 | -244, -110 | -197, -110 | -192, -110 | -222, -110 | -227, -110 |
| m? | -251, -125 | -249, -112 | -249, -112 | -252, -99 | -205, -68 | -240, -133 | -240, -123 | -178, -98 | -255, -134 | -252, -118 | -252, -119 | -240, -123 | -195, -94 | -189, -111 | -228, -92 | -228, -74 |
| m$\delta$ | 1, 9 | 0, -31 | 2, 6 | 3, -46 | -8, -2 | -4, 23 | -4, 13 | -38, -48 | 2, 21 | 6, 5 | 1, 11 | -4, 13 | -2, -16 | -3, 1 | 6, -18 | 1, -36 |
| it | *n̯H̯* | *n̝H̯* | *n̞H̯* | *n̟H̯* | *n̠H̯* | *n̡H̯* | *n̢H̯* | *ṇH̯* | *n̤H̯* | *n̥H̯* | *n̦H̯* | *ņH̯* | *n̨H̯* | *n̩H̯* | *n̪H̯* | *n̫H̯* |
| patch | *n̯H̯* | *n̝H̯* | *n̞H̯* | *n̟H̯* | *n̠H̯* | *n̡H̯* | *n̢H̯* | *ṇH̯* | *n̤H̯* | *n̥H̯* | *n̦H̯* | *ņH̯* | *n̨H̯* | *n̩H̯* | *n̪H̯* | *n̫H̯* |
| m | -274, -116 | -278, -143 | -269, -106 | -278, -145 | -227, -70 | -266, -110 | -266, -110 | -246, -146 | -276, -113 | -269, -113 | -273, -108 | -266, -110 | -219, -110 | -214, -110 | -244, -110 | -249, -110 |
| m? | -276, -127 | -271, -114 | -272, -114 | -271, -101 | -220, -74 | -268, -135 | -266, -125 | -198, -100 | -283, -136 | -277, -120 | -275, -121 | -266, -125 | -214, -96 | -213, -113 | -247, -94 | -243, -76 |
| m$\delta$ | 2, 11 | -7, -29 | 3, 8 | -7, -44 | -7, 4 | 2, 25 | 0, 15 | -48, -46 | 7, 23 | 8, 7 | 2, 13 | 0, 15 | -5, -14 | -1, 3 | 3, -16 | -6, -34 |
| sb | **n̯H̯** | **n̝H̯** | **n̞H̯** | **n̟H̯** | **n̠H̯** | **n̡H̯** | **n̢H̯** | **ṇH̯** | **n̤H̯** | **n̥H̯** | **n̦H̯** | **ņH̯** | **n̨H̯** | **n̩H̯** | **n̪H̯** | **n̫H̯** |
| patch | **n̯H̯** | **n̝H̯** | **n̞H̯** | **n̟H̯** | **n̠H̯** | **n̡H̯** | **n̢H̯** | **ṇH̯** | **n̤H̯** | **n̥H̯** | **n̦H̯** | **ņH̯** | **n̨H̯** | **n̩H̯** | **n̪H̯** | **n̫H̯** |
| m | -250, -116 | -249, -143 | -247, -106 | -249, -146 | -213, -70 | -244, -58 | | | | | | | | | | |
| m? | -263, -127 | -249, -121 | -249, -120 | -251, -108 | -192, -70 | -240, -89 | | | | | | | | | | |
| m$\delta$ | 13, 11 | 0, -22 | 2, 14 | 2, -38 | -21, 0 | -4, 31 | | | | | | | | | | |
| si | ***n̯H̯*** | ***n̝H̯*** | ***n̞H̯*** | ***n̟H̯*** | ***n̠H̯*** | ***n̡H̯*** | ***n̢H̯*** | ***ṇH̯*** | ***n̤H̯*** | ***n̥H̯*** | ***n̦H̯*** | ***ņH̯*** | ***n̨H̯*** | ***n̩H̯*** | ***n̪H̯*** | ***n̫H̯*** |
| patch | ***n̯H̯*** | ***n̝H̯*** | ***n̞H̯*** | ***n̟H̯*** | ***n̠H̯*** | ***n̡H̯*** | ***n̢H̯*** | ***ṇH̯*** | ***n̤H̯*** | ***n̥H̯*** | ***n̦H̯*** | ***ņH̯*** | ***n̨H̯*** | ***n̩H̯*** | ***n̪H̯*** | ***n̫H̯*** |
| m | -250, -116 | -249, -143 | -247, -106 | -249, -145 | -213, -70 | -244, -110 | | | | | | | | | | |
| m? | -265, -131 | -252, -132 | -247, -108 | -249, -108 | -184, -77 | -234, -89 | | | | | | | | | | |
| m$\delta$ | 15, 15 | 3, -11 | 0, 2 | 0, -37 | -29, 7 | -10, -21 | | | | | | | | | | |

## Vertical references, anchor $Y$ coordinates and reference, and clearance

A font has at least five vertical reference lines where glyph outlines live inside: descender height, baseline, x-height, ascender height, capital height. The values or ranges of these $Y$ coordinates are in the first five rows of Table 16. These vertical aspects are the gray horizontal lines of Figure 1. These vertical aspects are displayed in a FontForge glyph outline window as black horizontal guidelines with red or blue shaded bands for ranges. I denote these as $yra$, $yrc$, $yrx$, $yr0$, $yrd$ because they are $Y$ values, and r indicates a reference value. a, c, x, 0, d indicate the five vertical aspects. These values (and the ranges that set the bands) are in the SFD, in a line that starts with `BlueValues`, e.g. in Libertinus serif regular,

```
BlueValues 47 [-12 0 429 442 460 472 600 610 645 658 688 698]
OtherBlues 11 [-238 -227]
```

means $yra = 698, [688, 698]$; $yrc = 645, [645, 658]$; $yrx = 429, [429, 442]$; $yr0 = 0, [-12, 0]$; and $yrd = -232, [-238, -227]$.

In the anchor model for Libertinus, the $Y$ coordinate of an above-anchor ($ay$) or a below-anchor ($by$) is at a stable value depending on where the glyph outline lives relative to a vertical aspect. See the rows a and b in Tables 1–4 and find the common values according to the glyph's vertical aspects. There is no set name in font design for these above- or below-anchor $Y$ coordinate reference values by ascender aspect, so I will denote them $ayra$, $ayrc$, $ayrx$, $byr0$, $byrd$, These are the middle five rows of Table 16. For example, all glyphs that sit on the baseline have below-anchor $Y$ coordinate $by = -110$, and thus $byr0 = -110$. All capitals A–Z have $ay = 850$, and thus $ayrc = 850$. Any glyph that descends to the descender value $yrd = -232$ or descender range $[-238, -227]$ should have $by = byrd = -319$.

The clearance is defined as the difference between the anchor reference value and the vertical aspect $Y$ value. I will denote these clearances as $ay\delta a$, $ay\delta c$, $ay\delta x$, $by\delta 0$, $by\delta d$. So, $ay\delta a = ayra - yra$ etc. For example, capital letters A–Z have a capital height $yrc = 645$, and all capital letters A–Z have an above-anchor $Y$ coordinate at $ayrc = 850$. So the above-anchor clearance for capitals is $ay\delta c = ayrc - yec = 850 - 645 = 205$. Because of the weird anchor model for Libertinus, this clearance value is not the exact vertical distance between a base and a mark, but this clearance value is still useful.

A glyph may not be a full ascender or a full descender, and its bounding box may not live closely within two vertical aspects. In such a case, the anchor $Y$ coordinate is set by looking for the nearest, or most suitable, vertical aspect, and considering the clearance of that vertical aspect relative to the bounding box of the glyph.

Here is the heuristic algorithm for calculating the $Y$ coordinate of the above- or below-anchor ($ay$ or $by$) of a glyph that is not a full ascender or descender.

Consider a glyph that is not a full ascender. Find the nearest, or most suitable, vertical aspect – ascender $a$, capital $c$, or x-height $x$. Then $ay = Y_{\text{max}} + ay\delta[acx] - \epsilon$, where $Y_{\text{max}}$ is the maximum $Y$ coordinate of bounding box of the glyph, and $\epsilon$ is the correction for the vertical aspect range, usually for an overshoot.

Consider a glyph that is not a full descender. Find the nearest, or most suitable, vertical aspect – baseline 0 or desender $d$. Then $by = Y_{\text{min}} + by\delta[0d] + \epsilon$, where $Y_{\text{min}}$ in the minimum $Y$ coordinate of the bounding box of the glyph, and $\epsilon$ is the correction for the vertical aspect range, usually for an overshoot.

For the vertical aspect, the first single value is the optical value of the vertical aspect. In practice, this is the y-coordinate of the central control point of a (vertical) stem that meets that aspect. In the table, the range for the vertical aspects are the BlueValues set by the designer in the SFD and seen as guidelines in a FontForge glyph outline window view.

I need to keep track of serif and bowl corrections relative to the (single) vertical aspect value. In Libertinus serif regular, a glyph that meets an aspect with a stem needs a 2 unit correction. For example, the letter x meets the baseline and x-height aspects, with stems, but $y_{\text{min}} = 0 - 2 = -2$ and $y_{\text{max}} = 429 + 2 = 431$. For bowls, the corrections tend to be different for each aspect. For Libertinus serif regular, a glyph that is a bowl full-descender (usually) has a 6 unit overshoot, i.e. $y_{\text{min}} = -232 - 6 = -238$. A bowl on the baseline (usually) has a 10 unit overshoot, i.e. $y_{\text{min}} = 0 - 10 = -10$. A bowl at the x-height aspect (usually) has a 10 unit overshoot, i.e. $y_{\text{max}} = 429 + 10 = 439$. A bowl at the capital aspect (usually) has a 13 unit overshoot, i.e. $y_{\text{max}} = 645 + 13 = 658$. For full ascenders, there is no correction in practice, regardless of the shape of the glyph's ascender, because $y_{\text{max}} = 698$ for all such glyphs.

That is, these stem and bowl corrections are 2 and 6, 10, 10, 13.

For the other font styles (italic, semibold, semibold italic), these corrections are remarkably consistent for all aspects, even though e.g. italic small letters do not really have flat terminal stems. But I think that the correction principle is valid for glyphs that are not within the threshold of an expected vertical aspect. For a non-full descender, if $y_{\text{min}}$ is closest to the descender aspect, and the glyph looks like a bowl at this lowest point, then the correction is 6; otherwise I assume that the glyph must look like a flat stem and the correction is 2. But for a non-full descender that is closer to the baseline, the bowl correction is 10. Full descender bowls tend to be flatter because the lower threhold of the descender really is a hard lower bound.

Table 16: Vertical aspects, anchor $Y$ references, and clearances of Libertinus

| | v. aspect | regular | italic | semibold | semibold italic |
|---|---|---|---|---|---|
| $yra$ | ascender | $698, [688, 698]$ | $698, [688, 698]$ | $698, [690, 698]$ | $698, [696, 705]$ |
| $yrc$ | capital | $645, [645, 658]$ | $645, [645, 658]$ | $645, [645, 662]$ | $645, [645, 662]$ |
| $yrx$ | x-height | $429, [429, 442]$ | $429, [429, 442]$ | $434, [433, 447]$ | $434, [434, 447]$ |
| $yr0$ | baseline | $0, [-12, 0]$ | $0, [-12, 1]$ | $0, [-12, 1]$, | $0, [-20, 0]$ |
| $yrd$ | descender | $-232, [-238, -227]$ | $-232, [-238, -227]$ | $-232, [-238, -212]$ | $-232, [-239, -219]$ |
| $ayra$ | ascender | 885 | 890 | 885 | 890 |
| $ayrc$ | capital | 850 | 850 | 805 | 850 |
| $ayrx$ | x-height | 645 | 645 | 645 | 645 |
| $byr0$ | baseline | −110 | −110 | −110 | −110 |
| $byrd$ | descender | −319 | −319 | −319 | −319 |
| $ay\delta a$ | ascender | 187 | 192 | 187 | 192 |
| $ay\delta c$ | capital | 205 | 205 | 160 | 205 |
| $ay\delta x$ | x-height | 216 | 216 | 211 | 211 |
| $by\delta 0$ | baseline | 110 | 110 | 110 | 110 |
| $by\delta d$ | descender | 87 | 87 | 87 | 87 |

### Copied anchors

## Regular, above

(281, 850) ĖĚÊĖP̌P̂

(336, 850) ḰǨḰV̇V̌V̂

(346, 850) ṄŇN̂U̇ǓÛ

(209, 645) ȧǎâȩc̨ĉe̊ěêȣȝȝd̨ǧĝḧḧḧḧḧḧḫḫḫ

(109, -319) JJJjjj
(350, -110) ỌỌỌØØØỌỌỌ
(234, -110) ṢṢṢḅḅḅẹẹẹ
(196, -110) ạạạsssɨɨɨllllkkkkoooɵɵɵɸɸɸɻɻɻɾɾɾ

## Semibolditalic, below

*No clusters detected.*

Þ̰ Þ̱ Þ̲ Þ̳ Þ̴ Þ̵ Þ̶ Ƀ̰ Ƀ̱ Ƀ̲ Ƀ̳ Ƀ̴ Ƀ̵ Ƀ̶ Đ̰ Đ̱ Đ̲ Đ̳ Đ̴ Đ̵ Đ̶ Ɯ̰ Ɯ̱ Ɯ̲ Ɯ̳ Ɯ̴ Ɯ̵ Ɯ̶ j̰ j̱ j̲ j̳ j̴ j̵ j̶ ɘ̰ ɘ̱ ɘ̲ ɘ̳ ɘ̴ ɘ̵ ɘ̶ ġ ċ ġ̄ ċ̄ ḡ ċ̄ ḡ̈ ḭ i̱ i̲ ĭ̲ ī̲ ī̳ ɯ̰ ɯ̱ ɯ̲ ɯ̳ ɯ̴ ɯ̵ ɯ̶ ṁ ṁ ṃ ṁ̄ ṃ̄ ṁ̈ ṃ̈

ṇ̰ ṇ̱ ṇ̲ ṇ̳ ṇ̴ ṇ̵ ṇ̶ ṛ̰ ṛ̱ ṛ̲ ṛ̳ ṛ̴ ṛ̵ ṛ̶

J̰ J̱ J̲ J̳ J̴ J̵ J̶ j̰ j̱ j̲ j̳ j̴ j̵ j̶ Ạ̰ Ạ̱ Ạ̲ Ạ̳ Ạ̴ Ạ̵ Ạ̶ ạ̰ ạ̱ ạ̲ ạ̳ ạ̴ ạ̵ ạ̶ Ɓ̰ Ɓ̱ Ɓ̲ Ɓ̳ Ɓ̴ Ɓ̵ Ɓ̶ Ḍ̰ Ḍ̱ Ḍ̲ Ḍ̳ Ḍ̴ Ḍ̵ Ḍ̶ Ɖ̰ Ɖ̱ Ɖ̲ Ɖ̳ Ɖ̴ Ɖ̵ Ɖ̶ Ụ̰ Ụ̱ Ụ̲ Ụ̳ Ụ̴ Ụ̵ Ụ̶ ẹ̰ ẹ̱ ẹ̲ ẹ̳ ẹ̴ ẹ̵ ẹ̶ ạ̰ ạ̱ ạ̲ ạ̳ ạ̴ ạ̵ ạ̶

ɓ̰ ɓ̱ ɓ̲ ɓ̳ ɓ̴ ɓ̵ ɓ̶ ç̰ ç̱ ç̲ ç̳ ç̴ ç̵ ç̶ ḍ̰ ḍ̱ ḍ̲ ḍ̳ ḍ̴ ḍ̵ ḍ̶ ɖ̰ ɖ̱ ɖ̲ ɖ̳ ɖ̴ ɖ̵ ɖ̶ ɗ̰ ɗ̱ ɗ̲ ɗ̳ ɗ̴ ɗ̵ ɗ̶ ɘ̰ ɘ̱ ɘ̲ ɘ̳ ɘ̴ ɘ̵ ɘ̶ ɚ̰ ɚ̱ ɚ̲ ɚ̳ ɚ̴ ɚ̵ ɚ̶ ɛ̰ ɛ̱ ɛ̲ ɛ̳ ɛ̴ ɛ̵ ɛ̶ ɜ̰ ɜ̱ ɜ̲ ɜ̳ ɜ̴ ɜ̵ ɜ̶ ɝ̰ ɝ̱ ɝ̲ ɝ̳ ɝ̴ ɝ̵ ɝ̶ ọ̰ ọ̱ ọ̲ ọ̳ ọ̴ ọ̵ ọ̶ ɔ̰ ɔ̱ ɔ̲ ɔ̳ ɔ̴ ɔ̵ ɔ̶ ʒ̰ ʒ̱ ʒ̲ ʒ̳ ʒ̴ ʒ̵ ʒ̶

ɵ̰ ɵ̱ ɵ̲ ɵ̳ ɵ̴ ɵ̵ ɵ̶ ụ̰ ụ̱ ụ̲ ụ̳ ụ̴ ụ̵ ụ̶ ɥ̰ ɥ̱ ɥ̲ ɥ̳ ɥ̴ ɥ̵ ɥ̶ ɧ̰ ɧ̱ ɧ̲ ɧ̳ ɧ̴ ɧ̵ ɧ̶ ɦ̰ ɦ̱ ɦ̲ ɦ̳ ɦ̴ ɦ̵ ɦ̶ ɨ̰ ɨ̱ ɨ̲ ɨ̳ ɨ̴ ɨ̵ ɨ̶ ɪ̰ ɪ̱ ɪ̲ ɪ̳ ɪ̴ ɪ̵ ɪ̶ ɬ̰ ɬ̱ ɬ̲ ɬ̳ ɬ̴ ɬ̵ ɬ̶ ɭ̰ ɭ̱ ɭ̲ ɭ̳ ɭ̴ ɭ̵ ɭ̶ ɯ̰ ɯ̱ ɯ̲ ɯ̳ ɯ̴ ɯ̵ ɯ̶

ɳ̰ ɳ̱ ɳ̲ ɳ̳ ɳ̴ ɳ̵ ɳ̶ ɱ̰ ɱ̱ ɱ̲ ɱ̳ ɱ̴ ɱ̵ ɱ̶ ɲ̰ ɲ̱ ɲ̲ ɲ̳ ɲ̴ ɲ̵ ɲ̶ ŋ̰ ŋ̱ ŋ̲ ŋ̳ ŋ̴ ŋ̵ ŋ̶ ɴ̰ ɴ̱ ɴ̲ ɴ̳ ɴ̴ ɴ̵ ɴ̶ ȯ̰ ȯ̱ ȯ̲ ȯ̳ ȯ̴ ȯ̵ ȯ̶ ɷ̰ ɷ̱ ɷ̲ ɷ̳ ɷ̴ ɷ̵ ɷ̶ ɾ̰ ɾ̱ ɾ̲ ɾ̳ ɾ̴ ɾ̵ ɾ̶ ɽ̰ ɽ̱ ɽ̲ ɽ̳ ɽ̴ ɽ̵ ɽ̶ ɻ̰ ɻ̱ ɻ̲ ɻ̳ ɻ̴ ɻ̵ ɻ̶ ʃ̰ ʃ̱ ʃ̲ ʃ̳ ʃ̴ ʃ̵ ʃ̶

ʋ̰ ʋ̱ ʋ̲ ʋ̳ ʋ̴ ʋ̵ ʋ̶ ʏ̰ ʏ̱ ʏ̲ ʏ̳ ʏ̴ ʏ̵ ʏ̶ ʔ̰ ʔ̱ ʔ̲ ʔ̳ ʔ̴ ʔ̵ ʔ̶ ʕ̰ ʕ̱ ʕ̲ ʕ̳ ʕ̴ ʕ̵ ʕ̶

Copied above-anchors

ȧ ā ċ̄ ḡ̄ ė̄ ē ǯ̄ ǯ̄ ǧ̄ ɬ̄ ɬ̄ ɮ̄ ɭ̄ ɮ̄ ṫ̄ ṭ̄ ṙ̄ ṝ ṙ̄ ṝ ṙ̄ ṝ ṙ̄ ṝ ṡ̄ s̄ ʃ̄ ʃ̄ ʃ̄ ʃ̄ ɭ̄ ɭ̄ ʃ̄ ʃ̄ ɟ̄ ɟ̄ ʈ̄ ʈ̄ ʉ̄ ʉ̄ ʊ̄ ʊ̄ ʌ̄ ā ẁ ẅ ẏ̄ ȳ ẓ̄ ẓ̄ ẓ̄ ʒ̄ ʒ̄ ʔ̄ ʔ̄ ʕ̄ ɔ̄ ɔ̄ ċ̄ ḡ̄ ḃ̄ b̄ ḡ̄ ḡ̄ ḣ̄ h̄ j̄ j̄ ɫ̄ l̄

Copied below-anchors

ạ a̲ ṣ s̲ ɟ̲ ɟ̲ ɭ̲ ɭ̲ ɮ̲ ɵ̲ ɵ̲ ɸ̲ ɸ̲ ɻ̲ ɻ̲ ɽ̲ ɽ̲ ɾ̲ ɾ̲ ɹ̲ ɹ̲ ɻ̲ ʁ̲ ʁ̲ ʁ̲ ʁ̲ ʂ̲ ʂ̲ ʃ̲ ʃ̲ ʃ̲ ʃ̲ ɭ̲ ɭ̲ ʃ̲ ʃ̲ ɟ̲ ɟ̲ ʈ̲ ʈ̲ ʉ̲ ʉ̲ ʌ̲ ʌ̲ ẁ ẅ ẏ̲ ȳ ẓ̲ ẓ̲ ẓ̲ ʒ̲ ʒ̲ ʔ̲ ʔ̲ ʕ̲ ɔ̲ ɔ̲ ḃ̲ b̲ ḡ̲ ḡ̲ ḣ̲ h̲ j̲ j̲ ɫ̲ l̲

ạ a̲ ḅ b̲ ç̣ ç̲ ḍ̣ ḍ̲ ɗ̣ ɗ̲ ẹ̣ ẹ̲ ɚ̣ ɚ̲ ɛ̣ ɛ̲ ɜ̣ ɜ̲

### Base coverage

Libertinus serif regular has good above- and below-anchor base coverage for A–Z, a–z, and IPA letters, in the sense that most anchors are defined. But I will show later that some anchors are not set to good values. The base coverage gets worse for italic and semibold. Observe that many IPA letters reuse $ax = 209$ and $bx = 196$. These are good anchors for the letter a, but these were just copied to some IPA letters without regard to the letterform and are therefore bad anchors for those IPA letters.

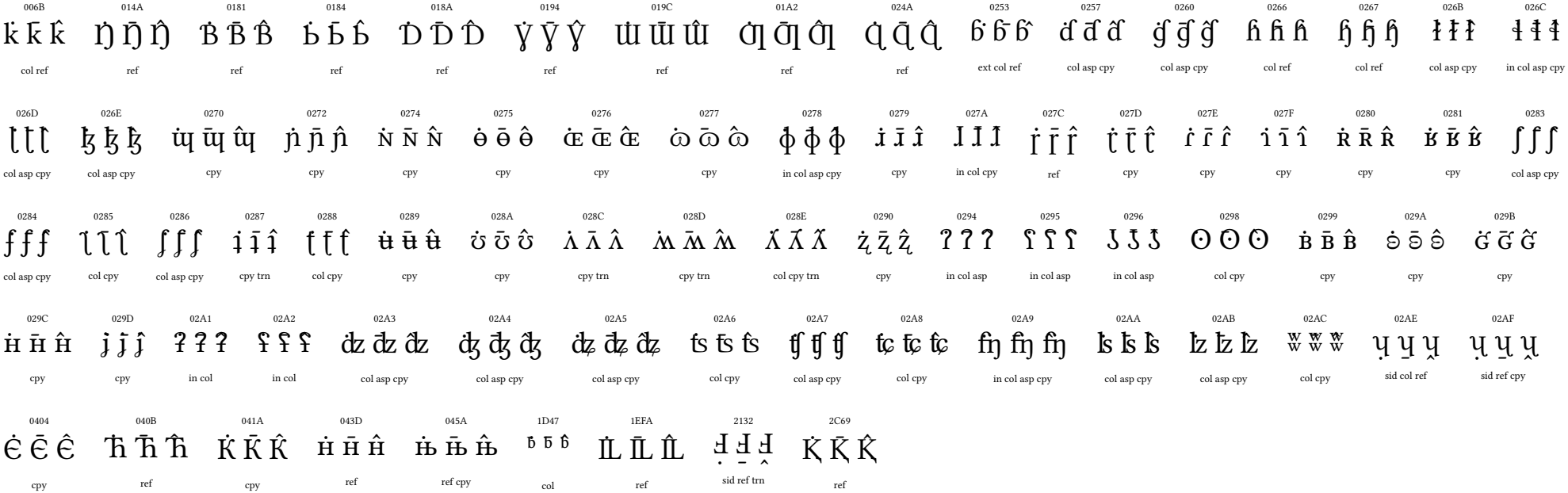
# Anchor audit

These are the designer-set above- and below-anchors that `run_anchor_audit.py` classifies as bad or suspect, in the original fonts. Each cell shows the codepoint, then the base with a few marks (dot, macron and circumflex above; dot and macron below), then tags for the tests that flagged the anchor. The combining grapheme joiner separates base and mark, so every mark is positioned by its anchor. The reasons in full are in `data/fontanchors_audit/`.

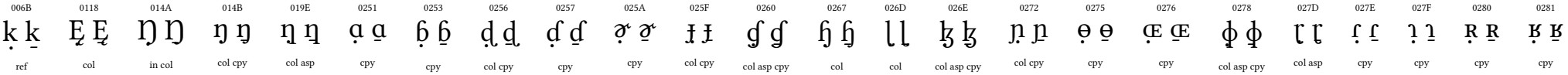
Tags: **sid** anchor on the wrong side of the glyph; **in** anchor inside the outline; **ext** anchor x outside the glyph; **ctr** far from the centre line; **ref** y far from the reference values; **asp** beside an ascender or descender; **clr** floats far from the glyph; **col** a mark overlaps or crowds the glyph; **cpy** copied from another glyph; **trn** turned letter not following its source; **sty** copied from another style; **pre** disagrees with precomposed glyphs; **pos** position differs between styles.

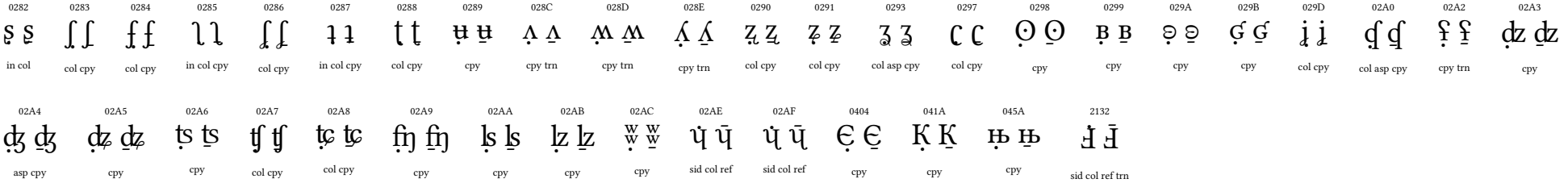
## Regular

### Bad, above-anchor (77)

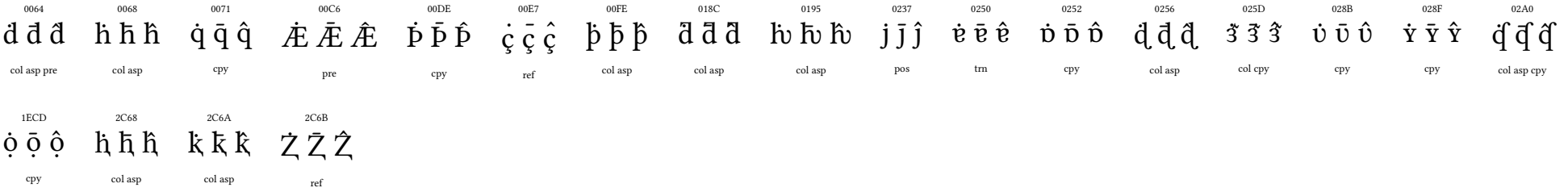



### Bad, below-anchor (62)

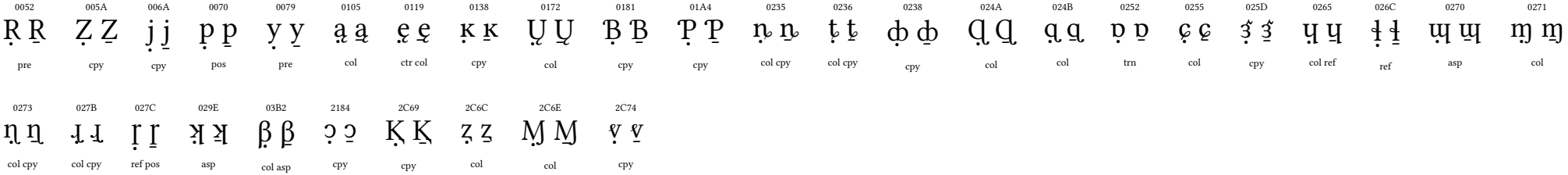





### Suspect, above-anchor (21)

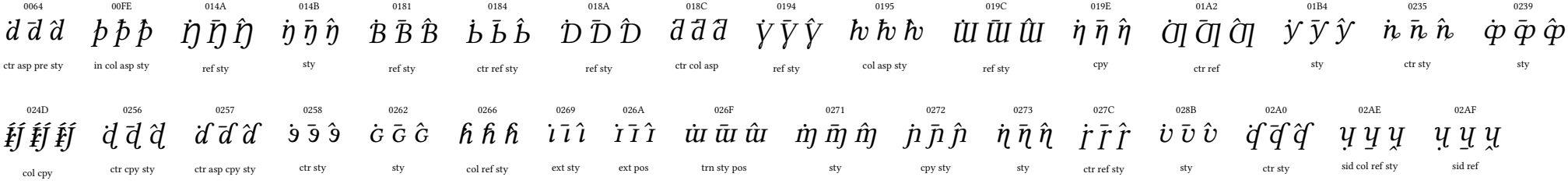



### Suspect, below-anchor (33)

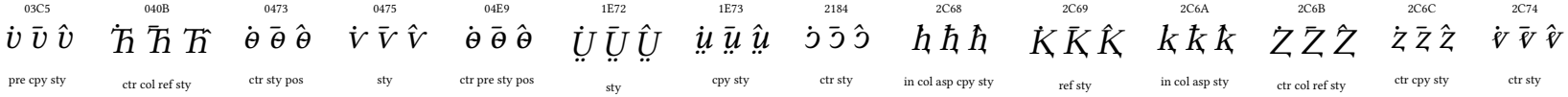



## Italic

### Bad, above-anchor (47)



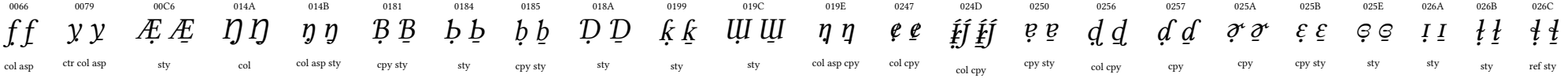
03C5 040B 0473 0475 04E9 1E72 1E73 2184 2C68 2C69 2C6A 2C6B 2C6C 2C74



pre cpy sty | ctr col ref sty | ctr sty pos | sty | ctr pre sty pos | sty | cpy sty | ctr sty | in col asp cpy sty | ref sty | in col asp sty | ctr col ref sty | ctr cpy sty | ctr sty

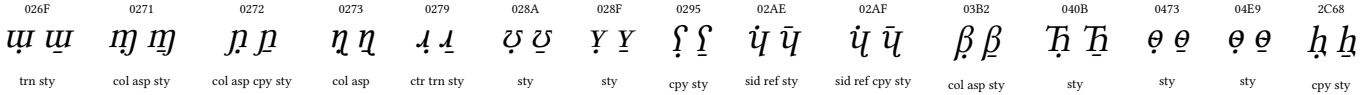
### Bad, below-anchor (38)

0066 0079 00C6 014A 014B 0181 0184 0185 018A 0199 019C 019E 0247 024D 0250 0256 0257 025A 025B 025E 026A 026B 026C



col asp | ctr col asp | sty | col | col asp sty | cpy sty | sty | cpy sty | sty | sty | sty | col asp cpy | col cpy | col cpy | cpy sty | col cpy | cpy | cpy | cpy sty | sty | sty | sty | ref sty

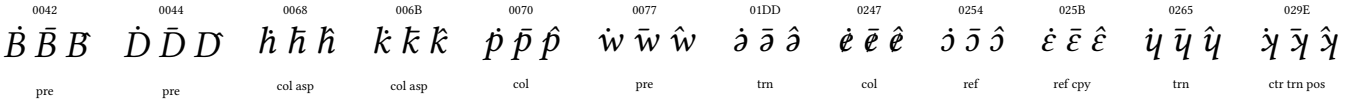
026F 0271 0272 0273 0279 028A 028F 0295 02AE 02AF 03B2 040B 0473 04E9 2C68



trn sty | col asp sty | col asp cpy sty | col asp | ctr trn sty | sty | sty | cpy sty | sid ref sty | sid ref cpy sty | col asp sty | sty | sty | sty | cpy sty

### Suspect, above-anchor (12)

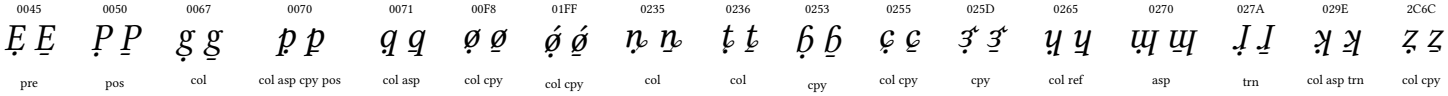
0042 0044 0068 006B 0070 0077 01DD 0247 0254 025B 0265 029E



pre | pre | col asp | col asp | col | pre | trn | col | ref | ref cpy | trn | ctr trn pos

### Suspect, below-anchor (17)

0045 0050 0067 0070 0071 00F8 01FF 0235 0236 0253 0255 025D 0265 0270 027A 029E 2C6C

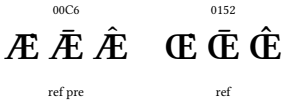


pre | pos | col | col asp cpy pos | col asp | col cpy | col cpy | col | col | cpy | col cpy | cpy | col ref | asp | trn | col asp trn | col cpy

## Semibold

### Bad, above-anchor (2)

00C6 0152



ref pre | ref

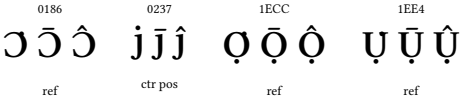
### Bad, below-anchor (1)

0279

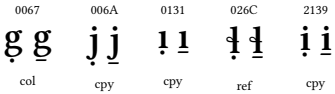


sty

### Suspect, above-anchor (4)



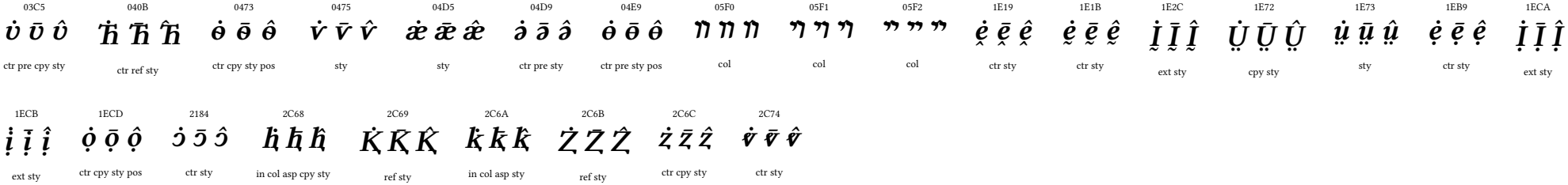

### Suspect, below-anchor (5)



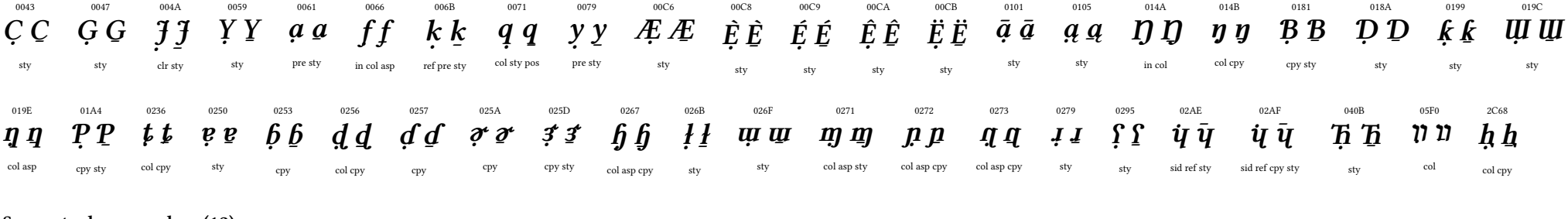

## Semibold italic

### Bad, above-anchor (122)

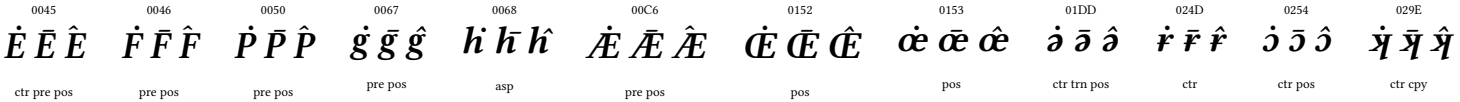





## Bad, below-anchor (44)




## Suspect, above-anchor (12)




## Suspect, below-anchor (27)



# Anchor survey, with patches

I will systematically survey above- and below anchors in the four font font styles, commenting on any features and patching any problems. Refer to Tables 1–4 for anchor values. Missing anchors are readily apparaent as blank entries. Large anchor-midpoint offset values ($ax\delta$, $bx\delta$) indicate possible problems, but these can often be explained by glyph symmetry.

## A–Z, regular, above- and below-anchor

The above-anchors of A–Z regular are good.

The above-anchor of D is just right of the major stem, not right enough to be centered over the bowl. For D, $ax = 306$. Consider the precomposed characters Ď (U+010E) and Ḋ (U+1E0A). The designer drew their glyphs with the mark centered at $X = 312$ and 296 respectively. So, this is a deliberate design decision to keep the mark closer to the stem, not centered over the bowl.

Similarly, the above anchor of F is just right of the major stem, not right enough to be centered over cross strokes. For F, $ax = 284$. For Ḟ (U+1E1E), the dot is centered at $X = 254$. So, the designer is deliberately choosing to keep the mark cloder to the stem.

The above-anchor of L is centered over the major stem, which is a valid design choice.

The below-anchors of A–Z regular are good. But J's below anchor is too low, and Q is missing a below-anchor. Even though J and Q arguably do not need below-anchors for linguistic notation, I will set this anchor by the heuristic algorithm.

#### Patch J regular, below-anchor

J's below-anchor is too low. For J, $by = -319$, but J is not a full descender; its bbox $Y_{\text{min}} = -172$, which is well above $yrd = -232$. Yes, the designer set $by = byrd = -319$ for J in all fout font styles, but this is arguably too low. According to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -172 - 87 + 5 = -254$, where $\epsilon = 5$ accounts for the overshoot of J's tail.

#### Patch Q regular, below-anchor

Q has bbox $Y_{\text{min}} = -209$. Again, according to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -209 - 87 + 5 = -291$, where $\epsilon = 5$ accounts for the overshoot of Q's tail. For $bx$, observe that O and Q have the same counter outline, literally. So, the below-anchor for Q should be at $bx = 350$, the same as that for O.

## a–z, regular, above- and below-anchor

b, f, j, and t are missing above-anchors. Some letters (n, p, r, v, w, x, z) should have $ay$ regularized to 645, not 646. For d, h, and k, the above-anchor is centered above the body, not the ascender; this is good design choice, but care must be taken to avoid collision between an above-mark and an ascender. For d and h, $ay$ is at the x-height anchor reference, and therefore wide marks collide with the ascender. For k, $ay$ is a bit higher, but not at the full ascender anchor references, and therefore the widest marks collide with the ascender.

The below-anchor of k is too low. Otherwise, the below-anchors of a–z are good.

#### Patch k regular, below-anchor

For k, $by - 214$. But it should be at $byr0 = -110$, like all other letters that sit on the baseline.

Here is this positioning of k in a patched version of the font (with o for comparison only).

Original: k̲ k̲o ok̲

Patched: k̲ k̲o ok̲

#### Patch b, d, f, h regular, above-anchor

For b, d, f, and h, set $ay = ayra = 885$.

For $ax$ for b, d, f, and h, there are at least two strategies. First, set $ax$ at the bowl apex. So, for f, set $ax = 279$ at the apex of the bowl. But for b, d, and h, you could set $ax$ centered horizontally over the bowl, in which case you also need to decide which two vertical aspects of the bowl should be used for the centering.

#### Patch j regular, above-anchor

j needs an above anchor for the rare case of an above-mark that is placed over the dot. i does have an above-anchor at $(137, 795)$, and the outlines of the top of

#### Patch n, p, r, v, w, x, z regular, above-anchor

In my patched version of the font, I regularize $ay$ to $ayrx = 645$ for all letters that sit at the x-height, including n, p, r, v, w, x, z.

### Above-anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

### Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.

To do.

### Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, cicrumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

A cirmumflex Â Ầ Ấ Ầ Ấ̉

O circumflex Ô Ồ Ố Ồ Ố̉

AE ligature Æ Æ̀ Æ Æ̀ Ǽ

S̊ T̊ V̊ S̊ T̊ V̊ S̊ T̊ V̊ S̊ T̊ V̊

S̊ T̊ V̊ S̊ T̊ V̊ S̊ T̊ V̊ S̊ T̊ V̊

## Regular, above-anchor, capitals, $X$ coordinate

Figure 5 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad $X$ coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor $X$ coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor $X$ coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 5 for an illustration of increasingly bad (spoofed) $X$ coordinates of N's above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

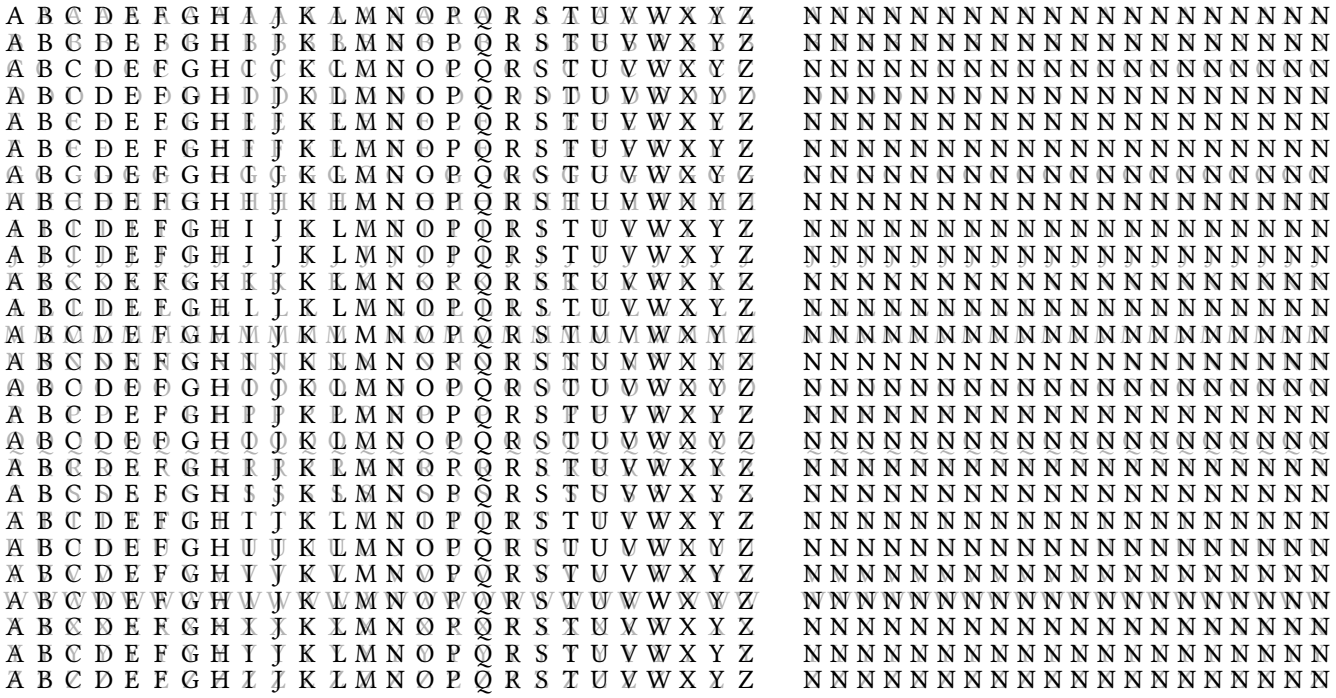


Figure 5: Above-anchor $X$ coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at the their above-anchor. The right panel shows how hard it is to discern bad $X$ coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with Ns above-anchor as currently set by the font, and then spoofing increasing bad values for the $X$ coordinate of N's above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed) $X$ coordinate of N's above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

## Comma-style marks

Libertinus has a some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

### Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` ď *ď* **ď** ***ď***

`\char"013D` Ľ *Ľ* **Ľ** ***Ľ***

`\char"013E` ľ *ľ* **ľ** ***ľ***

`\char"0148` ň *ň* **ň** ***ň***

`\char"0165` ť *ť* **ť** ***ť***

Libertinus even has a custom glyph for a small capital L with a caron, and it is comma-style.

`\textsc{l\char"030C}` ʟ̌

`\textsc{\char"013E}` ʟ̌

### Cedilla and Latvian *komatiņš*

Here are all precomposed characters with a cedilla.

| | |
|---|---|
| `\char"00C7` Ç *Ç* **Ç** ***Ç*** | `\char"00E7` ç *ç* **ç** ***ç*** |
| `\char"1E08` Ḉ *Ḉ* **Ḉ** ***Ḉ*** | `\char"1E08` ḉ *ḉ* **ḉ** ***ḉ*** |
| `\char"1E10` Ḑ *Ḑ* **Ḑ** ***Ḑ*** | `\char"1E11` ḑ *ḑ* **ḑ** ***ḑ*** |
| `\char"0228` Ȩ *Ȩ* **Ȩ** ***Ȩ*** | `\char"0229` ȩ *ȩ* **ȩ** ***ȩ*** |
| `\char"1E1C` Ḝ *Ḝ* **Ḝ** ***Ḝ*** | `\char"1E1D` ḝ *ḝ* **ḝ** ***ḝ*** |
| `\char"0122` Ģ *Ģ* **Ģ** ***Ģ*** | `\char"0123` ģ *ģ* **ģ** ***ģ*** |
| `\char"1E28` Ḩ *Ḩ* **Ḩ** ***Ḩ*** | `\char"1E29` ḩ *ḩ* **ḩ** ***ḩ*** |
| `\char"0136` Ķ *Ķ* **Ķ** ***Ķ*** | `\char"0137` ķ *ķ* **ķ** ***ķ*** |
| `\char"013B` Ļ *Ļ* **Ļ** ***Ļ*** | `\char"013C` ļ *ļ* **ļ** ***ļ*** |
| `\char"0145` Ņ *Ņ* **Ņ** ***Ņ*** | `\char"0146` ņ *ņ* **ņ** ***ņ*** |
| `\char"0156` Ŗ *Ŗ* **Ŗ** ***Ŗ*** | `\char"0157` ŗ *ŗ* **ŗ** ***ŗ*** |
| `\char"015E` Ş *Ş* **Ş** ***Ş*** | `\char"015F` ş *ş* **ş** ***ş*** |
| `\char"0162` Ţ *Ţ* **Ţ** ***Ţ*** | `\char"0163` ţ *ţ* **ţ** ***ţ*** |

These characters – Ģ ģ Ķ ķ Ļ ļ Ņ ņ Ŗ ŗ – use the Latvian *komatiņš* which is proper for Latvian orthography, but the result of a historical mistake in Unicode naming.

The Unicode characters of D and d with cedilla exist for scholarly and historical reasons, not Latvian orthography. Fonts design these Unicode characters differently. Libertinus designs them inconsistently: comma-style in regular and italic, and hook-style in (semi)bold and (semi)bold italic. I think the cedilla on a D or d should be hook-like, but I do not really know.

The other characters – c, e, h, s, t – with a cedilla should definitely be hook-like, and they are so in Libertinus.

Somewhat mysteriously, if the LaTeX source tries to combine any of these letters – c, d, e, g, h, k, l, n, r, s, t – with the combining cedilla, U+0327, then the precomposed character is produced.

c\char"0327 ç *ç* **ç** ***ç***
d\char"0327 ḑ *ḑ* **ḑ** ***ḑ***
e\char"0327 ȩ *ȩ* **ȩ** ***ȩ***
g\char"0327 ģ *ģ* **ģ** ***ģ***
h\char"0327 ḩ *ḩ* **ḩ** ***ḩ***
k\char"0327 ķ *ķ* **ķ** ***ķ***
l\char"0327 ļ *ļ* **ļ** ***ļ***
n\char"0327 ņ *ņ* **ņ** ***ņ***
r\char"0327 ŗ *ŗ* **ŗ** ***ŗ***
s\char"0327 ş *ş* **ş** ***ş***
t\char"0327 ţ *ţ* **ţ** ***ţ***
This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try `g\char"0327` or `\c{g}`.
This behavior is tolerable for the other bases – c, d, e, h, s, t.

## IJ

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.
IJ `\char"0132` U+0132 capital ligature IJ
ij `\char"0133` U+0133 small ligature ij
Libertinus has custom glyphs for these ligatures with an acute, and the small capital ligature with an acute. The GSUB table that activates these custom glyphs only works when the language is set to Dutch by a font loading command. For example,
```
\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}
```
ĲĲ `{\SerifRegularDutch \char"0132\char"0301}` Dutch capital ligature IJ with acute
ĳĳ `{\SerifRegularDutch \char"0133\char"0301}` Dutch small ligature ij with acute
ĳĳ `{\SerifRegularDutch \textsc{\char"0133\char"0301}}` Dutch small capital ligature IJ with acute
In an English LaTeX document, `ij` and `IJ` are kerned by the GPOS table to produce ij and IJ, so they look close together, but they are not substituted by the single glyph for the ligature.

## Lowercase i with tilde below and grave above, in italic and semibold italic

First, start with i with tilde below (precomposed).
ḭ *ḭ* **ḭ** ***ḭ*** `\char"1E2D` i with tilde below (precomposed)
You cannot simply put a grave on it because it collides with the dot.
ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"1E2D\char"0300` i with tilde below (precomposed) with grave
Start over with i with grave (precomposed).
ì *ì* **ì** ***ì*** `\char"00EC` i with grave (precomposed)
Try putting a grave below.
ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"00EC\char"0330` i with grave (precomposed) with tilde below

This fails, because normalization breaks down the input “i with grave (precomposed) with tilde below” into “i with grave with tilde below”, then into canonical order “i with tilde below with grave”, the recomposes it into “i with tilde below (precomposed) with grave”, which reintroduces i’s dot.

Next, try to block normalization with the combining grapheme joiner (CGJ).

ḭ̀ *ḭ̀* ḭ̀ *ḭ̀* `\char"00EC\cgj\char"0330` i with grave (precomposed) with combining grapheme joiner with tilde below

This looks bad because i with grave (precomposed) does not have a below-anchor and fallback shaping looks bad.

Finally, try starting with dotless i, which has OK above- and below-anchors.

ı̰̀ *ı̰̀* ı̰̀ *ı̰̀* `\char"0131\char"0330\char"0300` dotless i with tilde below with grave

Normalization does not affect the dotless i, so it looks the same with the CGJ.

ı̰̀ *ı̰̀* ı̰̀ *ı̰̀* `\char"0131\cgj\char"0330\char"0300` dotless i with CGJ with tilde below with grave

All together now, for comparison.

ḭìı̰ì *ḭìı̰ì* ḭìı̰ì *ḭìı̰ì* `\char"1E2D \char"00EC \char"0131\char"0330\char"0300` i with tilde below (precomposed); i with grave (precomposed); dotless i with tilde below with grave

The same strategy works with other above marks.

ı̰̏ *ı̰̏* ı̰̏ *ı̰̏* `\char"0131\char"0330\char"030F` dotless i with tilde below with double grave

ı̰́ *ı̰́* ı̰́ *ı̰́* `\char"0131\char"0330\char"0301` dotless i with tilde below with acute

ı̰̋ *ı̰̋* ı̰̋ *ı̰̋* `\char"0131\char"0330\char"030B` dotless i with tilde below with double acute

## Lowercase a with double grave, followed by floating double grave, in italic

In italic, we want to produce a (precomposed) vowel with a double grave, followed by floating double grave. Ideally, the two double graves should be aligned. We cannot use the combining double grave as the floater, because it has nothing to combine to, and we cannot control its height. So we use a spacing character, specifically the double prime character (‶ `\char"2036`), which looks like a double grave, in italic.

*ä*‶ `\textit{ä\char"2036}}`

Then we lower it by a few points.

*ä*‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

The distance to move the glyph can be calculated from its SFD splineset, or by trial and error.

Note that only a, e, and i have precomposed glyphs with double acute.

*ä*‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

*ë*‶ `\textit{ë\raisebox{-0.05em}{\char"2036}}`

*ë*‶ `\textit{ë\raisebox{-0.045em}{\char"2036}}`

*ɛ*‶ `\textit{\char"025B\raisebox{-0.05em}{\char"2036}}`

## Tilde below, double acute above, double grave above, on vowels, in italic and semibold italic

Consider the tilde below. There is the combining tilde below, U+0330, encoded in LaTeX as the codepoint `\char"0330`. Remember to put the base before the combining mark in LaTeX, e.g. `a\char"0330` a̰. Three vowels (e, i, u) have precomposed characters – lowercase and capital – encoded in LaTeX as these codepoints:

`\char"1E1A \char"1E1B \char"1E2C \char"1E2D \char"1E74 \char"1E75` Ḛ ḛ Ḭ ḭ Ṵ ṵ *Ḛ ḛ Ḭ ḭ Ṵ ṵ* **Ḛ ḛ Ḭ ḭ Ṵ ṵ** ***Ḛ ḛ Ḭ ḭ Ṵ ṵ***

E, e, I, i, U, or u followed by `\char"0330` will produce the precomposed character glyph. A, a, O, and o, as well as any other IPA vowel with a below-anchor, followed by `\char"0330` will be shaped by attaching the combining below tilde to that vowel at their below-anchor. Note the specific case of i. `i\char"0330 = \char"1E2D` ḭ = ḭ is a dotted i (a precomposed glyph).

Suppose you want a vowel with a tilde below, possibly also with a mark above, especially acute, grave, caron, double acute, or double grave. In the specific case of i, the above mark will crash into the dotted i unless you take care to encode the triplet starting with a dotless i, U+0131. For example, consider the combining acute U+0301.

`i\char"0330\char"0301 = \char"1E2D\char"0301` i̩ í̩ = ị ị́

`\char"0131\char"0330\char"0301` ı̩́ ị́

Recall that the canonical order for a triplet of a base and two combining marks (one below and one above) is base + combining below mark + combining above mark, not base + above + below.

Consider the wrong order:

`i\char"0301\char"0330` í̩ i\char"0301 í i\char"0330 í̩

`\char"00EC\char"0330` ì̩ `\char"00EC` ì ì̩

`\char"1E2D\char"0301` ḭ ḭ́ `\char"1E2D` ḭ ḭ́

But note that the position of the combining tilde below is bad in italic and semibold italic. This is because the below-anchor for the combining tilde below mark is bad. For tilde below, the precomposed character glyphs with i, e, and u are sometimes good and sometimes bad in italic and semibold italic.

Precomposed characters are blue and therefore cannot be changed unless their glyph outline is changed. Black combinations of vowels and marks are shaped by anchors and therefore can be changed by setting good anchors.

Use the combining grapheme joiner (CGJ) character U+034F, `\char"034F`, or `\cgj` if loaded by some package, to force the shaper to use combining marks instead of precomposed glyphs.

A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***

1 Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ *Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ* Á̩ É̩ Í̩ Ó̩ Ú̩ Ɛ̩ Ɔ̩ á̩ é̩ í̩ ó̩ ú̩ ɛ̩ ɔ̩ *Á̩ É̩ Í̩ Ó̩ Ú̩ Ɛ̩ Ɔ̩ á̩ é̩ í̩ ó̩ ú̩ ɛ̩ ɔ̩*

2 À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ *À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ* À̩ È̩ Ì̩ Ò̩ Ù̩ Ɛ̩ Ɔ̩ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩ ɔ̩ *À̩ È̩ Ì̩ Ò̩ Ù̩ Ɛ̩ Ɔ̩ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩ ɔ̩*

3 Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* Ă̩ Ĕ̩ Ĭ̩ Ŏ̩ Ŭ̩ Ɛ̩̆ Ɔ̩̆ ă̩ ĕ̩ ĭ̩ ŏ̩ ŭ̩ ɛ̩̆ ɔ̩̆ *Ă̩ Ĕ̩ Ĭ̩ Ŏ̩ Ŭ̩ Ɛ̩̆ Ɔ̩̆ ă̩ ĕ̩ ĭ̩ ŏ̩ ŭ̩ ɛ̩̆ ɔ̩̆*

4 Ắ Ế Ḯ Ố Ű Ɛ́ Ɔ́ ắ ế ḯ ố ű ɛ́ ɔ́ *Ắ Ế Ḯ Ố Ű Ɛ́ Ɔ́ ắ ế ḯ ố ű ɛ́ ɔ́* Ắ̩ Ế̩ Ḯ̩ Ố̩ Ű̩ Ɛ̩́ Ɔ̩́ ắ̩ ế̩ ḯ̩ ố̩ ű̩ ɛ̩́ ɔ̩́ *Ắ̩ Ế̩ Ḯ̩ Ố̩ Ű̩ Ɛ̩́ Ɔ̩́ ắ̩ ế̩ ḯ̩ ố̩ ű̩ ɛ̩́ ɔ̩́*

5 Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ɔ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀ *Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ɔ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀* Ằ̩ Ề̩ Ỉ̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ ằ̩ ề̩ ỉ̩ ồ̩ ừ̩ ɛ̩̀ ɔ̩̀ *Ằ̩ Ề̩ Ỉ̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ ằ̩ ề̩ ỉ̩ ồ̩ ừ̩ ɛ̩̀ ɔ̩̀*

6 Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣* Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣*

7 Ậ Ệ̂ Ị̂ Ộ Ự̂ Ɛ̣̂ Ɔ̣̂ ậ ệ̂ ị̂ ộ ự̂ ɛ̣̂ ɔ̣̂ *Ậ Ệ̂ Ị̂ Ộ Ự̂ Ɛ̣̂ Ɔ̣̂ ậ ệ̂ ị̂ ộ ự̂ ɛ̣̂ ɔ̣̂* Ậ Ệ̂ Ị̂ Ộ Ự̂ Ɛ̣̂ Ɔ̣̂ ậ ệ̂ ị̂ ộ ự̂ ɛ̣̂ ɔ̣̂ *Ậ Ệ̂ Ị̂ Ộ Ự̂ Ɛ̣̂ Ɔ̣̂ ậ ệ̂ ị̂ ộ ự̂ ɛ̣̂ ɔ̣̂*

7c Ậ́ Ệ́ Ị́ Ộ́ Ự́ Ɛ̣́ Ɔ̣́ ậ́ ệ́ ị́ ộ́ ự́ ɛ̣́ ɔ̣́ *Ậ́ Ệ́ Ị́ Ộ́ Ự́ Ɛ̣́ Ɔ̣́ ậ́ ệ́ ị́ ộ́ ự́ ɛ̣́ ɔ̣́* Ậ́ Ệ́ Ị́ Ộ́ Ự́ Ɛ̣́ Ɔ̣́ ậ́ ệ́ ị́ ộ́ ự́ ɛ̣́ ɔ̣́ *Ậ́ Ệ́ Ị́ Ộ́ Ự́ Ɛ̣́ Ɔ̣́ ậ́ ệ́ ị́ ộ́ ự́ ɛ̣́ ɔ̣́*

8 Ậ̀ Ệ̀ Ị̀ Ộ̀ Ự̀ Ɛ̣̀ Ɔ̣̀ ậ̀ ệ̀ ị̀ ộ̀ ự̀ ɛ̣̀ ɔ̣̀ *Ậ̀ Ệ̀ Ị̀ Ộ̀ Ự̀ Ɛ̣̀ Ɔ̣̀ ậ̀ ệ̀ ị̀ ộ̀ ự̀ ɛ̣̀ ɔ̣̀* Ậ̀ Ệ̀ Ị̀ Ộ̀ Ự̀ Ɛ̣̀ Ɔ̣̀ ậ̀ ệ̀ ị̀ ộ̀ ự̀ ɛ̣̀ ɔ̣̀ *Ậ̀ Ệ̀ Ị̀ Ộ̀ Ự̀ Ɛ̣̀ Ɔ̣̀ ậ̀ ệ̀ ị̀ ộ̀ ự̀ ɛ̣̀ ɔ̣̀*

9 Ặ Ệ̆ Ị̆ Ộ̆ Ự̆ Ɛ̣̆ Ɔ̣̆ ặ ệ̆ ị̆ ộ̆ ự̆ ɛ̣̆ ɔ̣̆ *Ặ Ệ̆ Ị̆ Ộ̆ Ự̆ Ɛ̣̆ Ɔ̣̆ ặ ệ̆ ị̆ ộ̆ ự̆ ɛ̣̆ ɔ̣̆* Ặ Ệ̆ Ị̆ Ộ̆ Ự̆ Ɛ̣̆ Ɔ̣̆ ặ ệ̆ ị̆ ộ̆ ự̆ ɛ̣̆ ɔ̣̆ *Ặ Ệ̆ Ị̆ Ộ̆ Ự̆ Ɛ̣̆ Ɔ̣̆ ặ ệ̆ ị̆ ộ̆ ự̆ ɛ̣̆ ɔ̣̆*

10 Ậ́ Ệ́ Ị́ Ộ́ Ự́ Ɛ̣́ Ɔ̣́ ậ́ ệ́ ị́ ộ́ ự́ ɛ̣́ ɔ̣́ *Ậ́ Ệ́ Ị́ Ộ́ Ự́ Ɛ̣́ Ɔ̣́ ậ́ ệ́ ị́ ộ́ ự́ ɛ̣́ ɔ̣́* Ậ́ Ệ́ Ị́ Ộ́ Ự́ Ɛ̣́ Ɔ̣́ ậ́ ệ́ ị́ ộ́ ự́ ɛ̣́ ɔ̣́ *Ậ́ Ệ́ Ị́ Ộ́ Ự́ Ɛ̣́ Ɔ̣́ ậ́ ệ́ ị́ ộ́ ự́ ɛ̣́ ɔ̣́*

11 Ậ̀ Ệ̀ Ị̀ Ộ̀ Ự̀ Ɛ̣̀ Ɔ̣̀ ậ̀ ệ̀ ị̀ ộ̀ ự̀ ɛ̣̀ ɔ̣̀ *Ậ̀ Ệ̀ Ị̀ Ộ̀ Ự̀ Ɛ̣̀ Ɔ̣̀ ậ̀ ệ̀ ị̀ ộ̀ ự̀ ɛ̣̀ ɔ̣̀* Ậ̀ Ệ̀ Ị̀ Ộ̀ Ự̀ Ɛ̣̀ Ɔ̣̀ ậ̀ ệ̀ ị̀ ộ̀ ự̀ ɛ̣̀ ɔ̣̀ *Ậ̀ Ệ̀ Ị̀ Ộ̀ Ự̀ Ɛ̣̀ Ɔ̣̀ ậ̀ ệ̀ ị̀ ộ̀ ự̀ ɛ̣̀ ɔ̣̀*

Patched

A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***

Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́ *Á É Í Ó Ú Ɛ́ Ɔ́ á é í ó ú ɛ́ ɔ́* Á̩ É̩ Í̩ Ó̩ Ú̩ Ɛ̩́ Ɔ̩́ á̩ é̩ í̩ ó̩ ú̩ ɛ̩́ ɔ̩́ *Á̩ É̩ Í̩ Ó̩ Ú̩ Ɛ̩́ Ɔ̩́ á̩ é̩ í̩ ó̩ ú̩ ɛ̩́ ɔ̩́*

À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* À̩ È̩ Ì̩ Ò̩ Ù̩ Ɛ̩̀ Ɔ̩̀ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩̀ ɔ̩̀ *À̩ È̩ Ì̩ Ò̩ Ù̩ Ɛ̩̀ Ɔ̩̀ à̩ è̩ ì̩ ò̩ ù̩ ɛ̩̀ ɔ̩̀*

Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ɔ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* Ă̩ Ĕ̩ Ĭ̩ Ŏ̩ Ŭ̩ Ɛ̩̆ Ɔ̩̆ ă̩ ĕ̩ ĭ̩ ŏ̩ ŭ̩ ɛ̩̆ ɔ̩̆ *Ă̩ Ĕ̩ Ĭ̩ Ŏ̩ Ŭ̩ Ɛ̩̆ Ɔ̩̆ ă̩ ĕ̩ ĭ̩ ŏ̩ ŭ̩ ɛ̩̆ ɔ̩̆*

Ắ Ế Ḯ Ố Ű Ɛ́ Ɔ́ ắ ế ḯ ố ű ɛ́ ɔ́ *Ắ Ế Ḯ Ố Ű Ɛ́ Ɔ́ ắ ế ḯ ố ű ɛ́ ɔ́* Ắ̩ Ế̩ Ḯ̩ Ố̩ Ű̩ Ɛ̩́ Ɔ̩́ ắ̩ ế̩ ḯ̩ ố̩ ű̩ ɛ̩́ ɔ̩́ *Ắ̩ Ế̩ Ḯ̩ Ố̩ Ű̩ Ɛ̩́ Ɔ̩́ ắ̩ ế̩ ḯ̩ ố̩ ű̩ ɛ̩́ ɔ̩́*

Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ɔ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀ *Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ɔ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀* Ằ̩ Ề̩ Ỉ̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ ằ̩ ề̩ ỉ̩ ồ̩ ừ̩ ɛ̩̀ ɔ̩̀ *Ằ̩ Ề̩ Ỉ̩ Ồ̩ Ừ̩ Ɛ̩̀ Ɔ̩̀ ằ̩ ề̩ ỉ̩ ồ̩ ừ̩ ɛ̩̀ ɔ̩̀*

Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣* Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣*

cgj Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣* Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣ *Ạ Ệ Ị Ọ Ự Ɛ̣ Ɔ̣ ạ ệ ị ọ ự ɛ̣ ɔ̣*

Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋***
cgj Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋***
Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ȩ̏ Ɔ̏ ą̏ ę̏ į̏ ǫ̏ ų̏ ȩ̏ ɔ̏ *Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ȩ̏ Ɔ̏ ą̏ ę̏ į̏ ǫ̏ ų̏ ȩ̏ ɔ̏* **Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ȩ̏ Ɔ̏ ą̏ ę̏ į̏ ǫ̏ ų̏ ȩ̏ ɔ̏** ***Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ȩ̏ Ɔ̏ ą̏ ę̏ į̏ ǫ̏ ų̏ ȩ̏ ɔ̏***
Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ȩ̏ Ɔ̏ ą̏ ę̏ į̏ ǫ̏ ų̏ ȩ̏ ɔ̏ *Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ȩ̏ Ɔ̏ ą̏ ę̏ į̏ ǫ̏ ų̏ ȩ̏ ɔ̏* **Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ȩ̏ Ɔ̏ ą̏ ę̏ į̏ ǫ̏ ų̏ ȩ̏ ɔ̏** ***Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ȩ̏ Ɔ̏ ą̏ ę̏ į̏ ǫ̏ ų̏ ȩ̏ ɔ̏***
Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀***
cgj Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ȩ̀ Ɔ̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ȩ̀ ɔ̀***
Ą̌ Ę̌ Į̌ Ǫ̌ Ų̌ Ȩ̌ Ɔ̌ ą̌ ę̌ į̌ ǫ̌ ų̌ ȩ̌ ɔ̌ *Ą̌ Ę̌ Į̌ Ǫ̌ Ų̌ Ȩ̌ Ɔ̌ ą̌ ę̌ į̌ ǫ̌ ų̌ ȩ̌ ɔ̌* **Ą̌ Ę̌ Į̌ Ǫ̌ Ų̌ Ȩ̌ Ɔ̌ ą̌ ę̌ į̌ ǫ̌ ų̌ ȩ̌ ɔ̌** ***Ą̌ Ę̌ Į̌ Ǫ̌ Ų̌ Ȩ̌ Ɔ̌ ą̌ ę̌ į̌ ǫ̌ ų̌ ȩ̌ ɔ̌***
cgj Ą̌ Ę̌ Į̌ Ǫ̌ Ų̌ Ȩ̌ Ɔ̌ ą̌ ę̌ į̌ ǫ̌ ų̌ ȩ̌ ɔ̌ *Ą̌ Ę̌ Į̌ Ǫ̌ Ų̌ Ȩ̌ Ɔ̌ ą̌ ę̌ į̌ ǫ̌ ų̌ ȩ̌ ɔ̌* **Ą̌ Ę̌ Į̌ Ǫ̌ Ų̌ Ȩ̌ Ɔ̌ ą̌ ę̌ į̌ ǫ̌ ų̌ ȩ̌ ɔ̌** ***Ą̌ Ę̌ Į̌ Ǫ̌ Ų̌ Ȩ̌ Ɔ̌ ą̌ ę̌ į̌ ǫ̌ ų̌ ȩ̌ ɔ̌***
Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋***
cgj Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ȩ̋ Ɔ̋ ą̋ ę̋ į̋ ǫ̋ ų̋ ȩ̋ ɔ̋***
Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈ *Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈* **Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈** ***Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈***
cgj Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈ *Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈* **Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈** ***Ą̈ Ę̈ Į̈ Ǫ̈ Ų̈ Ȩ̈ Ɔ̈ ą̈ ę̈ į̈ ǫ̈ ų̈ ȩ̈ ɔ̈***
Patched not grave
À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* **À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀** ***À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀***
Patched grave
À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀ *À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀* **À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀** ***À B̀È Ì Ò Ù Ɛ̀ Ɔ̀ à è ì ò ù ɛ̀ ɔ̀***

## Mark over space

I added a new en-width x-height PUA character, U+E100, with anchors. To float an above mark, say double grave U+030F over a space, write `\char"E100\char"030F` But in practice, the floating mark carries the same tonal specification as the preceding vowel, and therefore must align exactly with the height of that vowel's combining mark. To ensure this mark alignment, insert a combining grapheme joiner, `\cgj`, in the prior vowel + mark combo, to force anchor positioning without substituting a precomposed glyph there. Note that in Libertinus, the two graves in the U+030F glyph are not translates of an identical outline; this is a deliberate design choice by the designer.

Original

`-ke\char"030F\ \char"030F` -kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Patched without CGJ

`-ke\char"030F \char"E100\char"030F` -kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Patched with CGJ

`-ke\cgj\char"030F \char"E100\char"030F` -kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

## f with dot below

tiɣə̀ ṛfa

*tiɣə̀ ṛfa*

**tiɣə̀ ṛfa**

***tiɣə̀ ṛfa***

ṛ = `\char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.

f̣ = `f\char"0323` is f with U+0323 combining dot below.

Here are some letters with a dot below.

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ  ç f̧ g̦ j̨ p̧ q̧ x̧

*ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ  ç f̧ g̦ j̨ p̧ q̧ x̧*

**ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ  ç f̧ g̦ j̨ p̧ q̧ x̧**

***ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ  ç f̧ g̦ j̨ p̧ q̧ x̧***

The left group of letters – a b d e h i k l m n o r s t u v w y z – have a precomposed character. For the right group of letters – c f g j p q x – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by HarfBuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and HarfBuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but HarfBuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of `uni0307`, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references `dotbelowcomb`, combining dot below, U+0323. In semibold italic, `dotaccent` is used instead of `uni0307`. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In regular, `uni0307` has diameter 100, and `dotbelowcomb` has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take a bigger dot. In semibold, the size difference is reversed: `uni0307` has larger diameter 124, and `dotbelowcomb` has small diameter 108. Actually, semibold `dotbelowcomb` is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, `dotaccent` is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by $(\pm 6, \pm 7.5)$ from the center — and `dotbelowcomb` is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at $Y = -319$, so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of `dotbelowcomb`. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonbly faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

> For italic, the descender's U-pocket is defined by the control points at $(14, -238)$ and $(33, -205)$, giving a U-center around $x \approx 24$. Using the (symmetric) italic `dotbelowcomb` anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is $X \approx 30$.
>
> For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of $X = -33.5$. The semibold italic `dotbelowcomb` itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is $X \approx 62$.

Here are examples with the patched fonts, with f's below-anchor in italic at $(30, -319)$ and semibold italic at $(62, -319)$. This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

| Original | Patched italics (f only) |
| --- | --- |
| tiɣə ṛfa | tiɣə ṛfa |
| *tiɣə ṛfa* | *tiɣə ṛfa* |
| **tiɣə ṛfa** | **tiɣə ṛfa** |
| ***tiɣə ṛfa*** | ***tiɣə ṛfa*** |

Original (repeated from above)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ

*ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ*

**ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ**

***ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ***

Patched italics (f only)

*ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ*

***ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ģ j ƥ q ẋ***

## Ashkun

`a\char"0304\char"0303` ã *ã* **ã** ***ã***

`\char"0101\char"0303` ã *ã* **ã** ***ã***

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Patched

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`u\cgj\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\cgj\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Move box without patching font

`\char"016B\kern-4.132pt\raisebox{0.88pt}{\char"02DC}` ũ *ũ* **ũ** ***ũ***

## Tacks

a̰ḛḭo̰ṵa̰ḛḭo̰ṵ

a̰ḛḭo̰ṵa̰ḛḭo̰ṵ

## Upsilon and grave

Original, then patched, versions of upsilon and combining grave, after setting upsilon's above-anchor.

ὺ 𝜐́ **ὺ** **𝝊́**

ὺ 𝜐́ **ὺ** **𝝊́**

New upsilon above-anchor: regular $(253, 645)$; italic $(358, 645)$; semibold $(267, 645)$; semibold italic $(369, 645)$

Revised combining grave above-anchor: semibold italic $(0, 683)$. Moved from $(-142, 683)$, which is far left of the optical center of the mark, which has the effect of attaching the mark too far right.

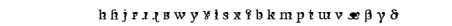
Spoof to move modifier grave (which has the exact same outline as combining grave) to an anchored position, if upsilon had an above-anchor at $(358, 645)$ (keeping combining grave's above-anchor as is). Note upsilon is U+028A and modifier grave is U+02CB.

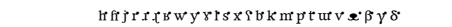
```
\textit{\char"028A\kern-4.66pt\raisebox{-0.38pt}{\char"02CB}}
```

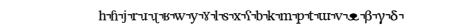
*ʊ̀*

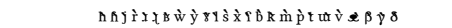
## Patch superscript consonant anchors

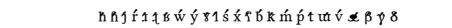
Libertinus serif regular

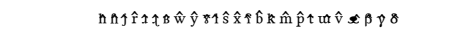




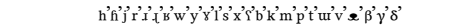


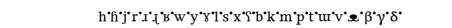


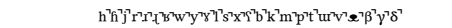


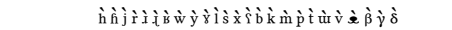


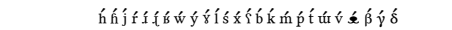
Libertinus serif regular, patched

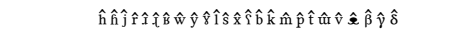


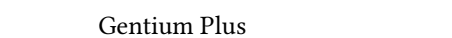




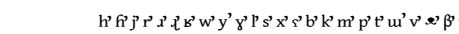


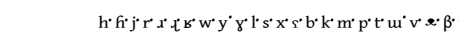


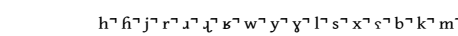


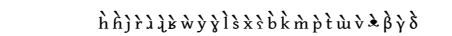


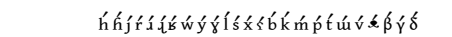
Gentium Plus

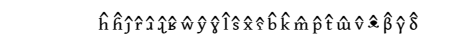




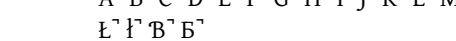












A B C D E F G H I J K L M N O P Q R S T U V W X Y Z a b c d e f g h i j k l m n o p q r s t u v w x y z Æ æ Œ œ Ø ø Đ ð Þ þ ß Ĺ Đ đ Ħ ħ ı j Ł ł Ƀ Ƃ
À Ḃ Ċ Ḋ È Ḟ Ġ Ḣ Ì J̇ Ǩ L̇ Ṁ Ṅ Ò Ṗ Q̇ Ṙ Ṡ Ṫ Ù V̇ Ẁ Ẋ Ẏ Ż à ḃ ċ ḋ è ḟ ġ ḣ ì j̇ ǩ l̀ ṁ ṅ ò ṗ q̇ ṙ ṡ ṫ ù v̇ ẁ ẋ ẏ ż Æ æ Œ œ Ø ø Đ ð Þ þ ß Ĺ Đ đ Ħ ħ ì j̇ Ł ł Ƀ Ƃ

## Dot removal

To do. Check GSUB to see if dot removal for i and j is complete.

## Precomposed characters

To do. Check GSUB if an available precomposed character is encoded by a doublet of the base and a combining mark, or a triplet of the base and the two combining marks, in either mark order, then the precomposed character should be be produced by the font's GSUB table.

# Marks

Here is a catalog of combining marks in Unicode, grouped by the seven GPOS anchor class indices in Libertinus (0–6).

## Above marks – GPOS anchor 0

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿

## Below marks – GPOS anchor 2

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿

## Left angle – GPOS anchor 3

x̕ x̕ x⃐ x⃐

## Below-right marks – GPOS anchor 4

x̙ x̚ x̛
x̙ x̚ x̛
x̙ x̚ x̛
x̙ x̚ x̛

## Cedilla – GPOS anchor 5

x̧ x̧ x̧ x̧

## Overlay marks – GPOS anchor 6

o̶ o̶ o̶ ø̶
o̶ o̶ ø̶ ø̶
o̶ o̶ o̶ o̶ ø̶
o̶ o̶ o̶ o̶ ø̶

## Double marks

x͡x x͡x x͞x x͞x x̃x x̃x x͡x x͡x
x͡x x͡x x͞x x͞x x̃x x̃x x͡x x͡x
x̋x x̋x x̏x x̂x x̌x x̆x ẍx
x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x

## Greek marks

x̀x́x̃x̄x̌ẋẍx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̋x̋x̏ẍx̨
x̀x́x̃x̄x̌ẋẍx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̋x̋x̏x̋x̨
x̀x́x̃x̄x̌ẋx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̋x̋x̏ẍx̣
x̀x́x̃x̄x̌ẋẍx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̋x̋x̏ẍx̣

## Adding missing marks

Semibold and semibold italic lack many of the combining marks of regular and italic. The patch adds them, with mark anchors, in the range U+0332–U+036F. No mark is drawn from scratch: each is an outline that the designer drew in another style. A semibold mark is the regular outline (an above-mark is moved right by 12 units, with its anchor, as the designer's semibold above-marks are). A semibold italic mark is the italic outline moved right by 12 units, with the anchor of regular, as in the semibold italic anchor model. In semibold, U+0358–U+0363 were copies of the capital accents, and they are replaced. The patch also adds `dotaccent.cap` to italic and semibold italic: the dot of the precomposed İ, at the height it has in the precomposed dotted capitals.

The designer does not thicken below-marks between weights, but thickens above-marks by about 12 percent (mean stroke thickness, measured as twice the area of the outline divided by its perimeter: tilde 40 to 45, caron 36 to 42). So the added above-marks are made bolder. Where a mark is, or contains, another mark, the bold style's own mark is used. The rest of the outline is pushed outward by 7.5 percent of its stroke thickness on each side, about 2 to 3 units. These algorithmically emboldened marks are shown in the table on the next page, on a few bases that have above-anchors in all four patched fonts, with the designer's tilde and caron for comparison of weight. The regular and italic columns show the outlines that were copied.

Also added, but not shown in the table: the low line and double low line (U+0332, U+0333), the tilde, stroke and solidus overlays (U+0334–U+0337), the below-marks U+0339–U+033C, U+0347–U+0349, U+034D, U+034E, U+0353–U+0356, U+0359, U+035A, the double overline (U+033F), the dot above right (U+0358), the double marks (U+035C–U+0362) and the combining letters (U+0363–U+036F). These are copied without emboldening, since the designer's lines, below-marks and modifier letters are not thicker in the bold styles; the dot above right is the bold style's own dot. The double marks have no anchors, as in regular. In semibold, the overlays and the dot above right have no anchors, because the semibold lookup table has no anchor class for them. The half rings above (U+0351, U+0357) already exist in semibold, drawn to the right of the origin, so they sit beside the base; they are left as they are. Not added: the custom-width low lines and overlines of the next section, U+0344, U+0345 and U+0483.

| | | regular | semibold | italic | semibold italic | |
|---|---|---|---|---|---|---|
| U+0303 | tilde (designer's, for comparison) | ãñũÕ | **ãñũÕ** | *ãñũÕ* | ***ãñũÕ*** | |
| U+030C | caron (designer's, for comparison) | ǎňǔǑ | **ǎňǔǑ** | *ǎňǔǑ* | ***ǎňǔǑ*** | |
| U+033D | x above | x̌ňǔǑ | **x̌ňǔǑ** | *x̌ňǔǑ* | ***x̌ňǔǑ*** | emboldened |
| U+033E | vertical tilde | a̾n̾u̾O̾ | **a̾n̾u̾O̾** | *a̾n̾u̾O̾* | ***a̾n̾u̾O̾*** | emboldened |
| U+034A | not tilde above | ãn̋ũÕ̋ | **ãn̋ũÕ̋** | *ãn̋ũÕ̋* | ***ãn̋ũÕ̋*** | emboldened |
| U+034B | homothetic above | ãn̑ũÕ̑ | **ãn̑ũÕ̑** | *ãn̑ũÕ̑* | ***ãn̑ũÕ̑*** | own tilde; dots emboldened |
| U+034C | almost equal to above | ãn̆ũÕ̆ | **ãn̆ũÕ̆** | *ãn̆ũÕ̆* | ***ãn̆ũÕ̆*** | emboldened |
| U+0350 | right arrowhead above | ãn̓ũÕ̓ | **ãn̓ũÕ̓** | *ãn̓ũÕ̓* | ***ãn̓ũÕ̓*** | emboldened |
| U+0351 | left half ring above | ãn̔ũÕ̔ | **ãn̔ũÕ̔** | *ãn̔ũÕ̔* | ***ãn̔ũÕ̔*** | emboldened (semibold italic only) |
| U+0352 | fermata | ãn̗ũÕ̗ | **ãn̗ũÕ̗** | *ãn̗ũÕ̗* | ***ãn̗ũÕ̗*** | own inverted breve; dot emboldened |
| U+0357 | right half ring above | a̾n̾u̾O̾ | **a̾n̾u̾O̾** | *a̾n̾u̾O̾* | ***a̾n̾u̾O̾*** | emboldened (semibold italic only) |
| U+035B | zigzag above | ãn̛ũỠ | **ãn̛ũỠ** | *ãn̛ũỠ* | ***ãn̛ũỠ*** | emboldened |
| U+0340 | grave tone mark | àǹùÒ | **àǹùÒ** | *àǹùÒ* | ***àǹùÒ*** | own grave |
| U+0341 | acute tone mark | áńúÓ | **áńúÓ** | *áńúÓ* | ***áńúÓ*** | own acute |
| U+0346 | bridge above | ãn̪ũÕ̪ | **ãn̪ũÕ̪** | *ãn̪ũÕ̪* | ***ãn̪ũÕ̪*** | own bridge below, raised |

## Overline and low line

U+0305 combining overline and U+0332 combining low line are applied by GSUB lookups that substitute a custom-width line mark according to width classes. GSUB lookup 51 effectively classifies the glyphs by width, and GSUB lookups 52–82 provide the appropriate custom-width mark for the base width class. Semibold and semibold italic do not have any combining low lines; the patch adds U+0332 and U+0333 with below-anchors, but not the custom-width variants. Custom-width line marks do not have anchors. They are positioned by HarfBuzz fallback shaping. The overlines are above the ascender line, so they are above all bases. The low lines are below the baseline, but above the descender line, and so they cut though any descenders. If a new glyph is added, and you want to support custom-width line marks for it, then remember to classify the new glyph in GSUB lookup 51.



## Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

### Horn

The Vietnamese alphabet has two horn vowels – Ơơ Ưư– and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

Ờớ Ởở Ỡỡ Ợợ Ừứ Ửử Ữữ Ựự

*Ờớ Ởở Ỡỡ Ợợ Ừứ Ửử Ữữ Ựự*

**Ờớ Ởở Ỡỡ Ợợ Ừứ Ửử Ữữ Ựự**

***Ờớ Ởở Ỡỡ Ợợ Ừứ Ửử Ữữ Ựự***

There is a Unicode character for the combining horn mark: ’ `\char"031B` U+031B, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

### Comma above right

There is a Unicode character for the combining comma above right mark: ’ `\char"0315` U+0315.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *k̓wkt* [k̓$^{wkx}$t] ‘to scrape’ (Nuxalk)

(2) *p’ak* [p’$^{kx}$ak] ‘he/she cut it’ (Upper Chehalis)

In (1), the superscript consonant cluster is $^{wkx}$ and it is glottalized. In (2), the superscript consonant $^{x}$ is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

### Dot above right

There is a Unicode character for the combining dot above right mark: ˙ `\char"0358` U+0358 dot above right

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by HarfBuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.

Libertinus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [kʾʷᵏˣˀt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʾʷᵏˣʻt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [kʾʷᵏˣʼt]

Gentium Plus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [kʾʷᵏˣˀt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʾʷᵏˣʻt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [kʾʷᵏˣʼt]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ʰ’ɦ’ʲ’ʳ’ɹ’ɻ’ʁ’ʷ’ʸ’ʏ’ˡ’ˢ’ˣ’ʕ’ᵇ’ᵏ’ᵐ’ᵖ’ᵗ’ɯ’ᵛ’ꭥ’ᵝ’ᵞ’ᵟ’

ʰ‘ɦ‘ʲ‘ʳ‘ɹ‘ɻ‘ʁ‘ʷ‘ʸ‘ʏ‘ˡ‘ˢ‘ˣ‘ʕ‘ᵇ‘ᵏ‘ᵐ‘ᵖ‘ᵗ‘ɯ‘ᵛ‘ꭥ‘ᵝ‘ᵞ‘ᵟ‘

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʾʷᵏˣʻt]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ʰ’ɦ’ʲ’ʳ’ɹ’ɻ’ʁ’ʷ’ʸ’ʏ’ˡ’ˢ’ˣ’ʕ’ᵇ’ᵏ’ᵐ’ᵖ’ᵗ’ɯ’ᵛ’ꭥ’ᵝ’ᵞ’ᵟ’

ʰ‘ɦ‘ʲ‘ʳ‘ɹ‘ɻ‘ʁ‘ʷ‘ʸ‘ʏ‘ˡ‘ˢ‘ˣ‘ʕ‘ᵇ‘ᵏ‘ᵐ‘ᵖ‘ᵗ‘ɯ‘ᵛ‘ꭥ‘ᵝ‘ᵞ‘ᵟ‘

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [kʾʷᵏˣʻt]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at $Y = 630$. So their anchor will be at $Y = 630 - 62 + 80 = 648$. The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at $X = \text{xMax} + 26 + 12$. These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.

# Bases

Here are some subgroups of bases.

## Latin

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ
*ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ*
**ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ**
***ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƃ***

## IPA

ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒ2ǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɿɬɮɭɮɰɯɱɲɳɴɵœɷɸɹɻɺɽɾʁʂʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀ
*ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒ2ǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɿɬɮɭɮɰɯɱɲɳɴɵœɷɸɹɻɺɽɾʁʂʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀ*
**ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒ2ǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɿɬɮɭɮɰɯɱɲɳɴɵœɷɸɹɻɺɽɾʁʂʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀ**
***ɒɔɕɗɖɑɋɜəɛɟfɠɣɦɬɪʞɯɲɵɭʒ2ǁǂ!ɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɣɤɥɦɧɨɪɿɬɮɭɮɰɯɱɲɳɴɵœɷɸɹɻɺɽɾʁʂʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀ***

## Superscript consonants

ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛᵝᵞᵟ
*ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛᵝᵞᵟ*
**ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛᵝᵞᵟ**
***ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵠᵛᵝᵞᵟ***

## Greek

ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
*ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω*
**ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω**
***ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω***

## Cyrillic

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ
*АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ*
**АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ**
***АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ***

## Precomposed capital vowels

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈỈỊÒÓÔÕÖŌŎŐǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲǓȔȖỤỦỨỪỬỮỰ
*ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈỈỊÒÓÔÕÖŌŎŐǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲǓȔȖỤỦỨỪỬỮỰ*
**ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈỈỊÒÓÔÕÖŌŎŐǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲǓȔȖỤỦỨỪỬỮỰ**
***ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈỈỊÒÓÔÕÖŌŎŐǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲǓȔȖỤỦỨỪỬỮỰ***

## Precomposed small vowels

àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųǖȕȗụủứừửữự
*àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųǖȕȗụủứừửữự*
**àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųǖȕȗụủứừửữự**
***àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųǖȕȗụủứừửữự***

## Precomposed capital consonants

ḂḆÇĆĈČĊÇḈĎḐḒḌḎḐḊĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴĶḲĹḶĽḺḸḼḾṀṂŃṆŇṊṄṈƝǑṔṖṕŖŘṘṚṞŚŜŞŠṠṢṤṦṨṬṰṪṮṱṲṴƲŨṾỤỦẂẀẄẂẊẌÝỲŶỸỴỶỾŸỵŹŻŽẐẔẒẔ
*ḂḆÇĆĈČĊÇḈĎḐḒḌḎḐḊĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴĶḲĹḶĽḺḸḼḾṀṂŃṆŇṊṄṈƝǑṔṖṕŖŘṘṚṞŚŜŞŠṠṢṤṦṨṬṰṪṮṱṲṴƲŨṾỤỦẂẀẄẂẊẌÝỲŶỸỴỶỾŸỵŹŻŽẐẔẒẔ*
**ḂḆÇĆĈČĊÇḈĎḐḒḌḎḐḊĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴĶḲĹḶĽḺḸḼḾṀṂŃṆŇṊṄṈƝǑṔṖṕŖŘṘṚṞŚŜŞŠṠṢṤṦṨṬṰṪṮṱṲṴƲŨṾỤỦẂẀẄẂẊẌÝỲŶỸỴỶỾŸỵŹŻŽẐẔẒẔ**
***ḂḆÇĆĈČĊÇḈĎḐḒḌḎḐḊĜĞĠĢḠǦĤḢḤḦḪḨĴḰǨḴĶḲĹḶĽḺḸḼḾṀṂŃṆŇṊṄṈƝǑṔṖṕŖŘṘṚṞŚŜŞŠṠṢṤṦṨṬṰṪṮṱṲṴƲŨṾỤỦẂẀẄẂẊẌÝỲŶỸỴỶỾŸỵŹŻŽẐẔẒẔ***

## Precomposed small consonants

ƀƃɓƅḃḅḇƈćĉċčçḉƈďđḋḍḏḑḓḍḟĝğġģḡǧǥĥħḣḥḧḩḫẖĵǰķḱǩḳḵƙĺļľŀḷḽḻłḿṁṃṅṇňŉṋṉṇṋṅŋɲȵɵɋɋṕƥƿƾɍŗřŕȓȑṛṟśŝşšṡṣṥṧṩßſẛʂţťŧṫṭṯṱʈưũṽỵẁŵẃẅẋẍýÿŷỳẏỹƴźżžẑẓẕẓȝƨƽpǯ
*ƀƃɓƅḃḅḇƈćĉċčçḉƈďđḋḍḏḑḓḍḟĝğġģḡǧǥĥħḣḥḧḩḫẖĵǰķḱǩḳḵƙĺļľŀḷḽḻłḿṁṃṅṇňŉṋṉṇṋṅŋɲȵɵɋɋṕƥƿƾɍŗřŕȓȑṛṟśŝşšṡṣṥṧṩßſẛʂţťŧṫṭṯṱʈưũṽỵẁŵẃẅẋẍýÿŷỳẏỹƴźżžẑẓẕẓȝƨƽpǯ*
**ƀƃɓƅḃḅḇƈćĉċčçḉƈďđḋḍḏḑḓḍḟĝğġģḡǧǥĥħḣḥḧḩḫẖĵǰķḱǩḳḵƙĺļľŀḷḽḻłḿṁṃṅṇňŉṋṉṇṋṅŋɲȵɵɋɋṕƥƿƾɍŗřŕȓȑṛṟśŝşšṡṣṥṧṩßſẛʂţťŧṫṭṯṱʈưũṽỵẁŵẃẅẋẍýÿŷỳẏỹƴźżžẑẓẕẓȝƨƽpǯ**

***ƀƃɓƅḃḅḇƈćĉċčçḉƈďđḋḍḏḑḓḍḟĝğġģḡǧǥĥħḣḥḧḩḫẖĵǰķḱǩḳḵƙĺļľŀḷḽḻłḿṁṃṅṇňŉṋṉṇṋṅŋɲȵɵɋɋṕƥƿƾɍŗřŕȓȑṛṟśŝşšṡṣṥṧṩßſẛʂţťŧṫṭṯṱʈưũṽỵẁŵẃẅẋẍýÿŷỳẏỹƴźżžẑẓẕẓȝƨƽpǯ***

## No marks: rare, digraph etc.

ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
*ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj*
**ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj**
***ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj***

## Small capitals

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
*A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ*
**A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ**
***A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ***
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
*AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ*
**AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ**
***AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ***

## Precomposed characters of anchor-relevant bases and marks

àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
*àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ*
**àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ**
***àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ***

## Base and mark shaping – Linguistics notation

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic.

The grids use Latin in IPA bases and combining marks that are used in linguistics notation.

A black color means GPOS anchor positioning.

A blue color means a precomposed glyph.

A red color means HarfBuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A faded color – gray, light blue, or pink – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This only happens in Libertinus Serif Semibold and Semibold Italic for three below marks: o͇ (U+0333 double macron below), o͚ (U+033A inverted bridge below), and o͛ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

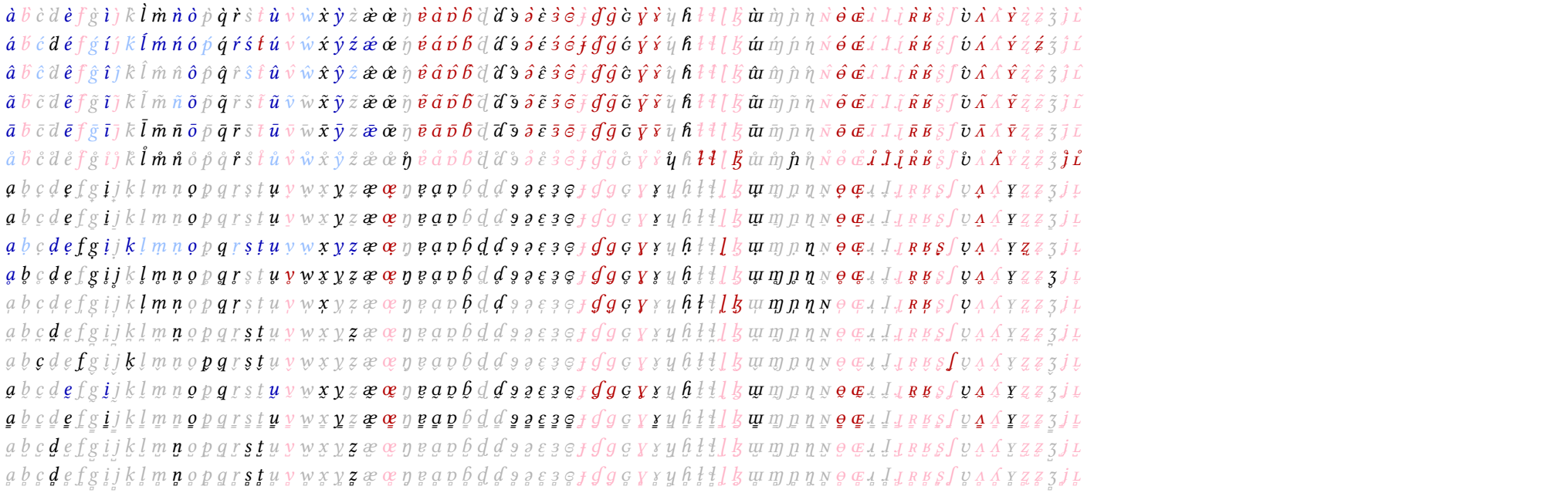
## Regular, Bases requiring anchors for common marks, Common combining marks



## Regular patched, Bases requiring anchors for common marks, Common combining marks



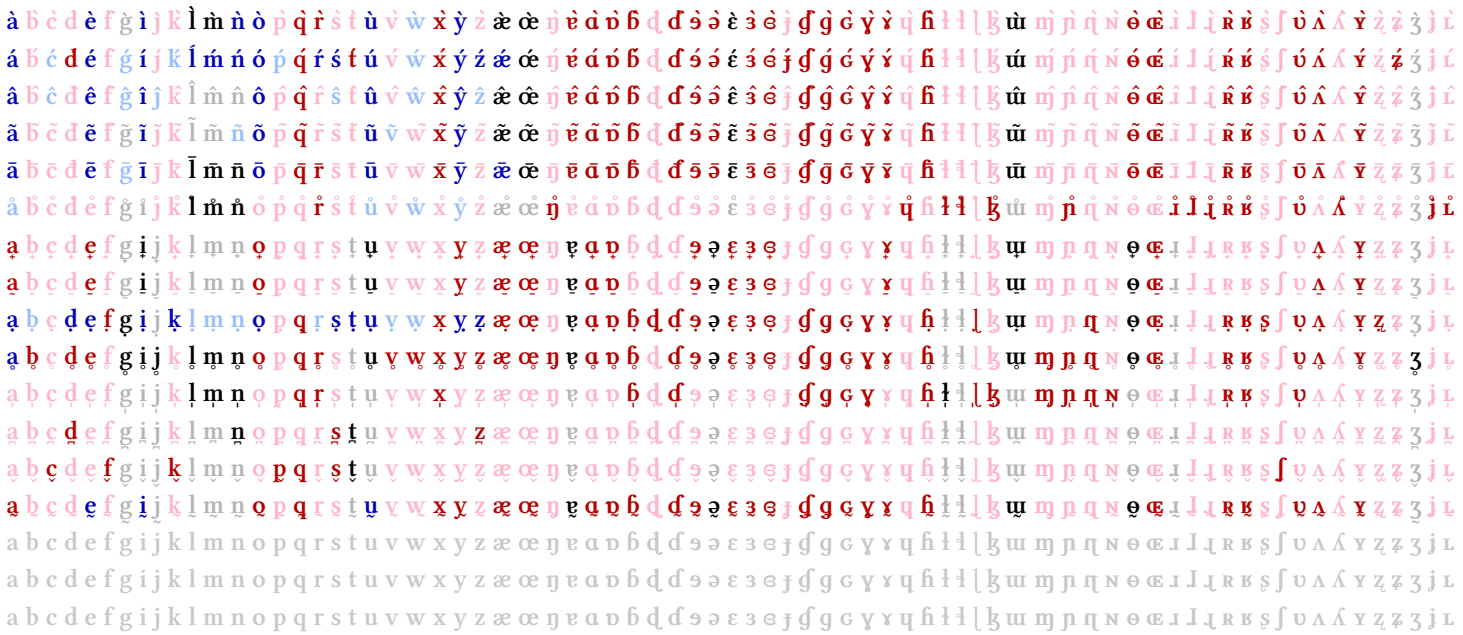
## Italic, Bases requiring anchors for common marks, Common combining marks



## Italic patched, Bases requiring anchors for common marks, Common combining marks



## Semibold, Bases requiring anchors for common marks, Common combining marks



## Semibold patched, Bases requiring anchors for common marks, Common combining marks



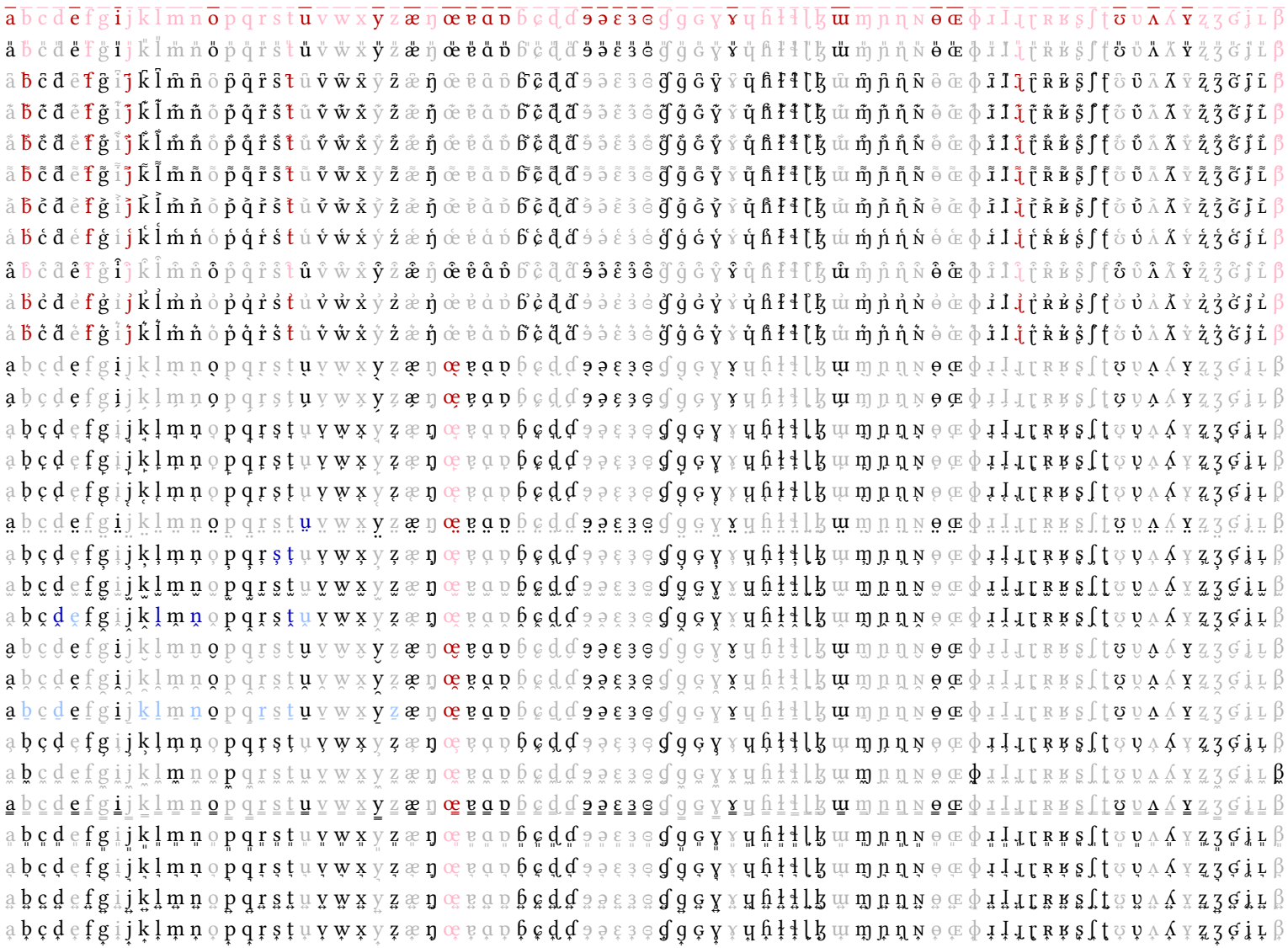
## Semibold italic, Bases requiring anchors for common marks, Common combining marks



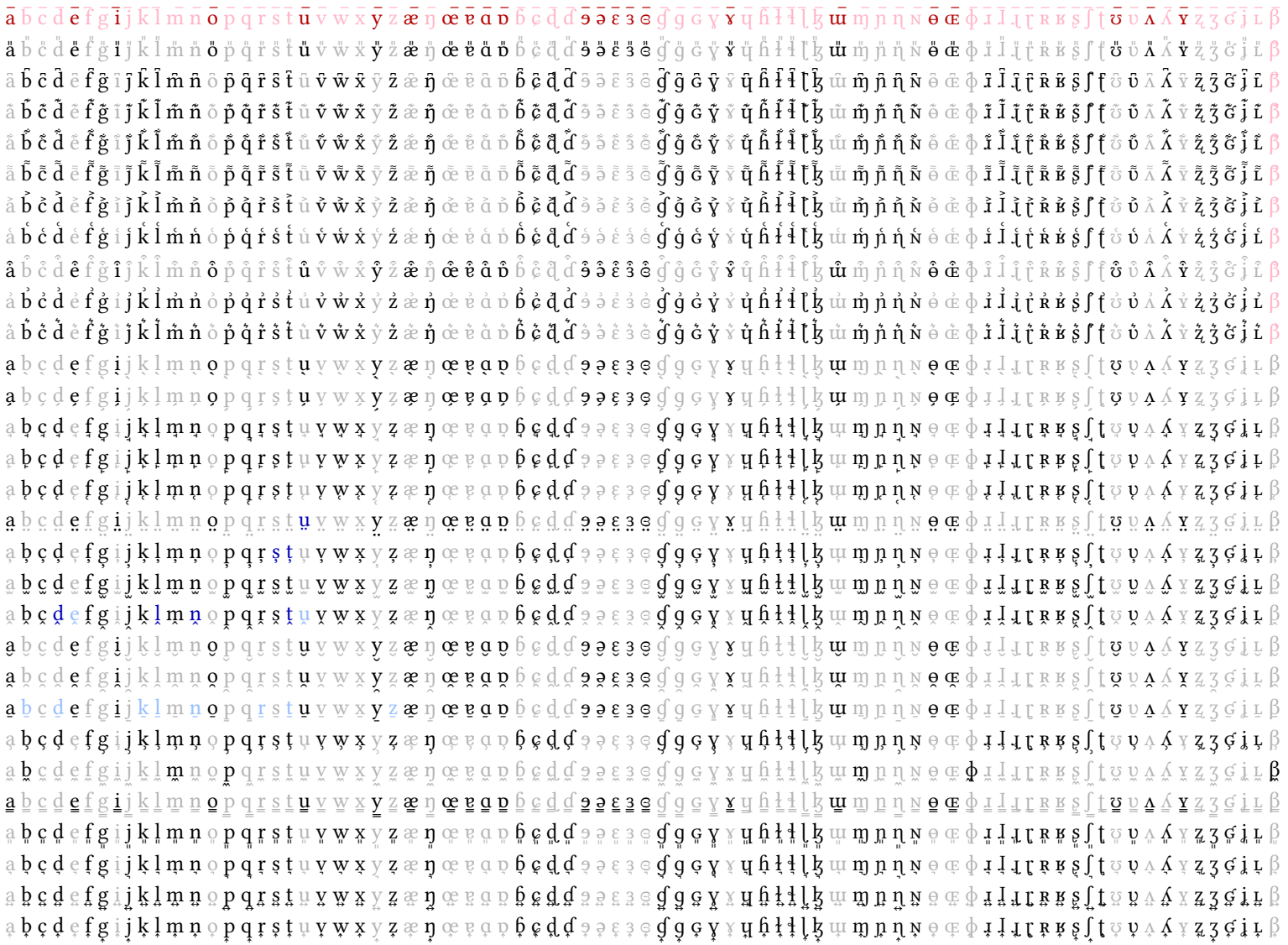
## Semibold italic patched, Bases requiring anchors for common marks, Common combining marks



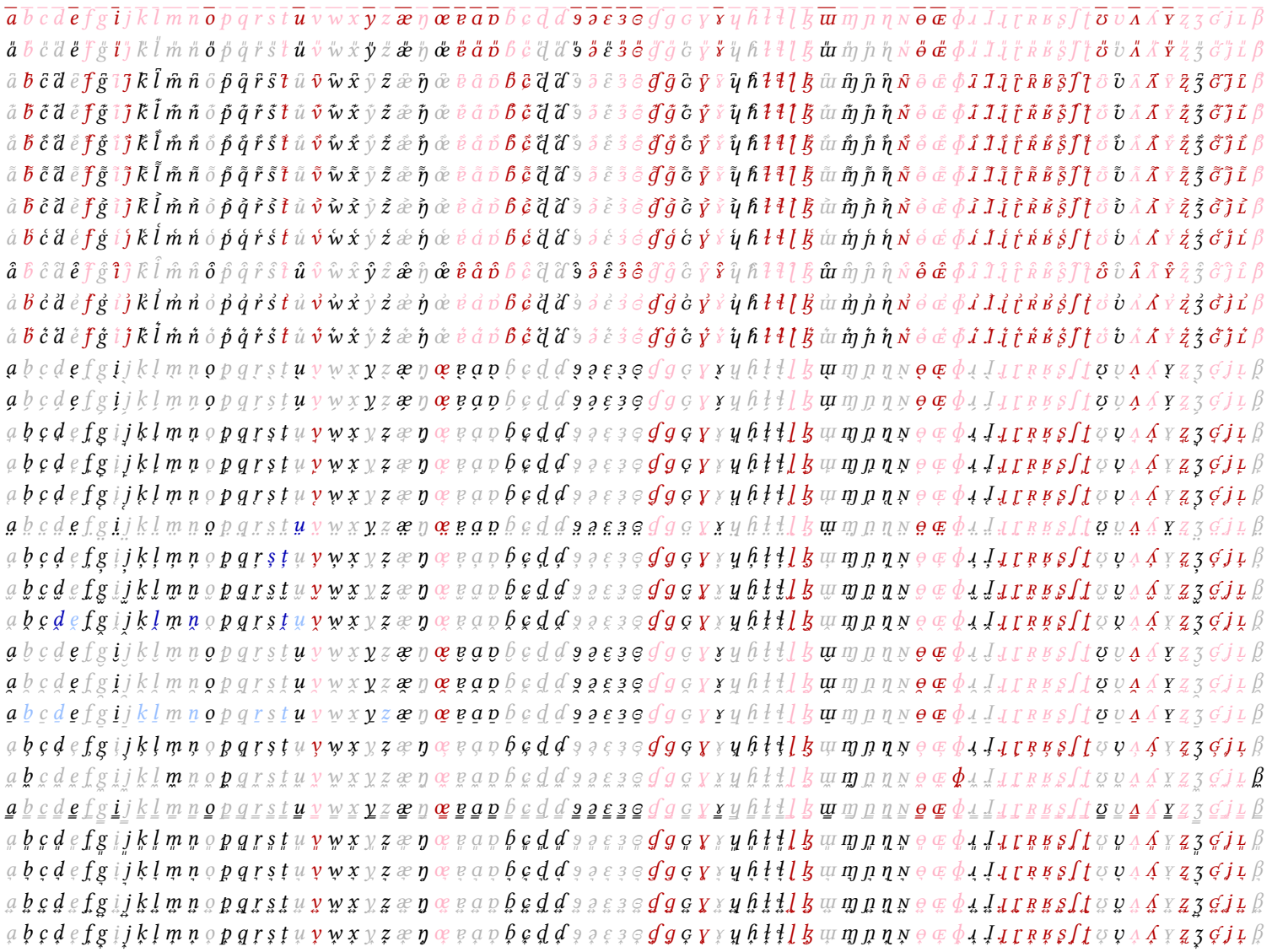
## Regular, Bases requiring anchors for rare marks, Rare combining marks



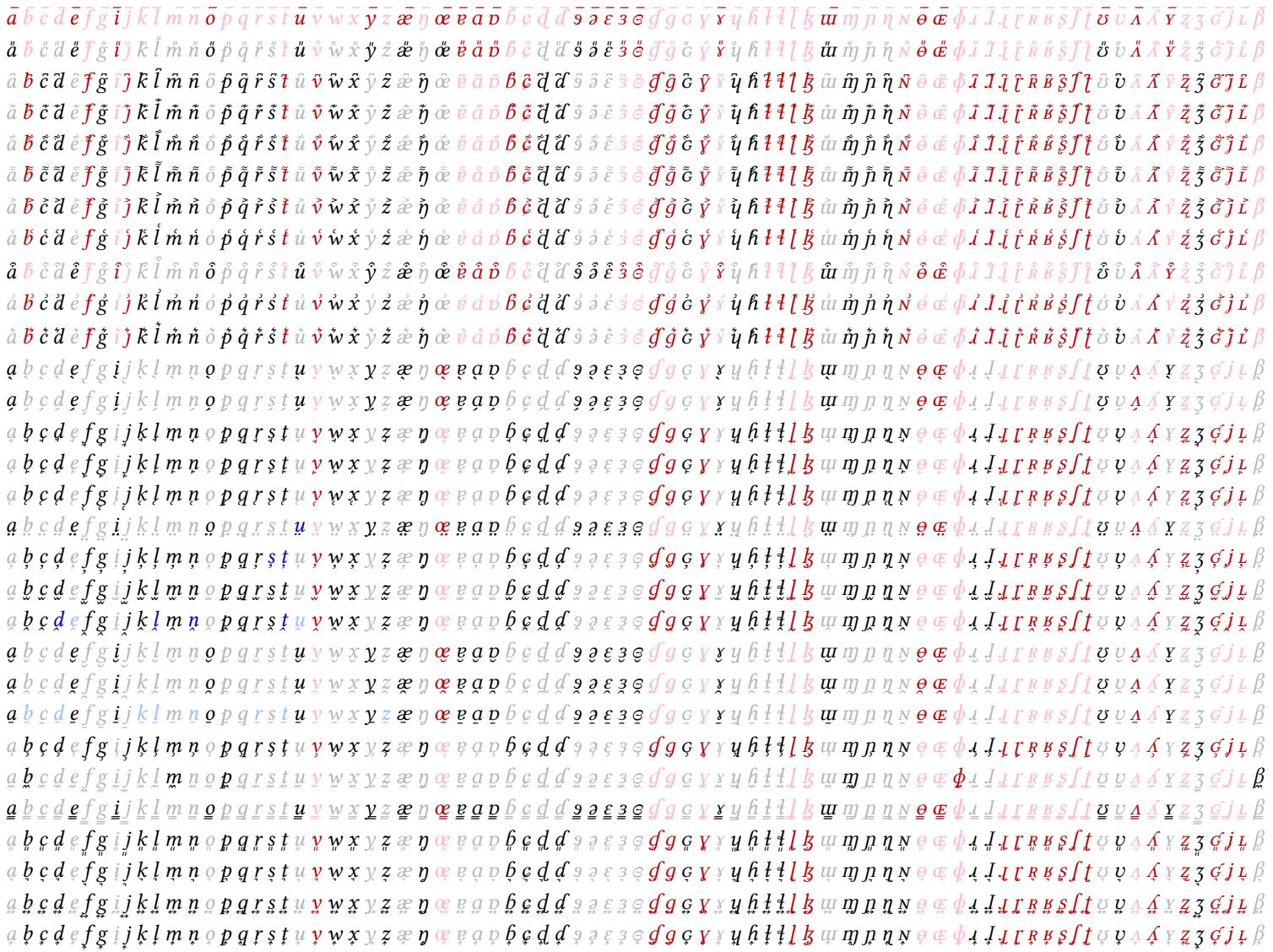
## Regular patched, Bases requiring anchors for rare marks, Rare combining marks



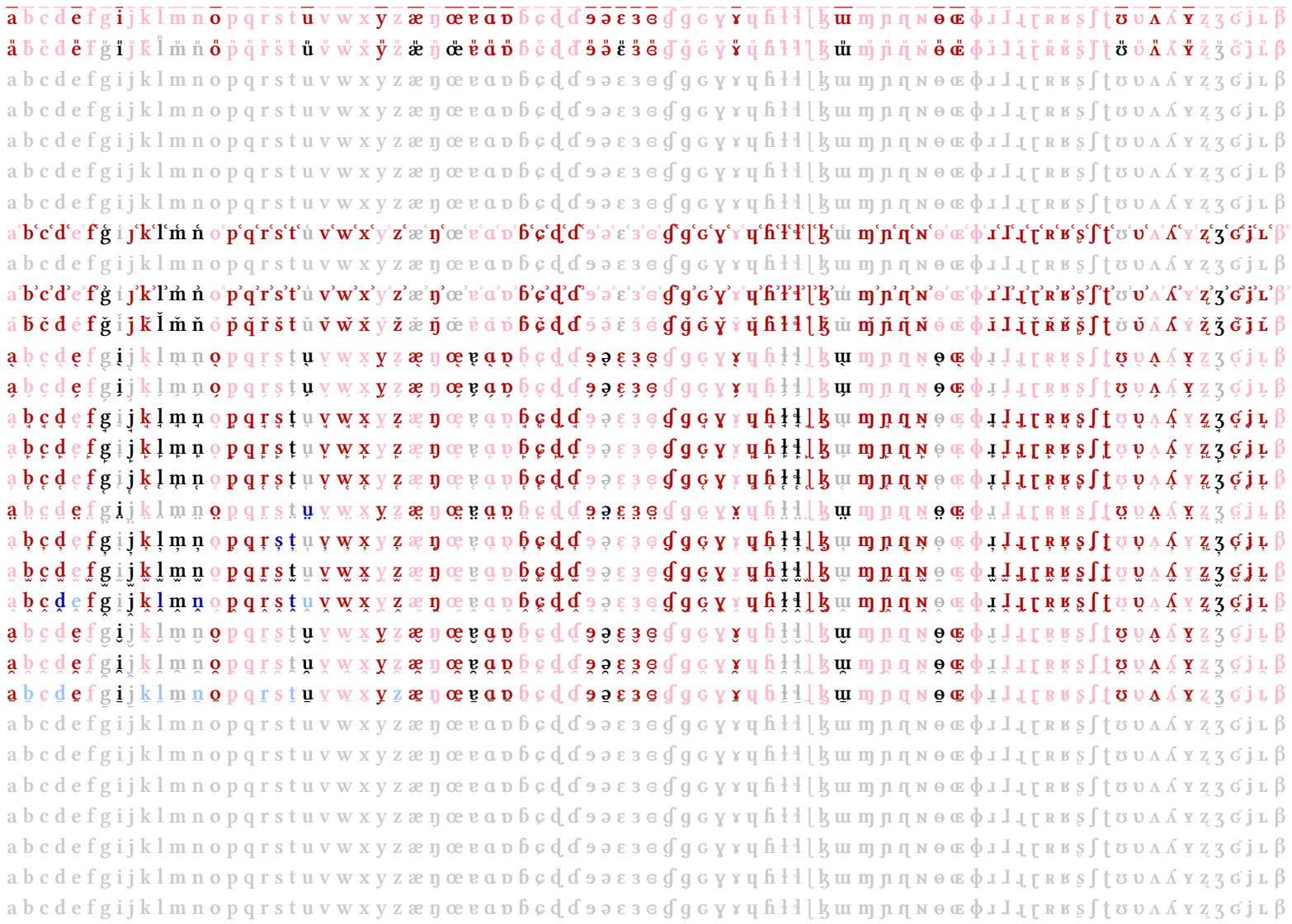
## Italic, Bases requiring anchors for rare marks, Rare combining marks



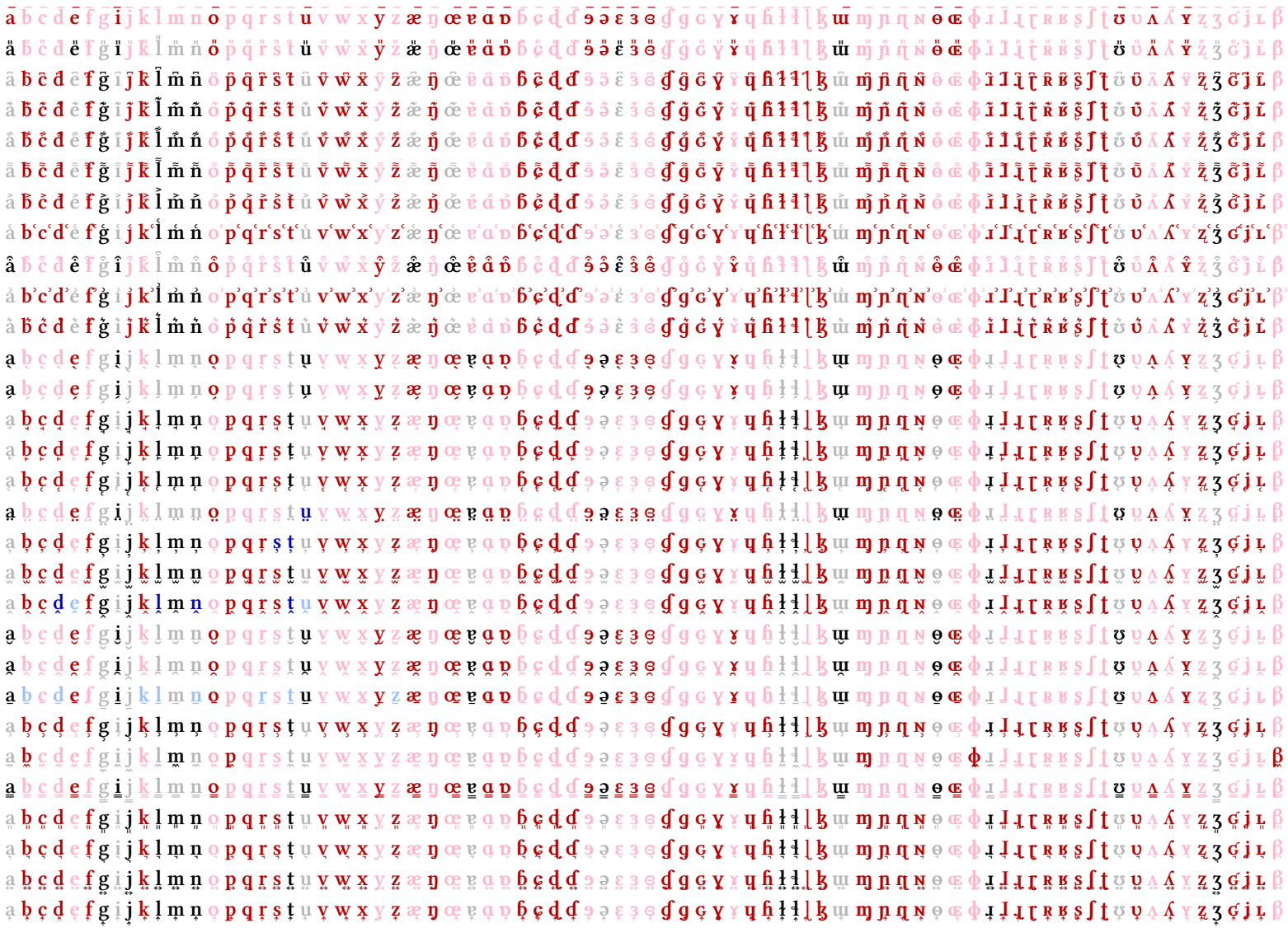
## Italic patched, Bases requiring anchors for rare marks, Rare combining marks



## Semibold, Bases requiring anchors for rare marks, Rare combining marks



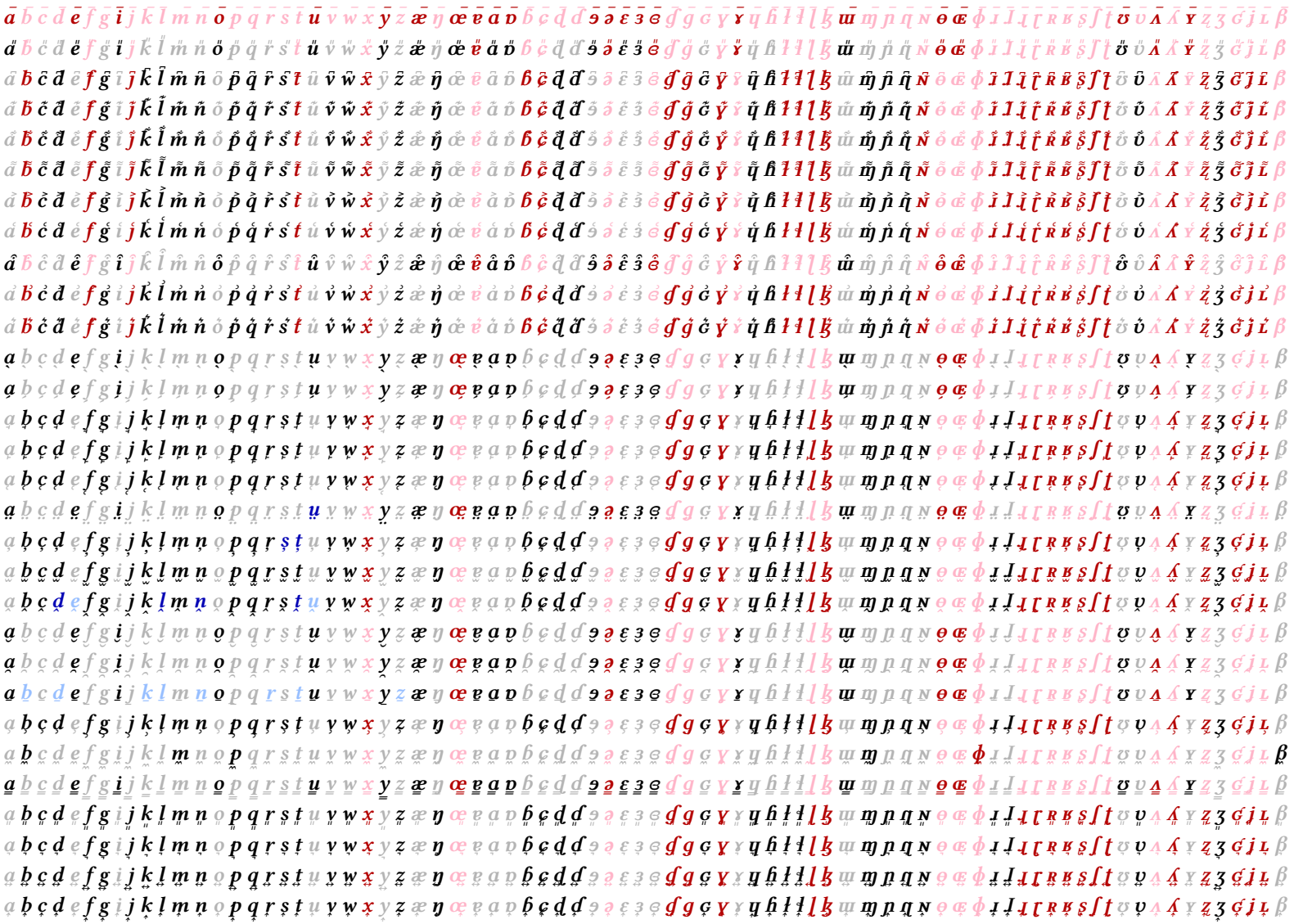
## Semibold patched, Bases requiring anchors for rare marks, Rare combining marks



## Semibold italic, Bases requiring anchors for rare marks, Rare combining marks



## Semibold italic patched, Bases requiring anchors for rare marks, Rare combining marks



## Base and mark shaping – Complete

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. The grids use more complete sets of Latin and IPA bases and above and below marks.

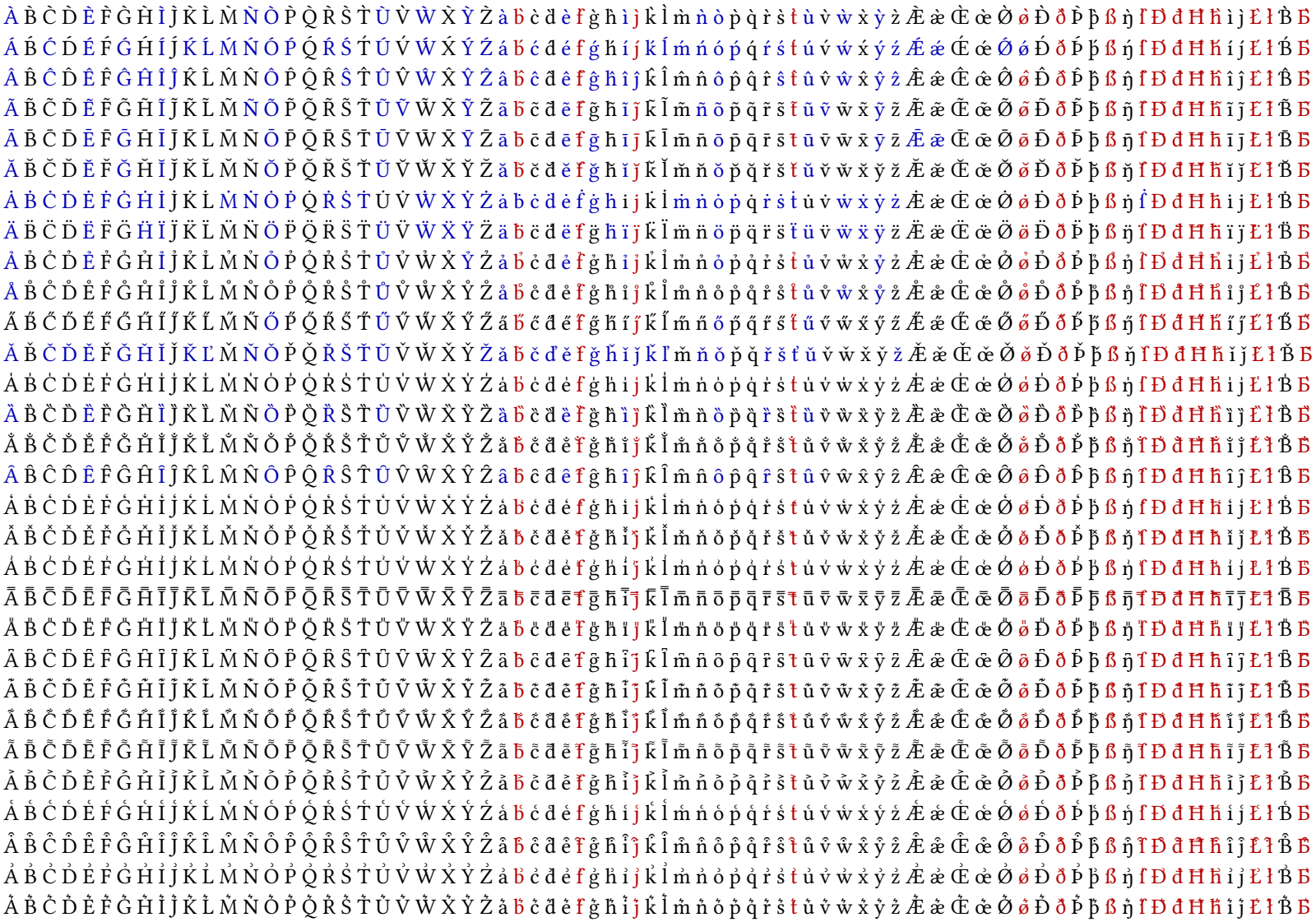
Blue indicates that XeLaTeX uses a precomposed Unicode character glyph.

Black indicates that the mark is positioned using GPOS anchors.

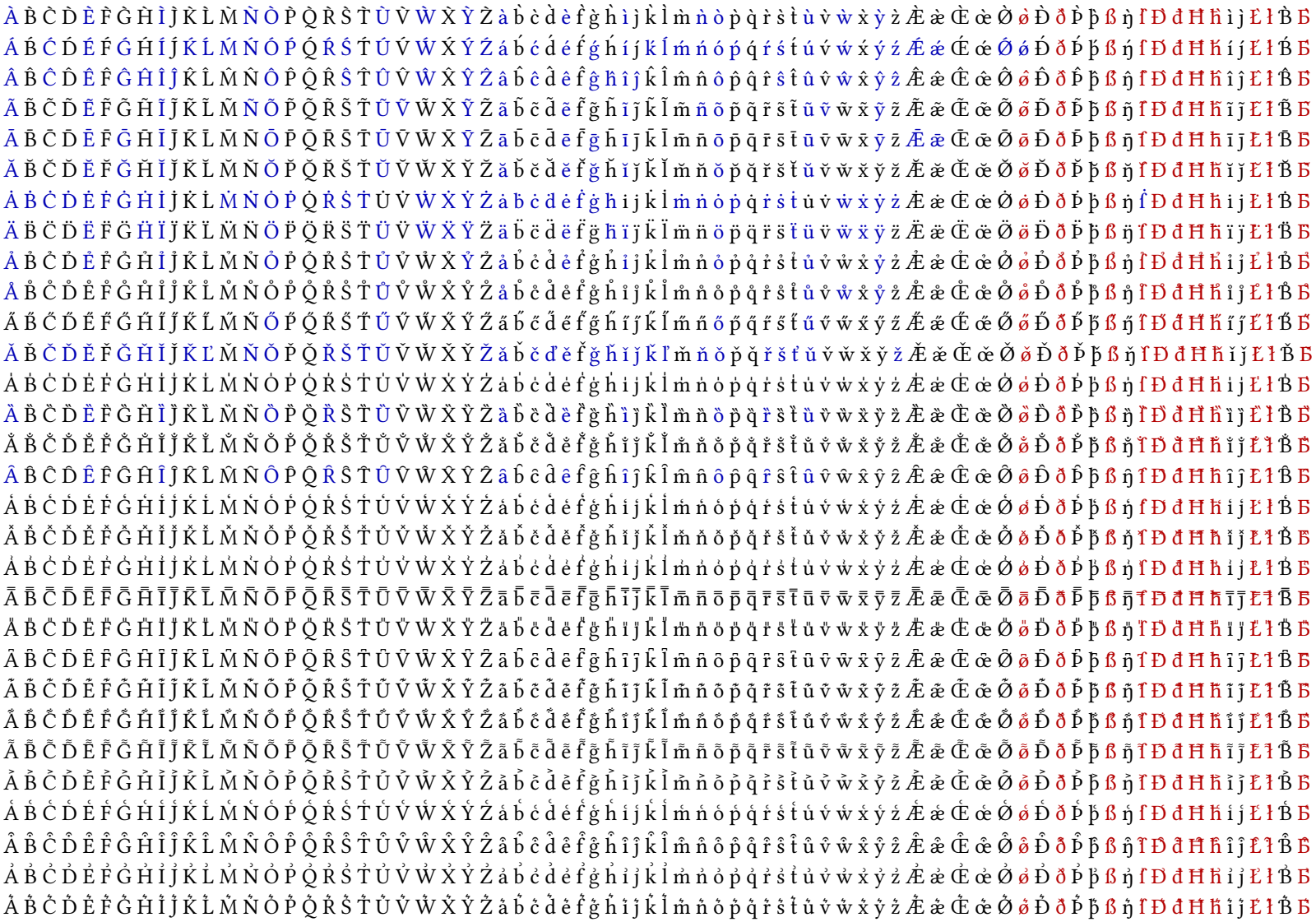
Red indicates HarfBuzz fallback shaping: no precomposed glyph and no GPOS anchors are used.

Light gray indicates that either the base or the mark glyph is missing from the font. In this case only the base glyph is shown.

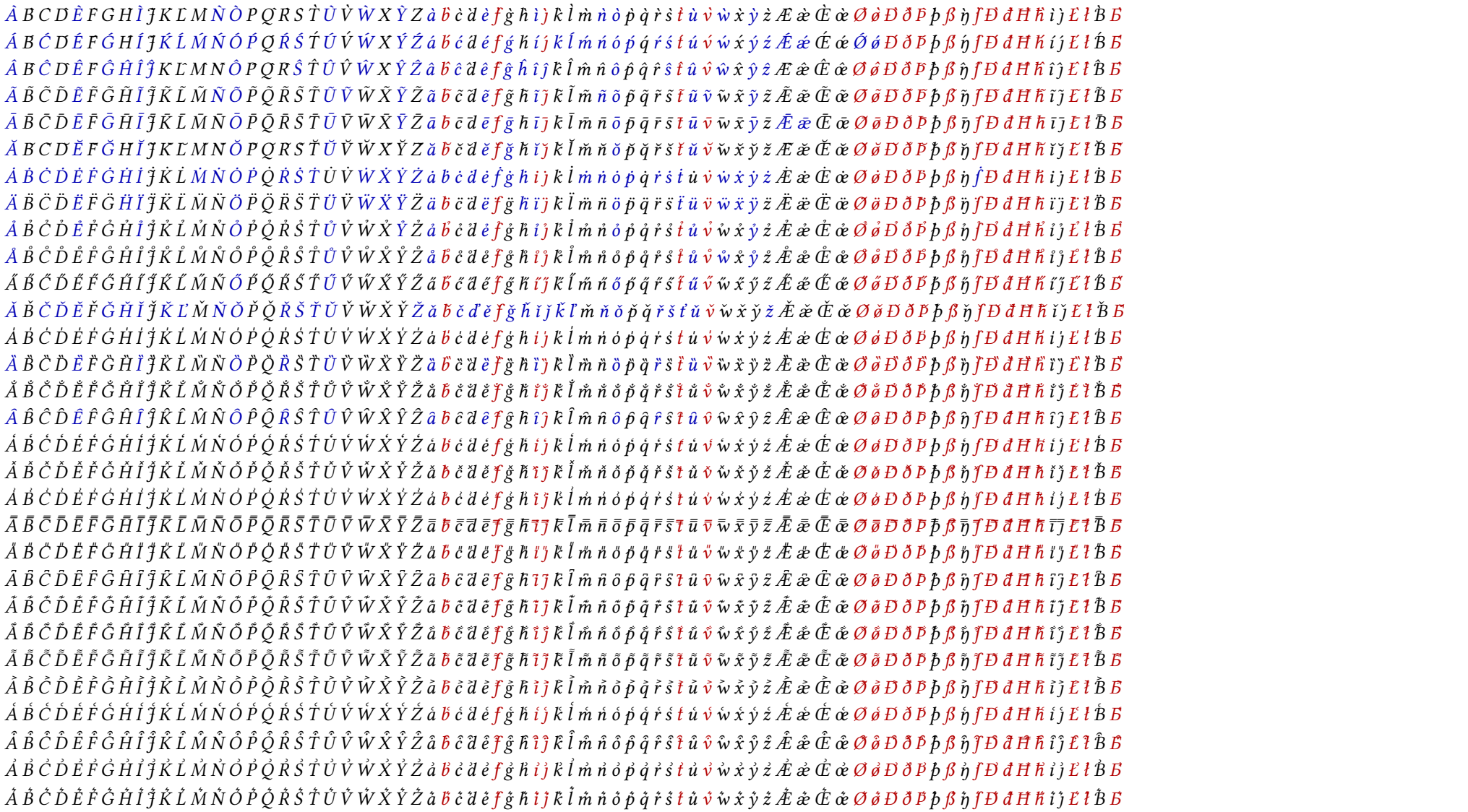
## Regular, Latin, Above



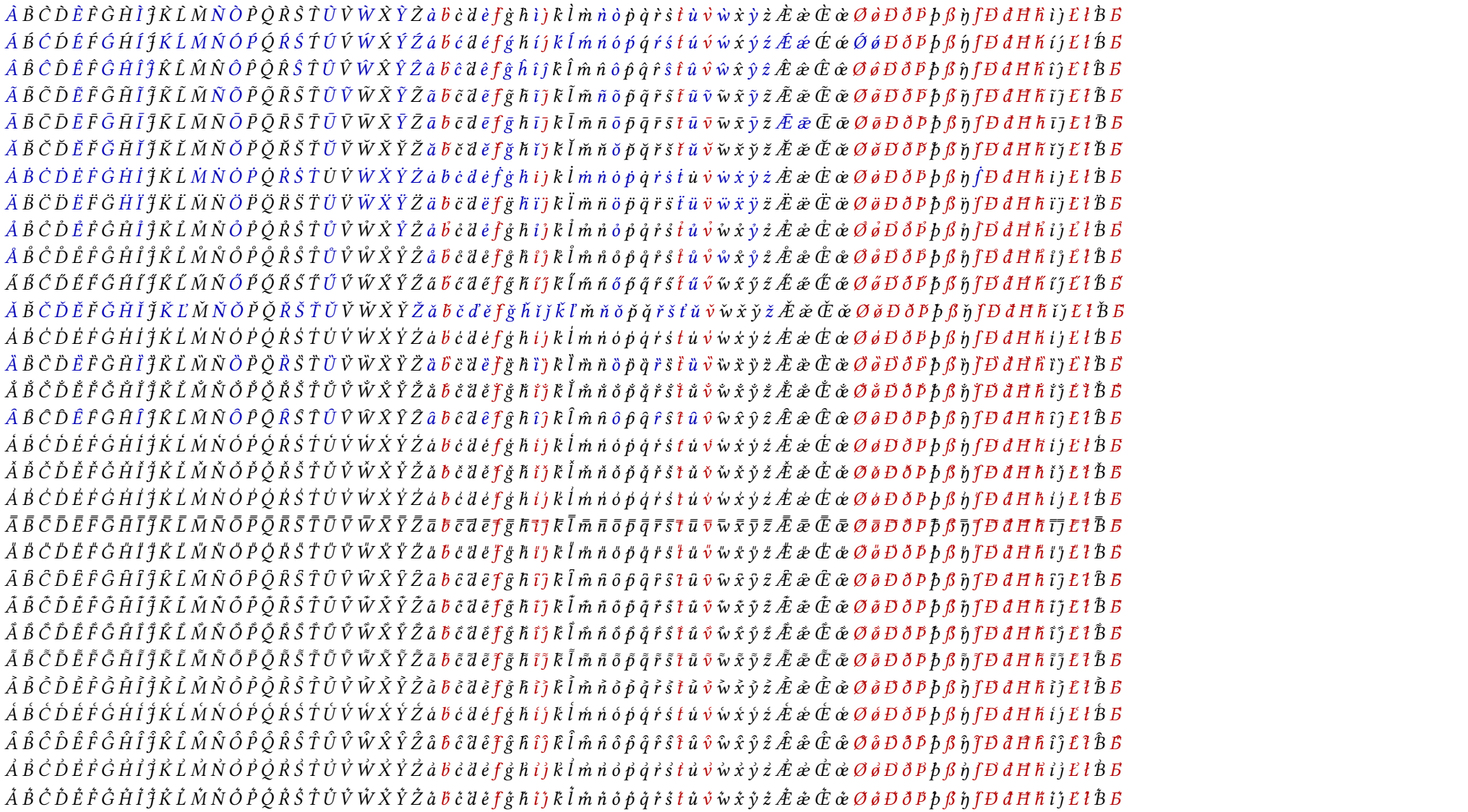
## Regular patched, Latin, Above



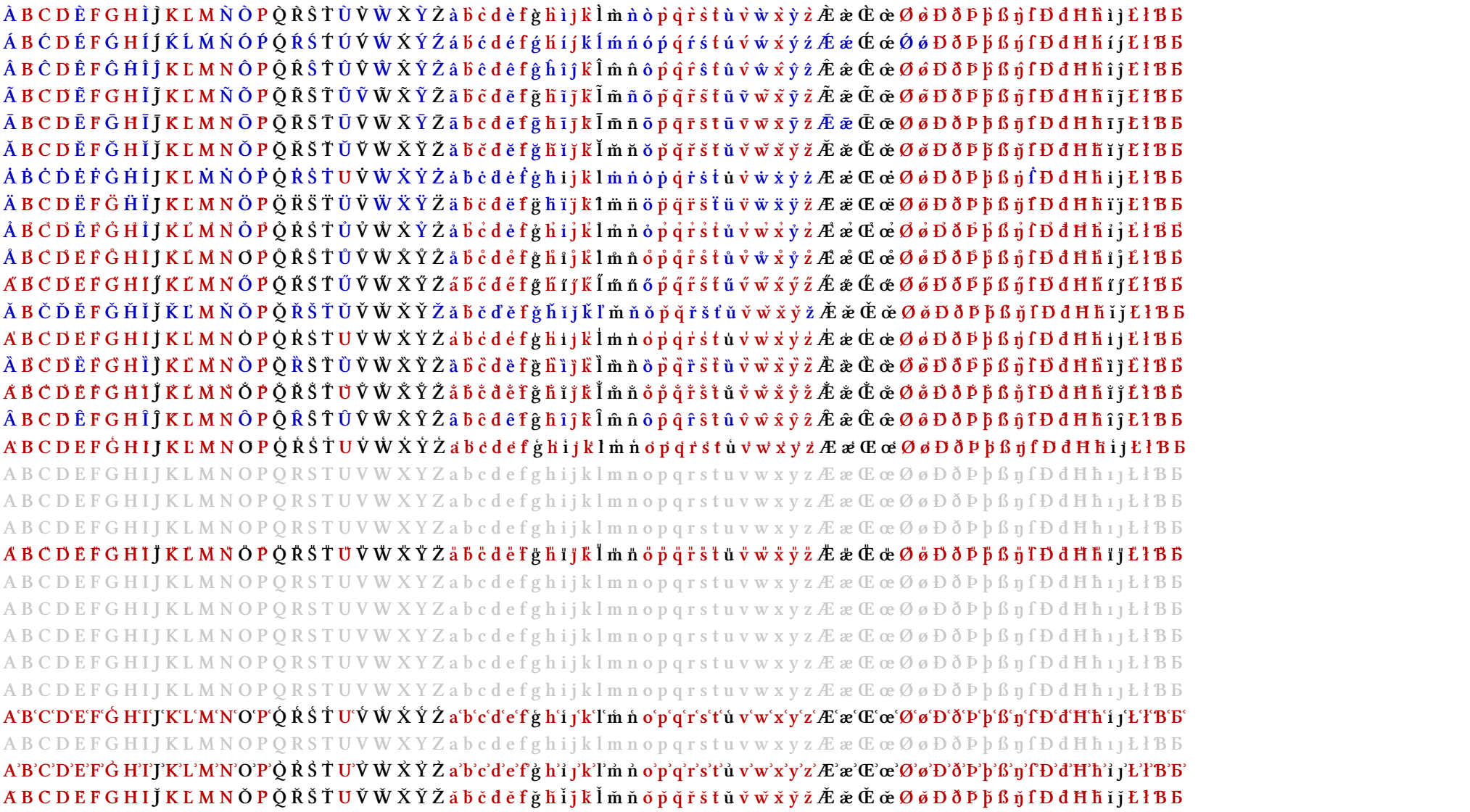
## Italic, Latin, Above



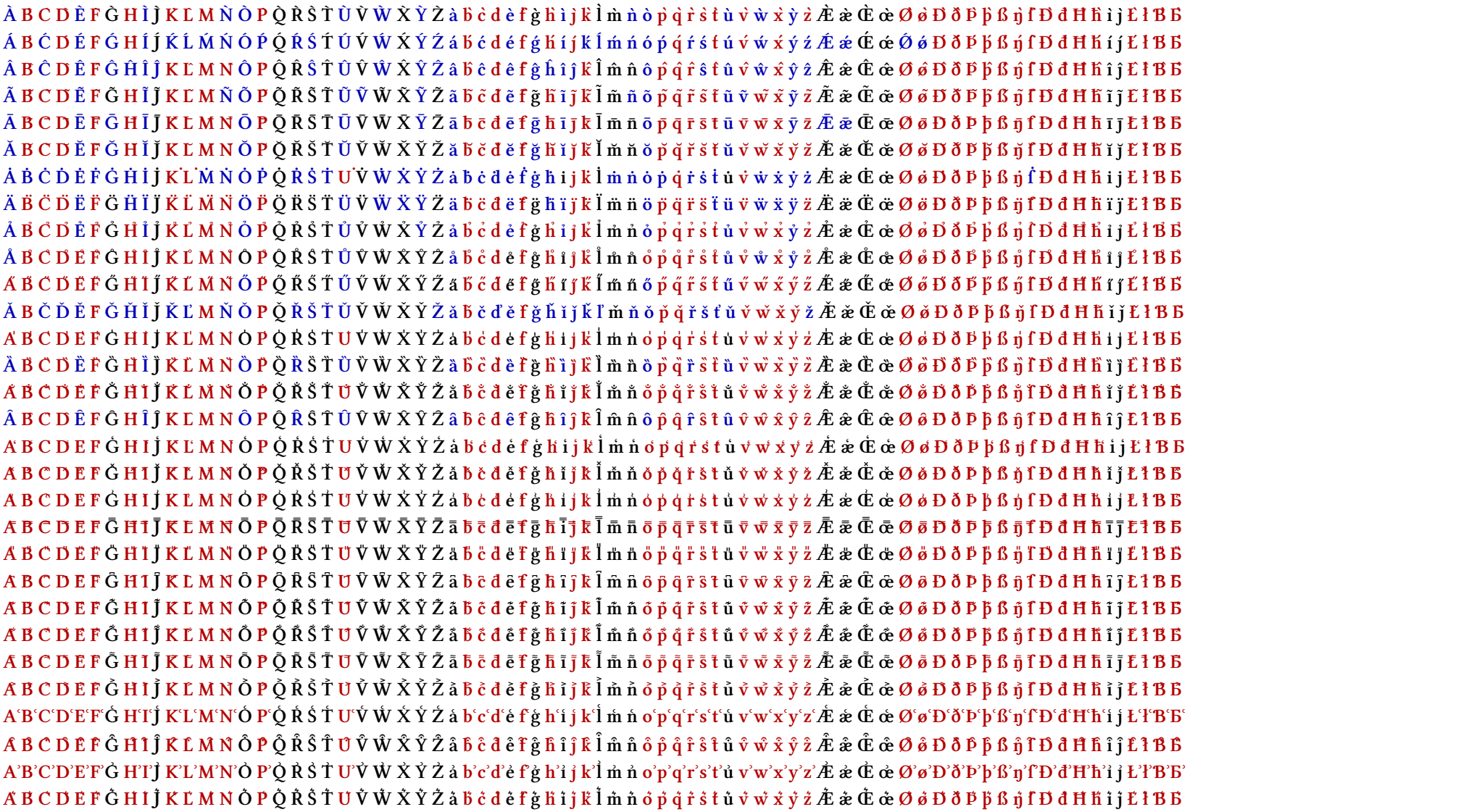
## Italic patched, Latin, Above



## Semibold, Latin, Above



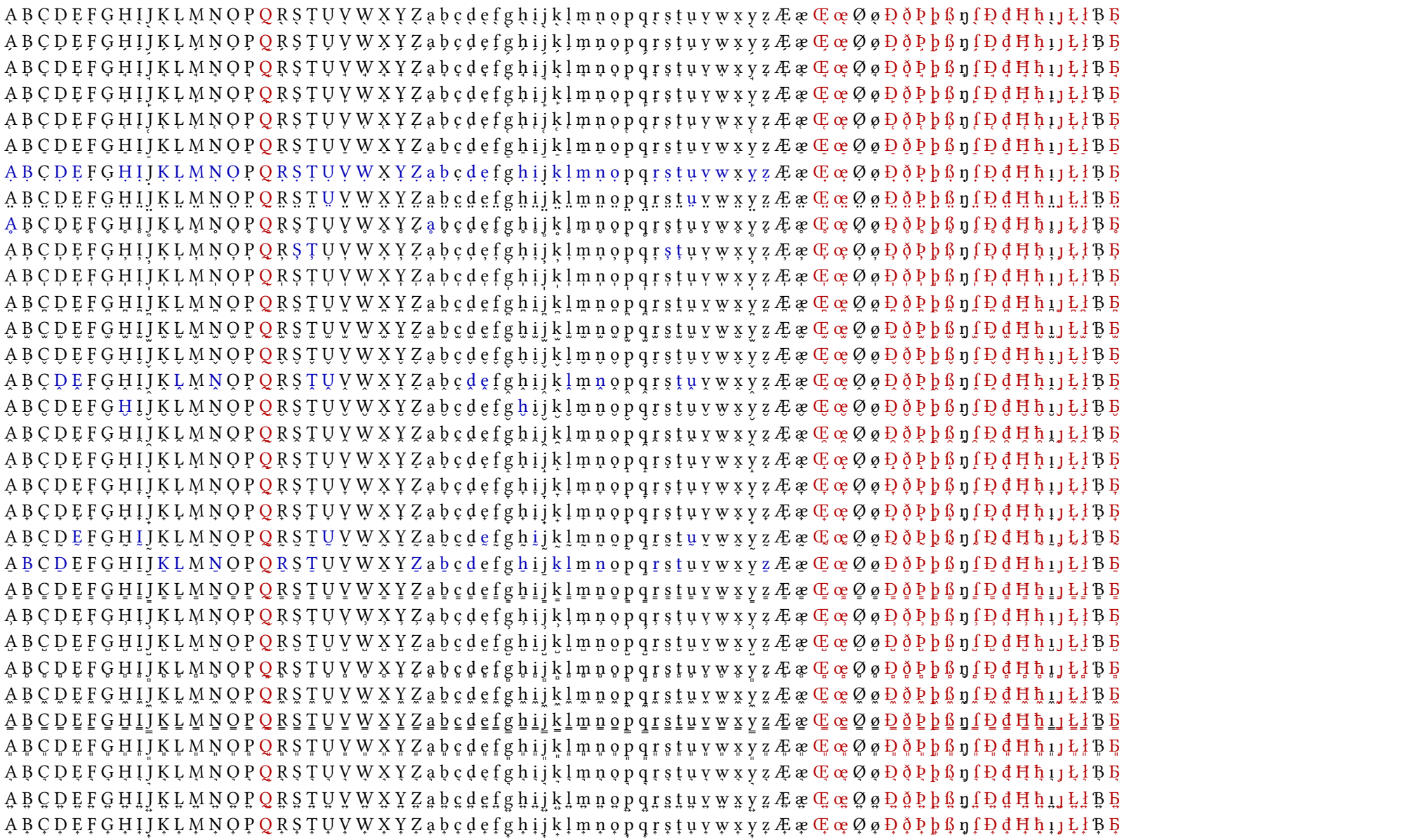
## Semibold patched, Latin, Above



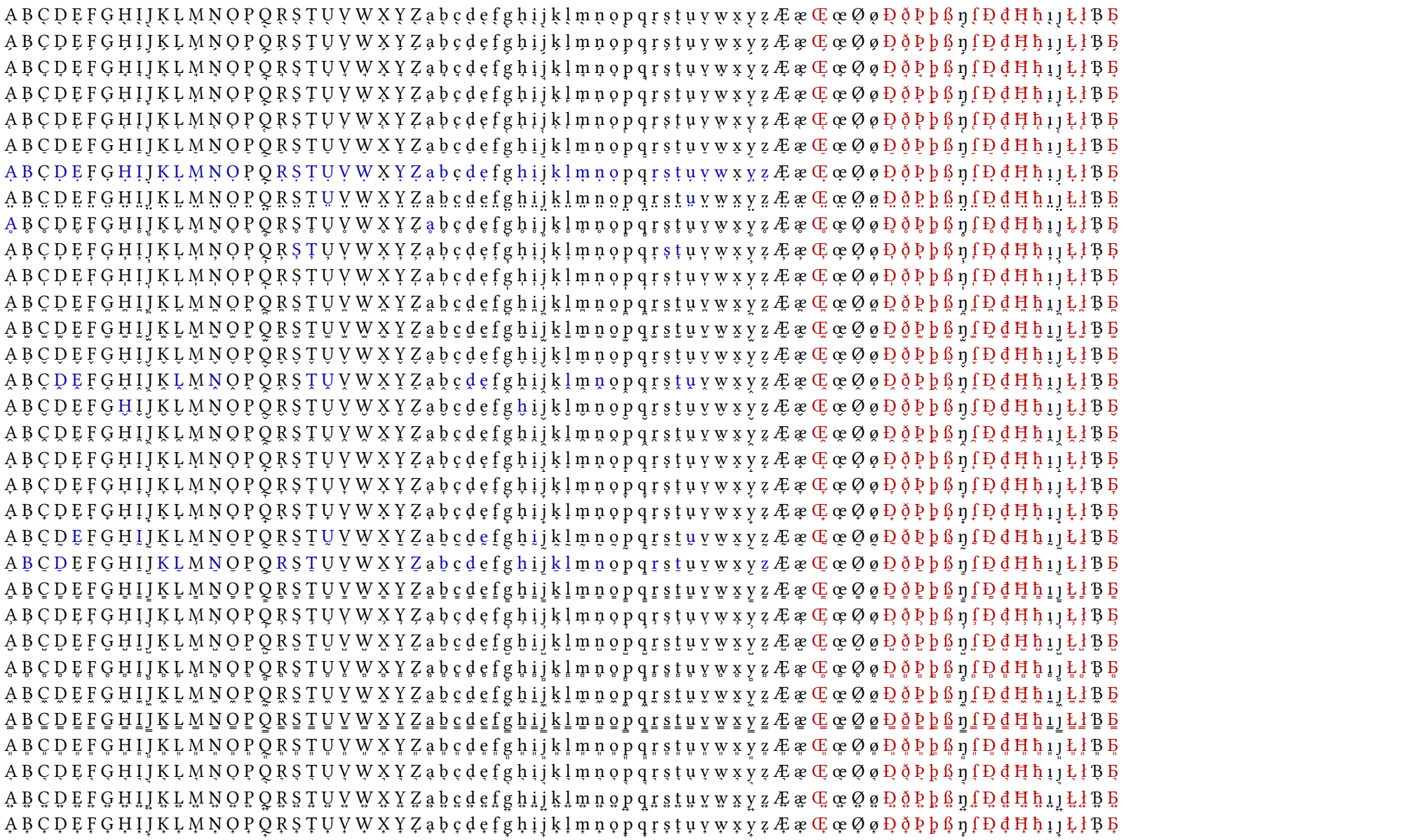
## Semibold italic, Latin, Above

## Semibold italic patched, Latin, Above

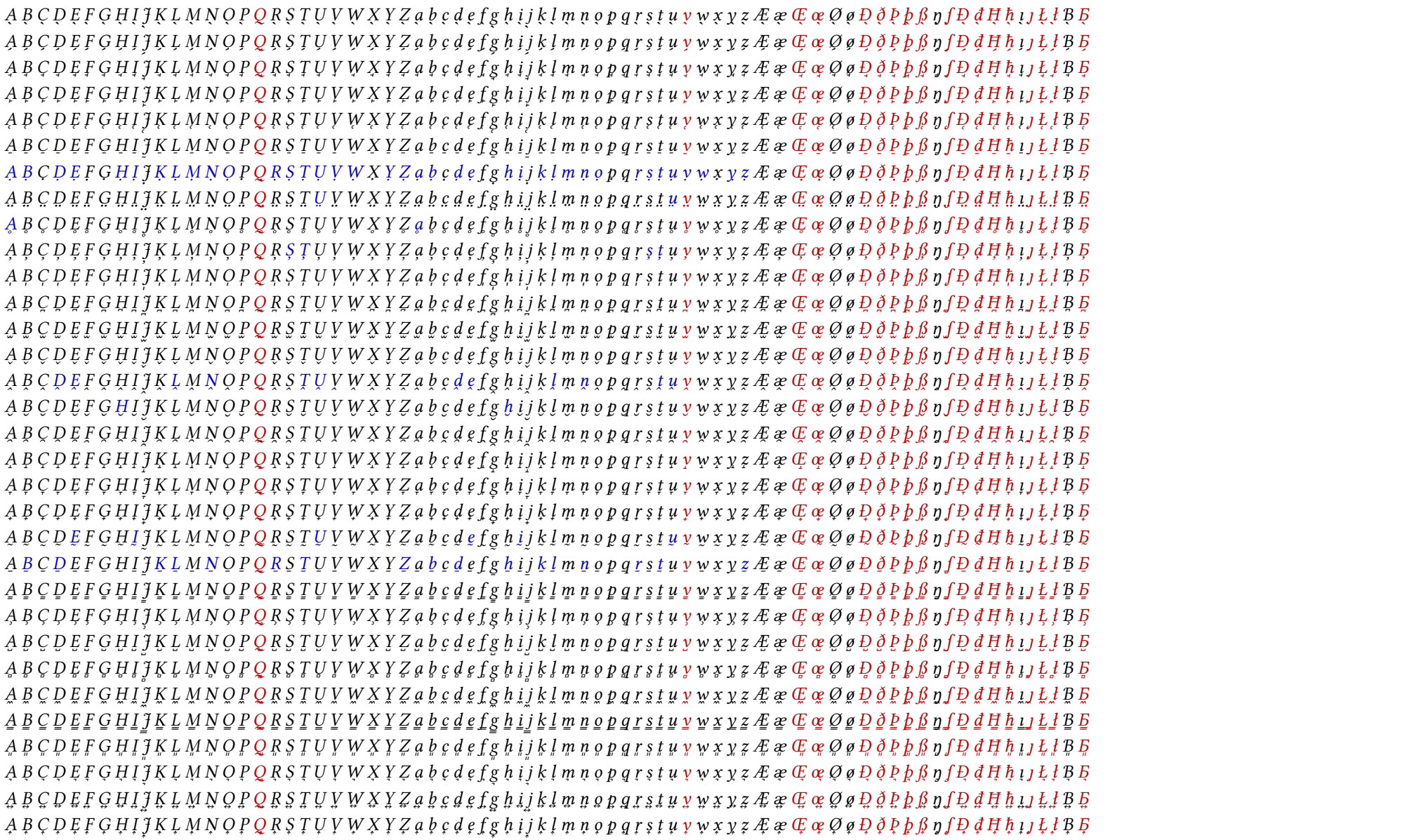
## Regular, Latin, Below



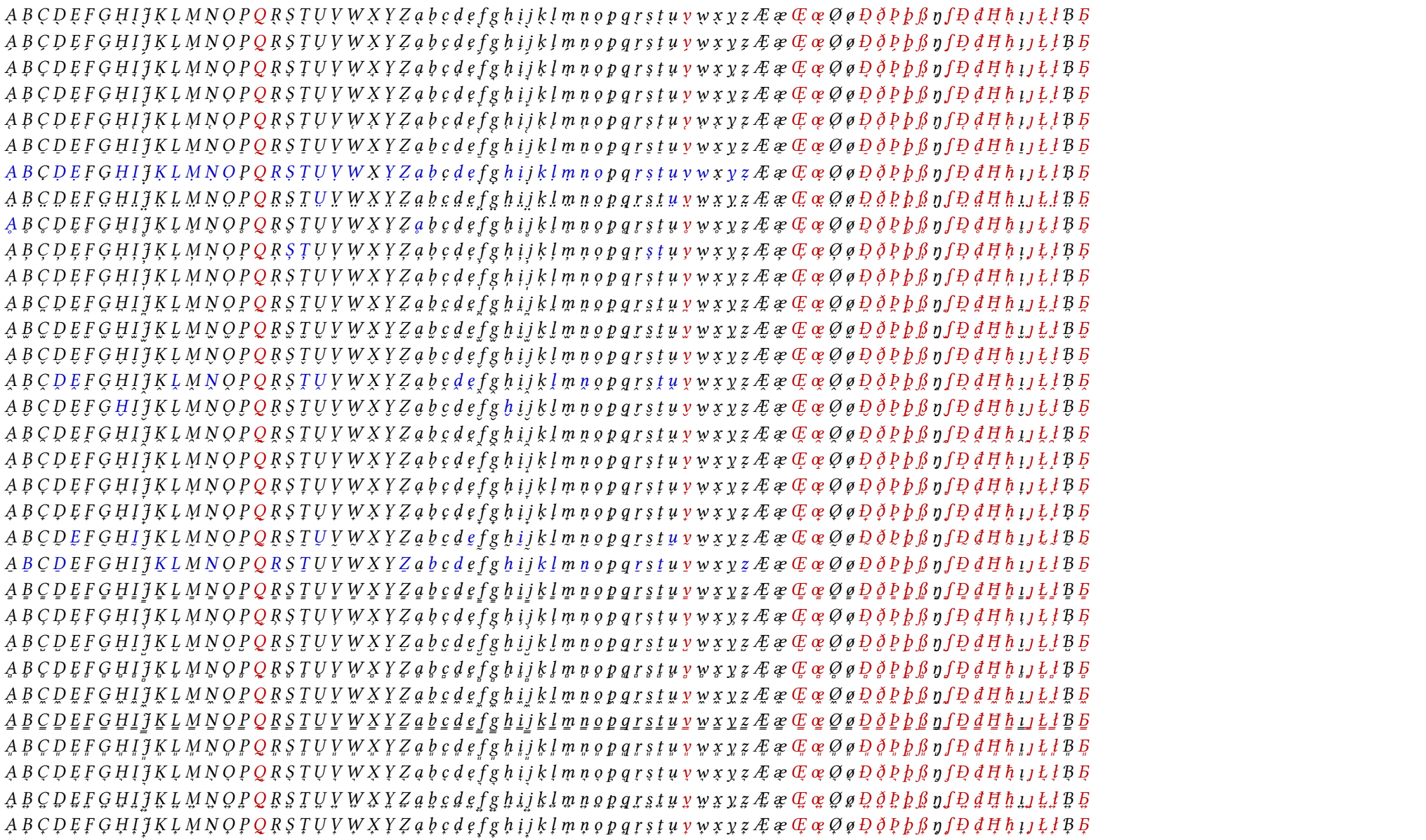
## Regular patched, Latin, Below



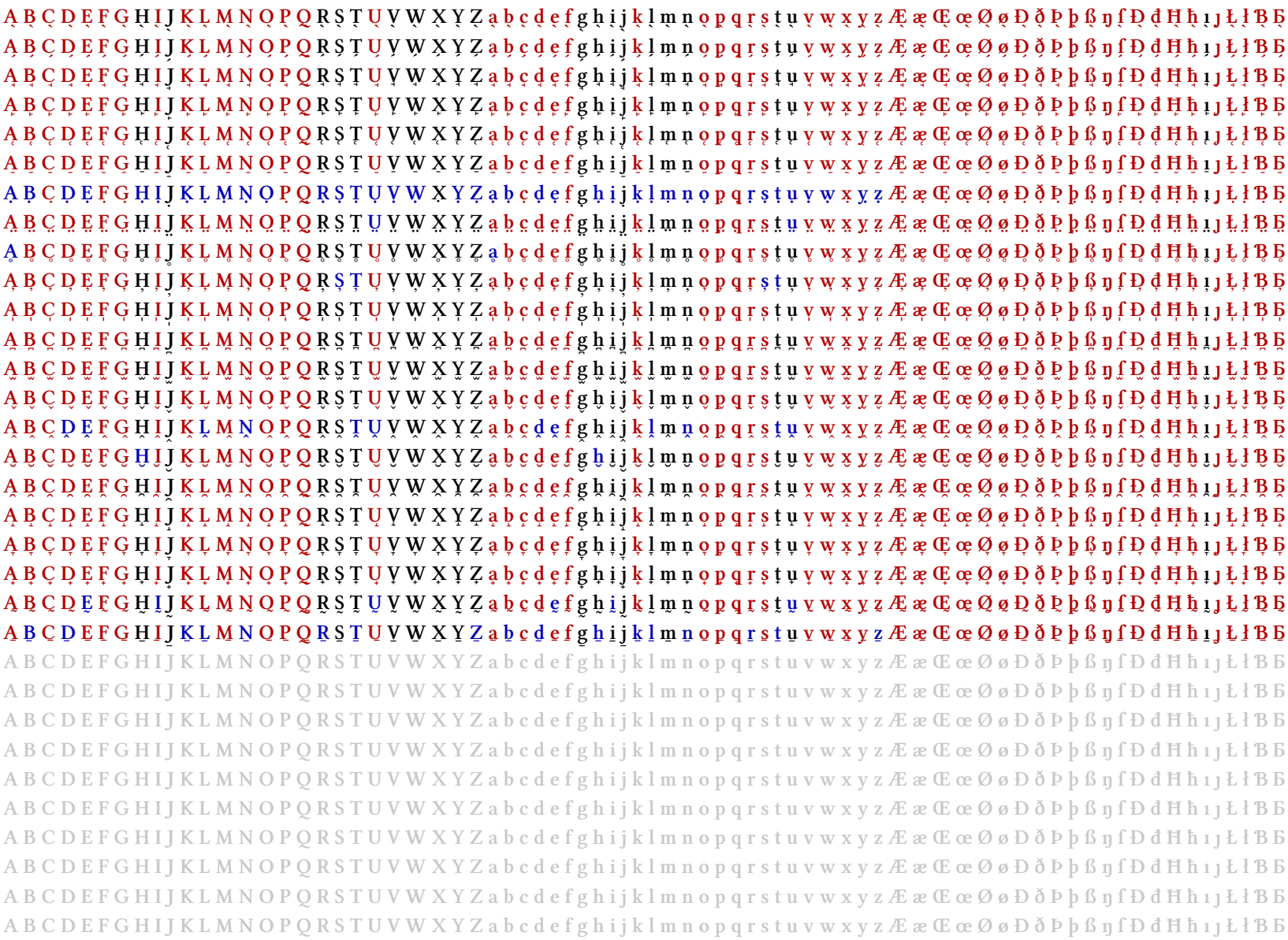
## Italic, Latin, Below



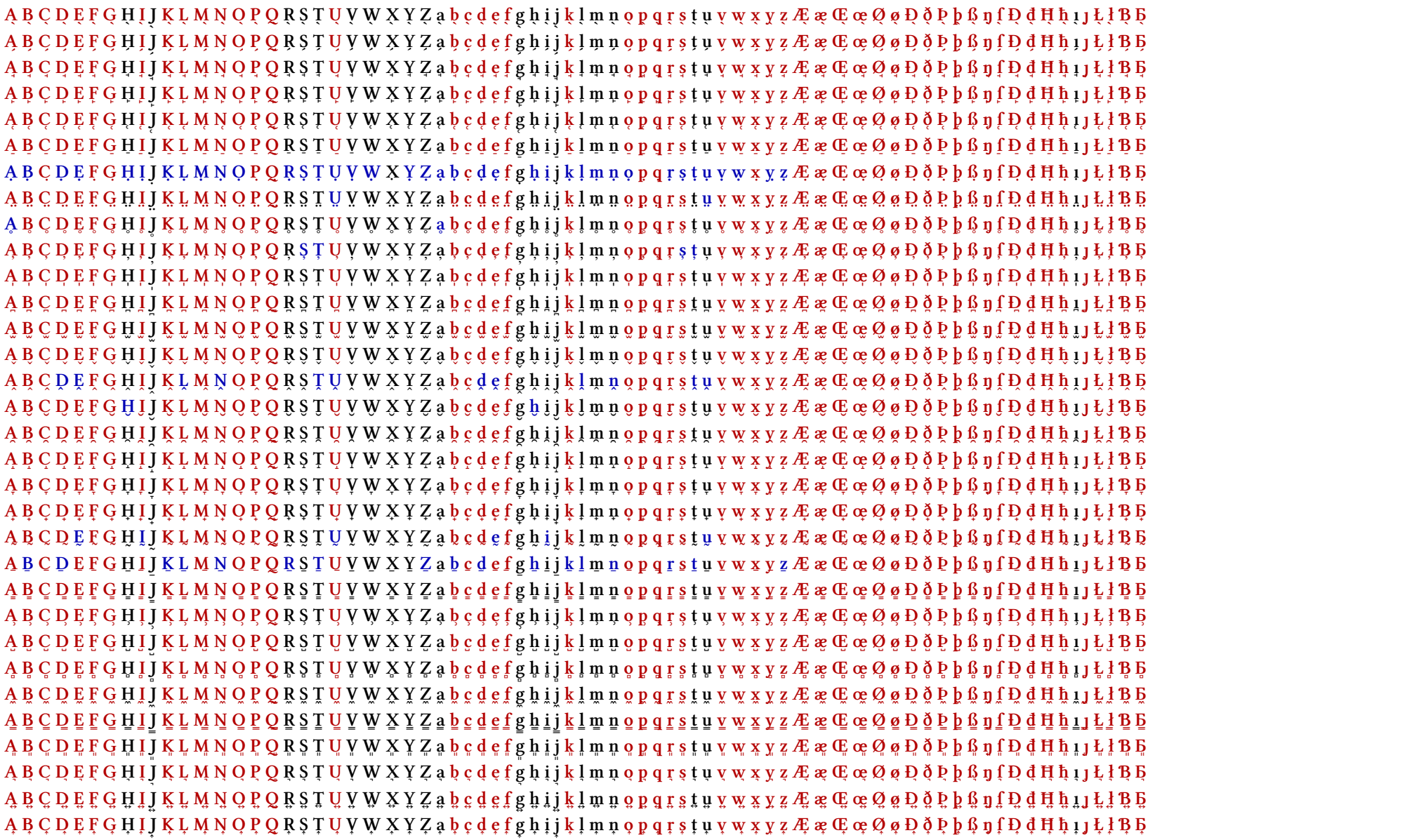
## Italic patched, Latin, Below



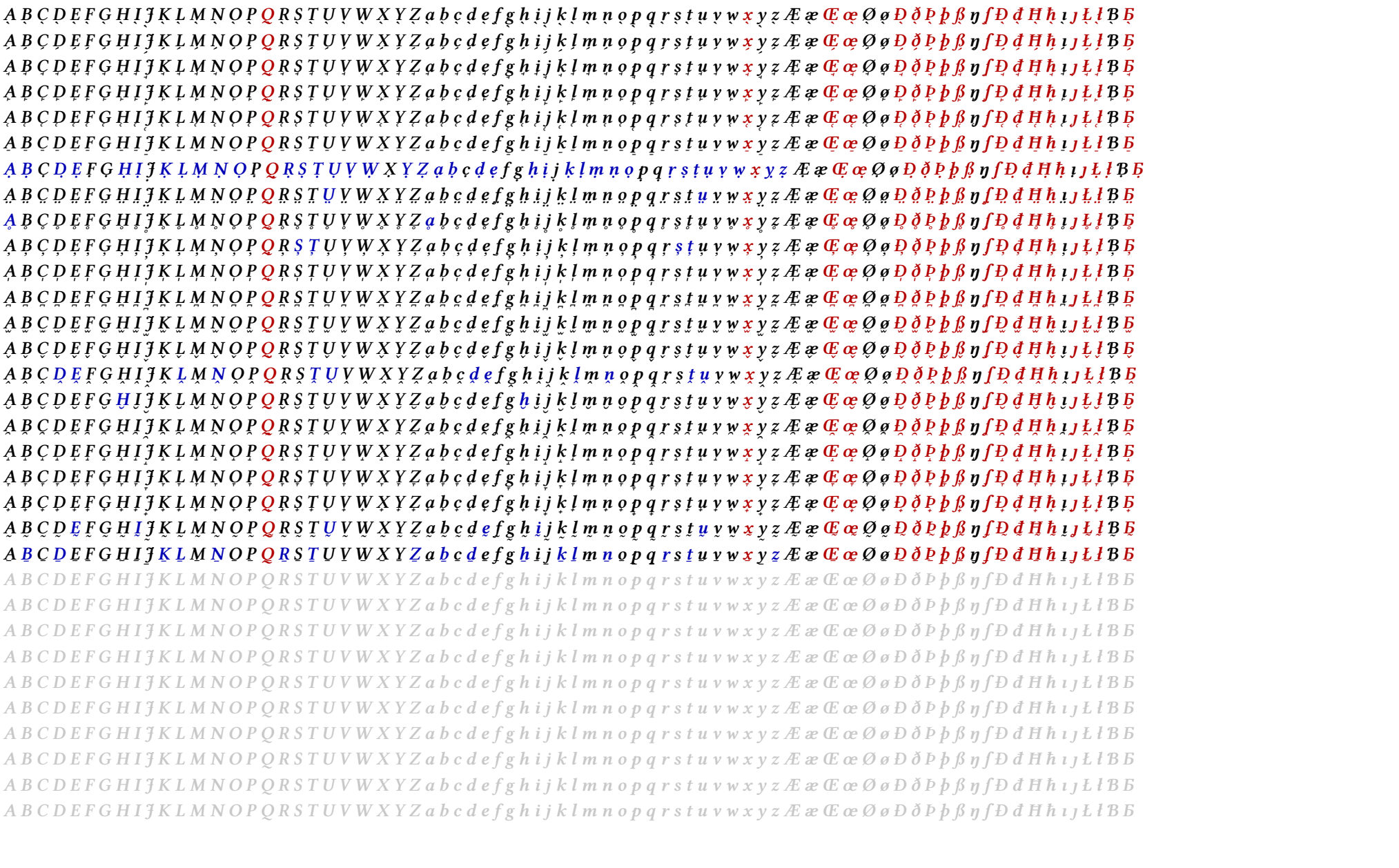
## Semibold, Latin, Below



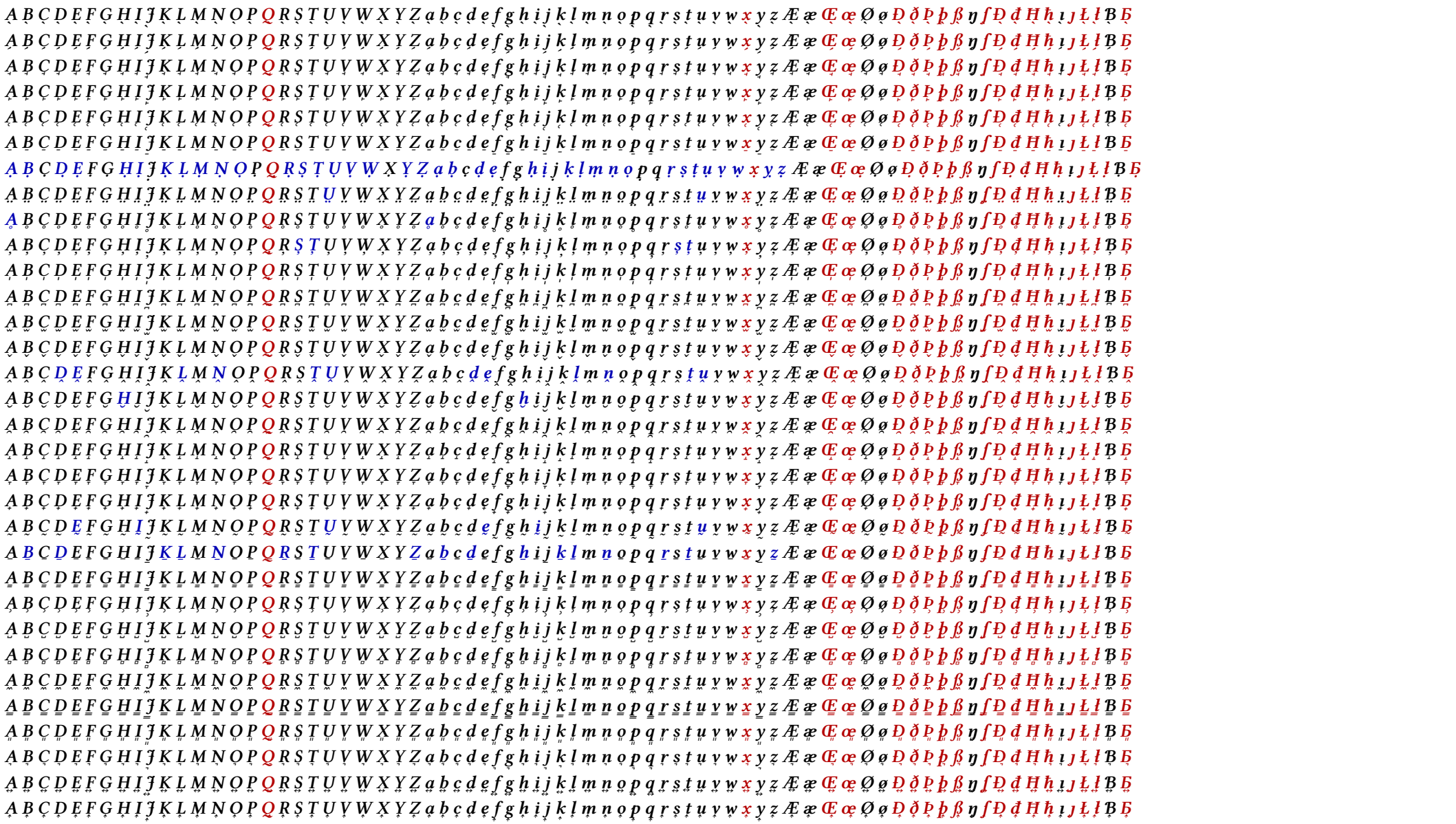
## Semibold patched, Latin, Below



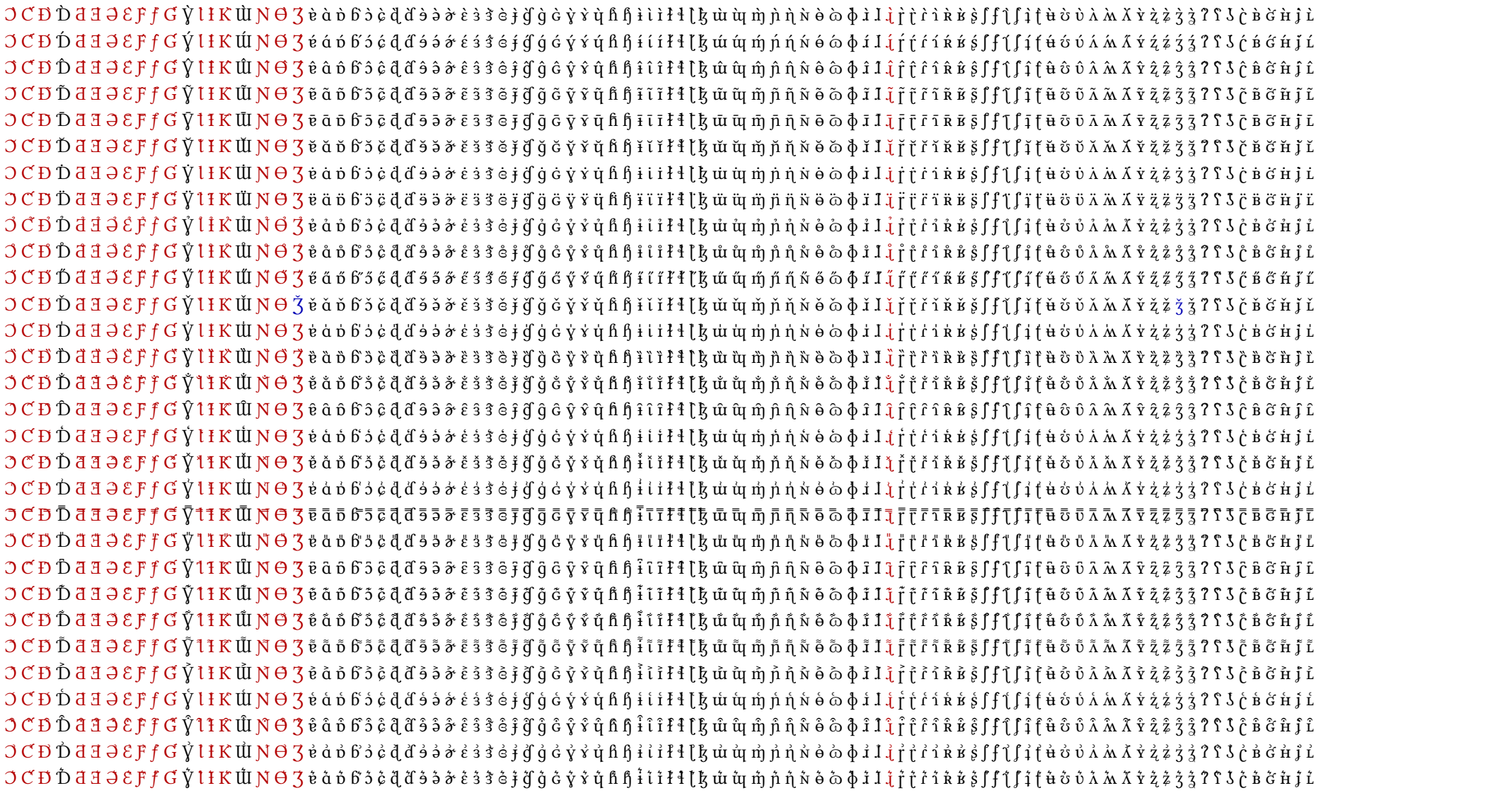
## Semibold italic, Latin, Below



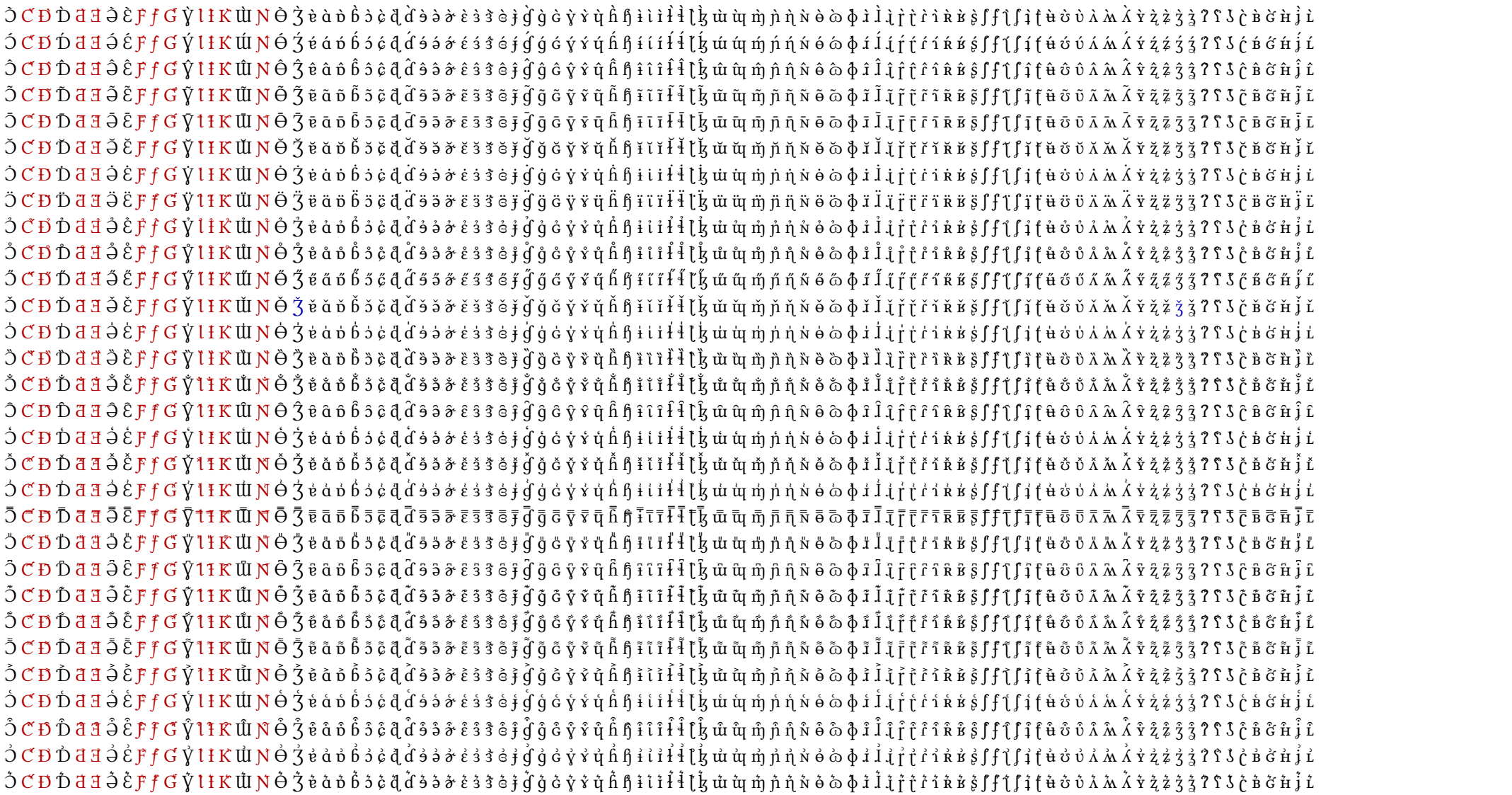
## Semibold italic patched, Latin, Below



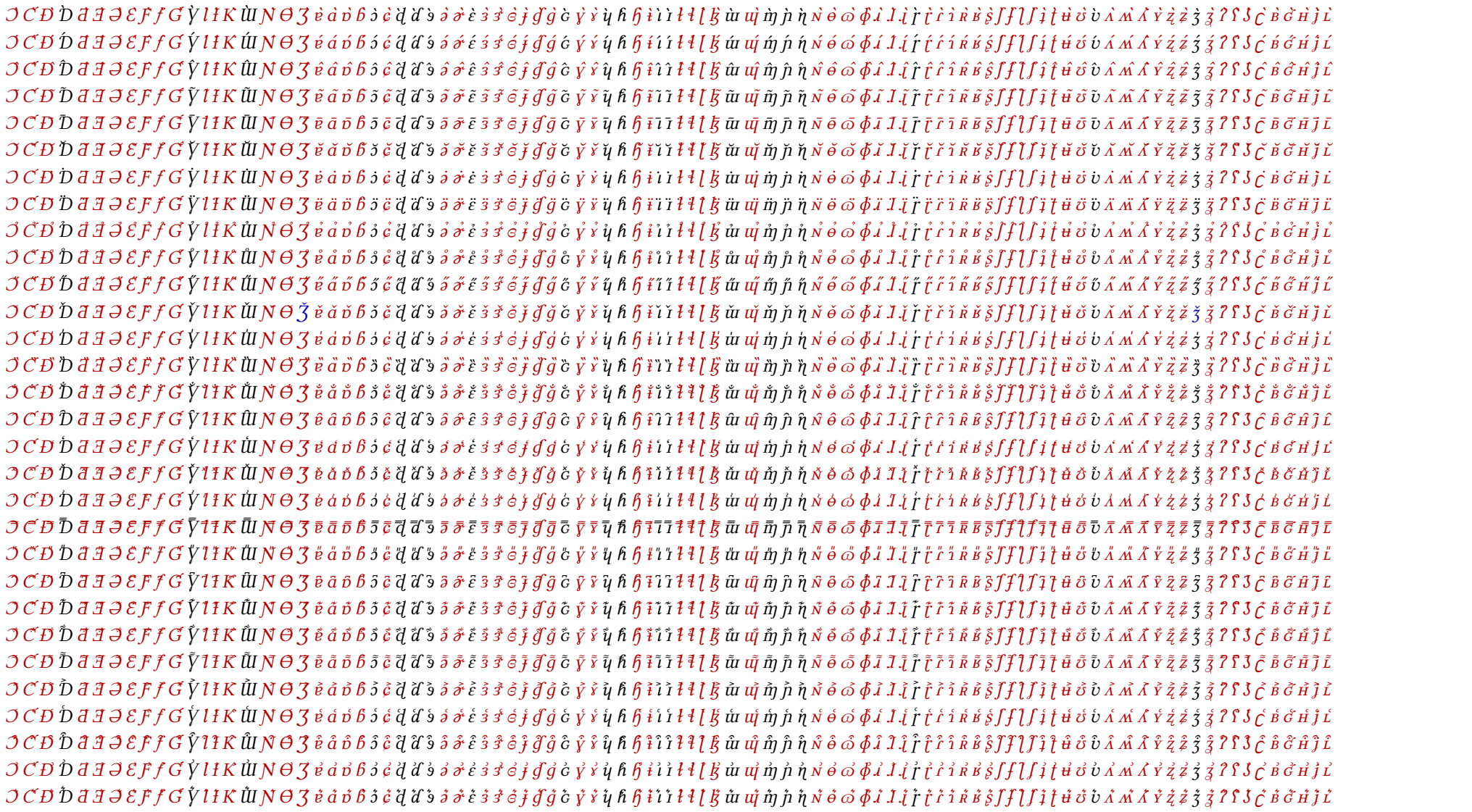
## Regular, IPA, Above



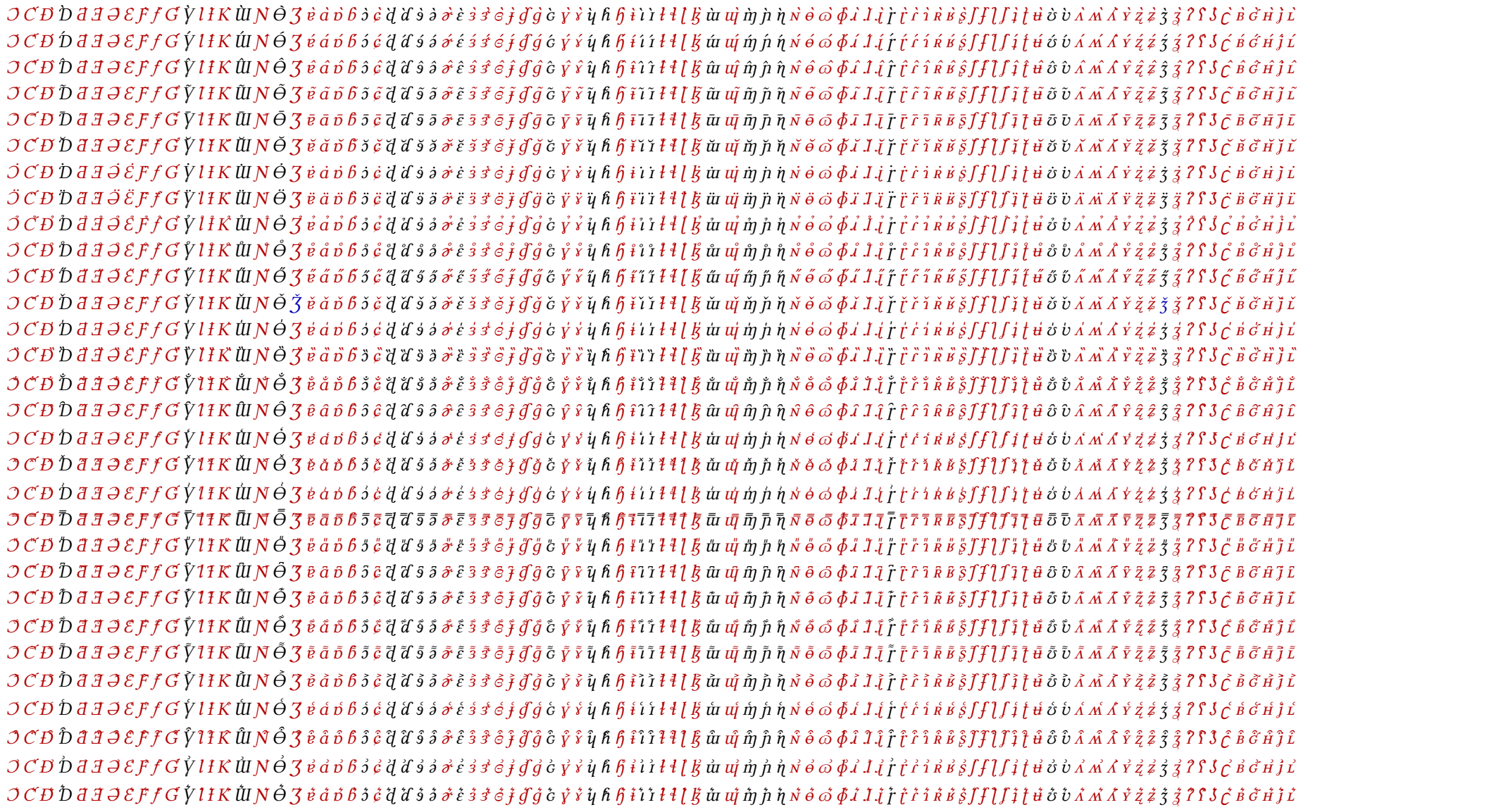
## Regular patched, IPA, Above



## Italic, IPA, Above



## Italic patched, IPA, Above



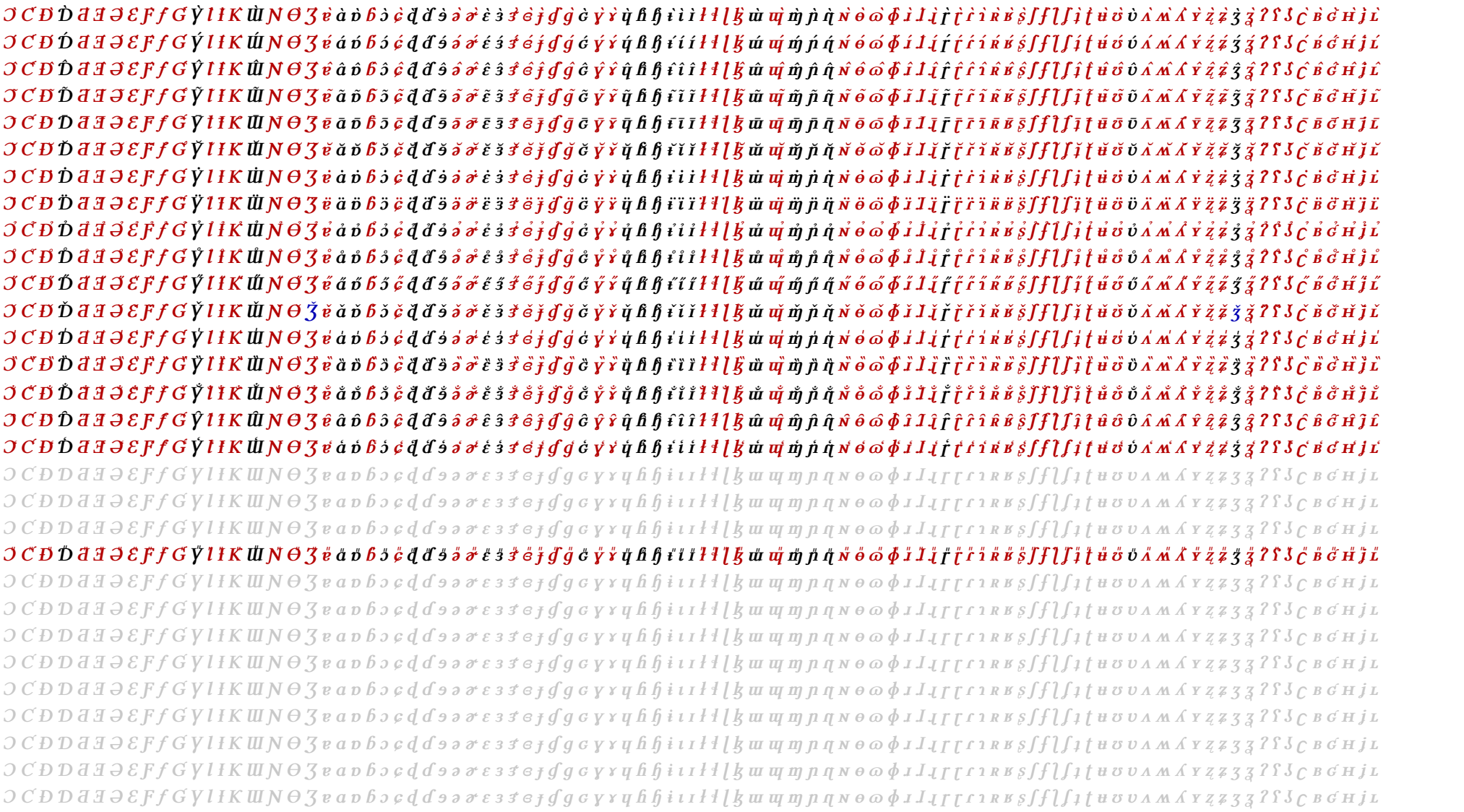
## Semibold, IPA, Above



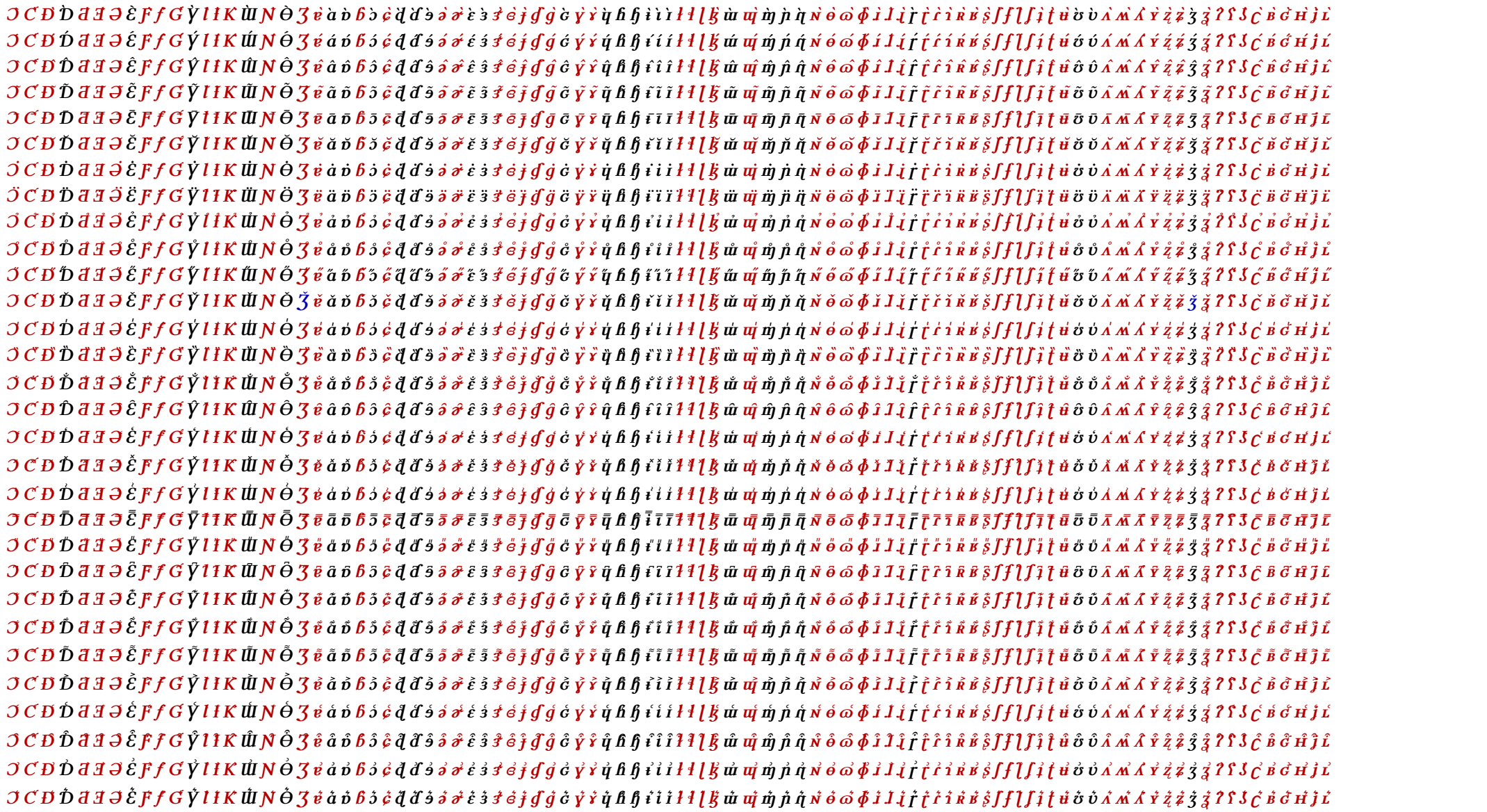
## Semibold patched, IPA, Above



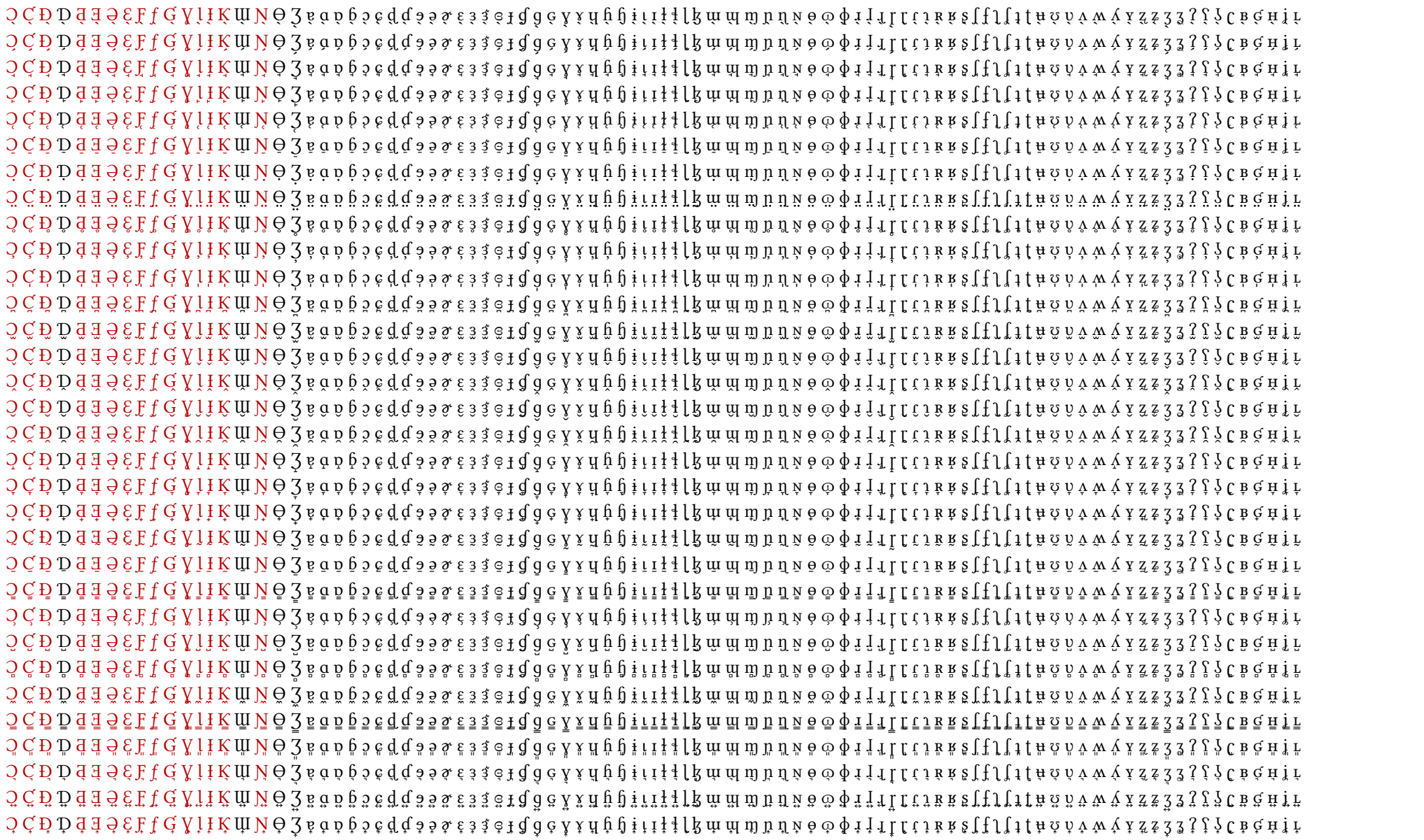
## Semibold italic, IPA, Above



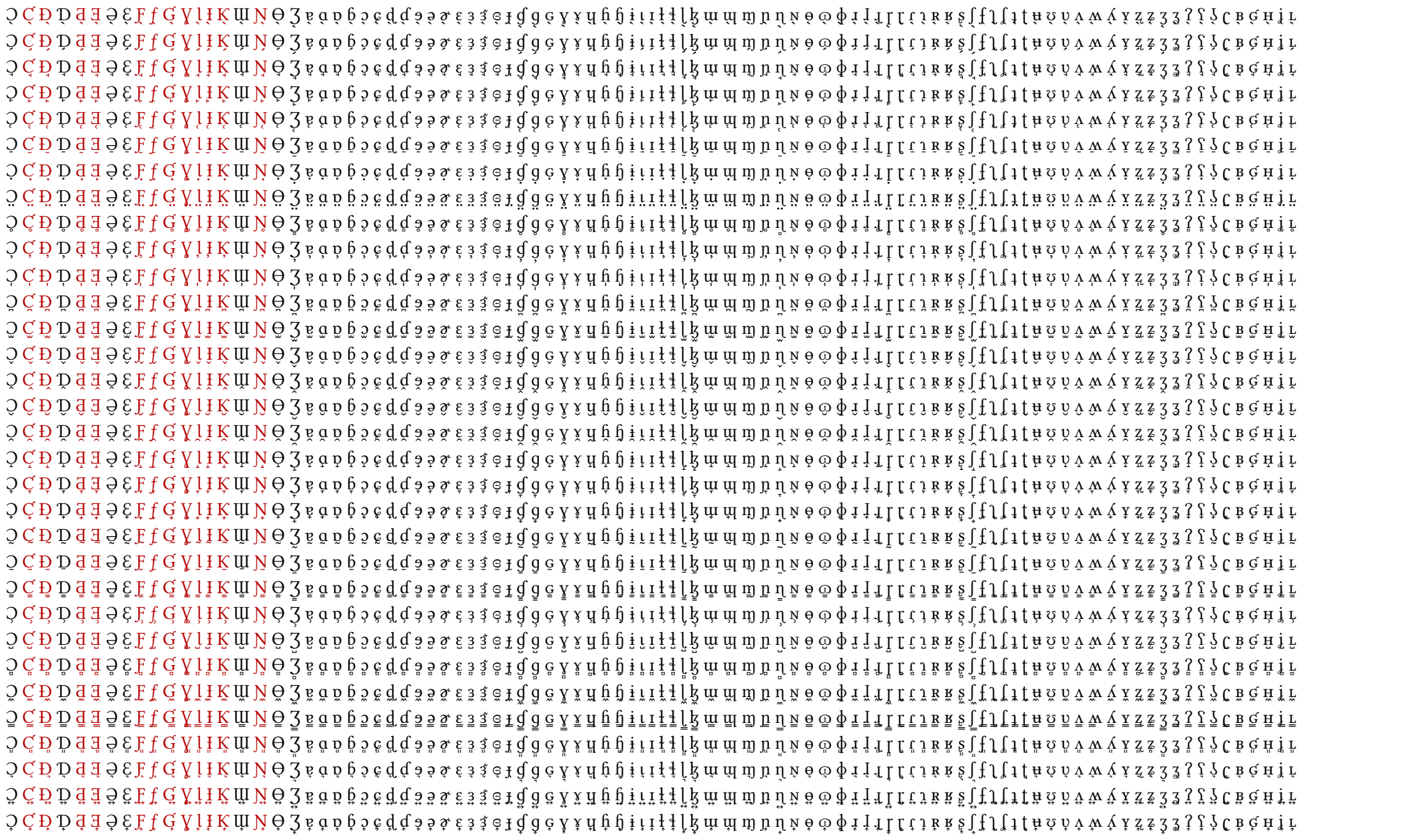
## Semibold italic patched, IPA, Above



## Regular, IPA, Below

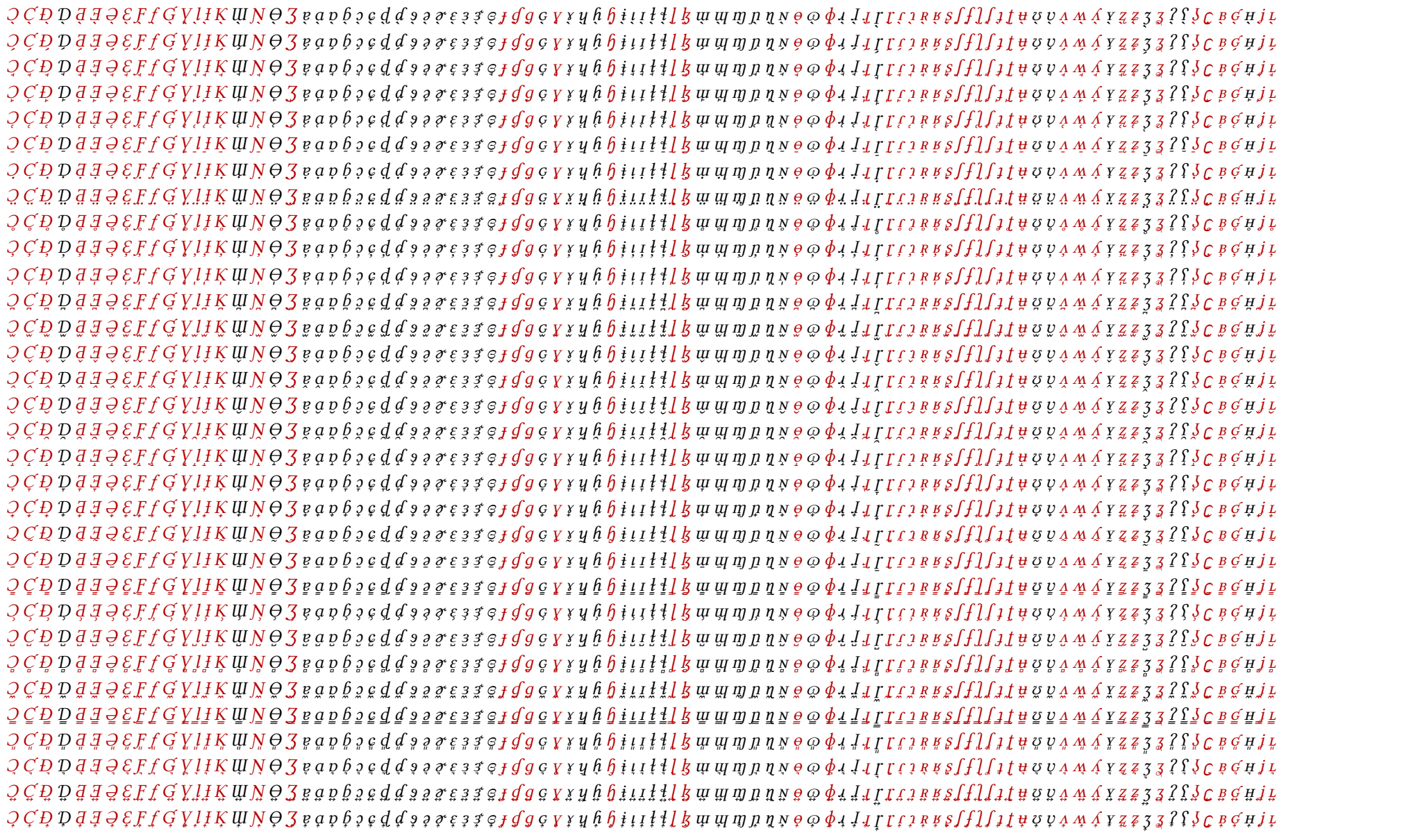


## Regular patched, IPA, Below

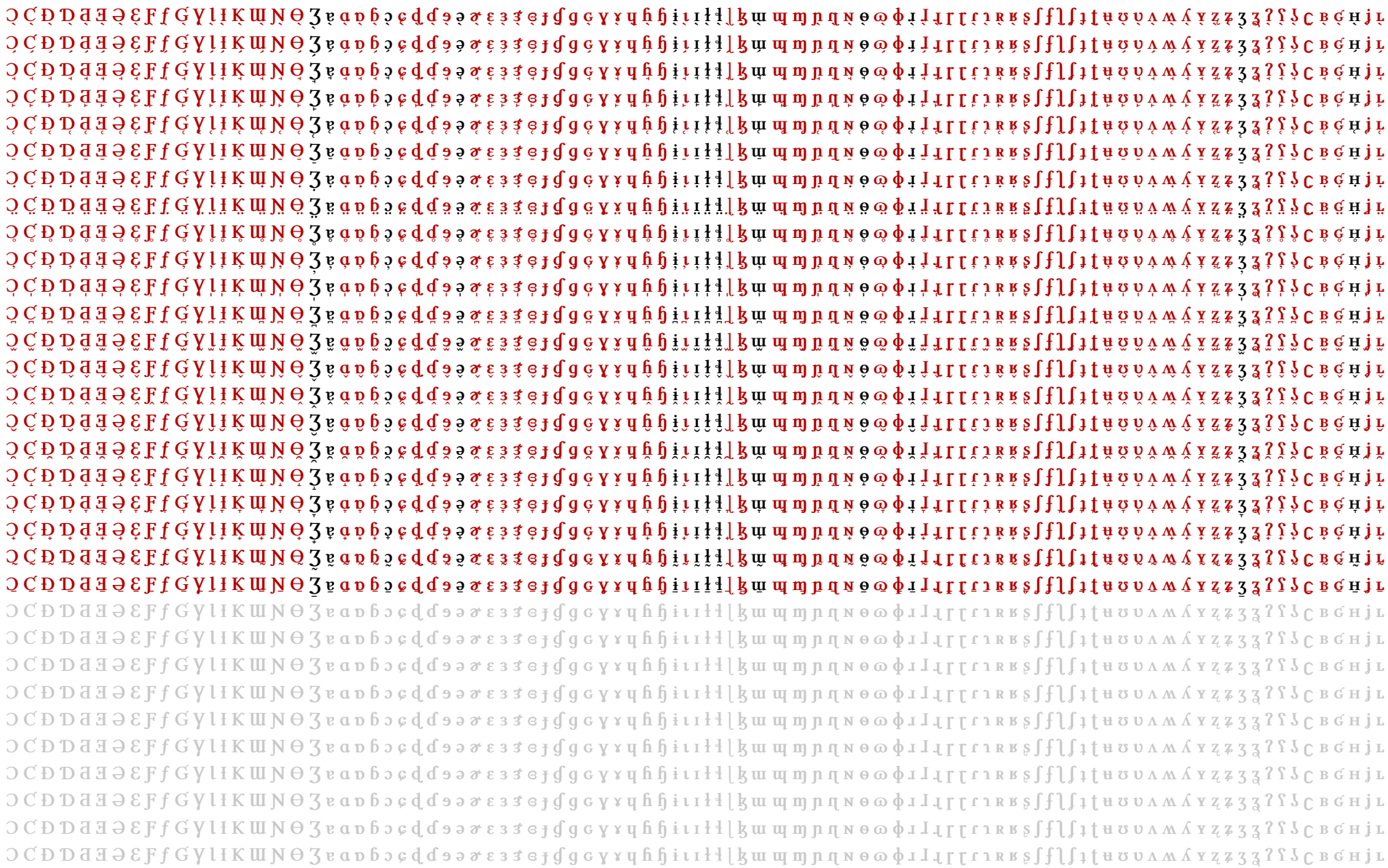


## Italic, IPA, Below

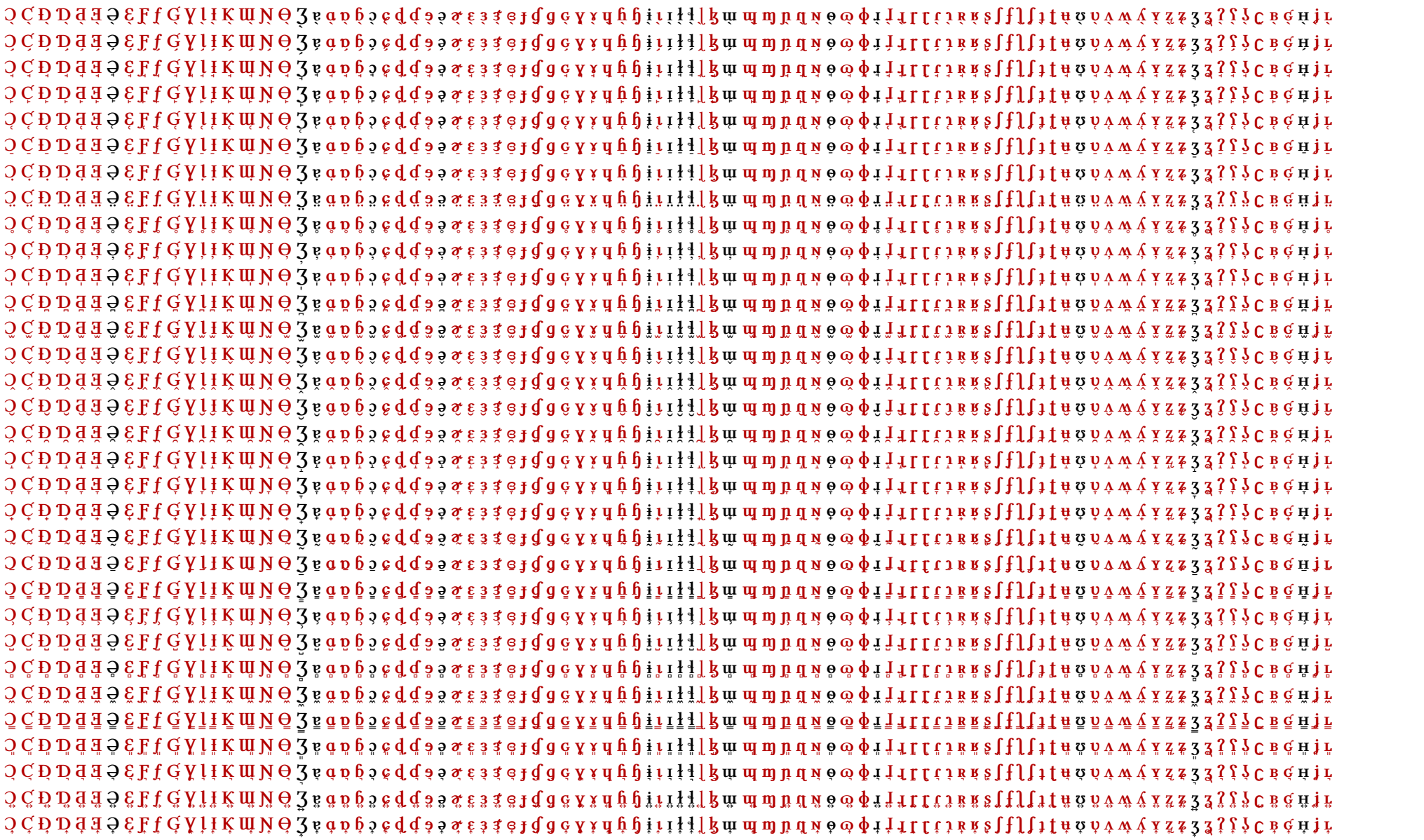
## Italic patched, IPA, Below



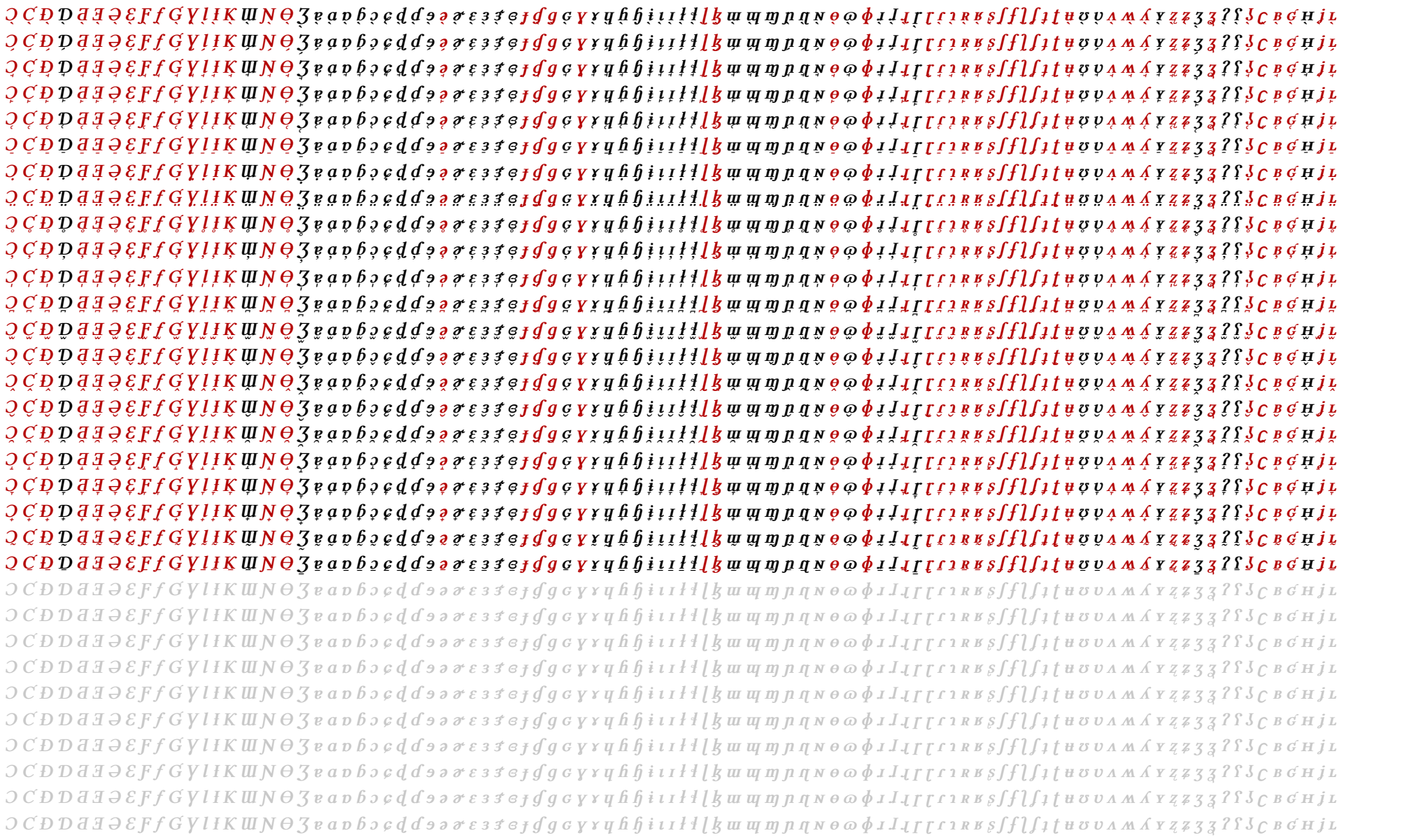
## Semibold, IPA, Below



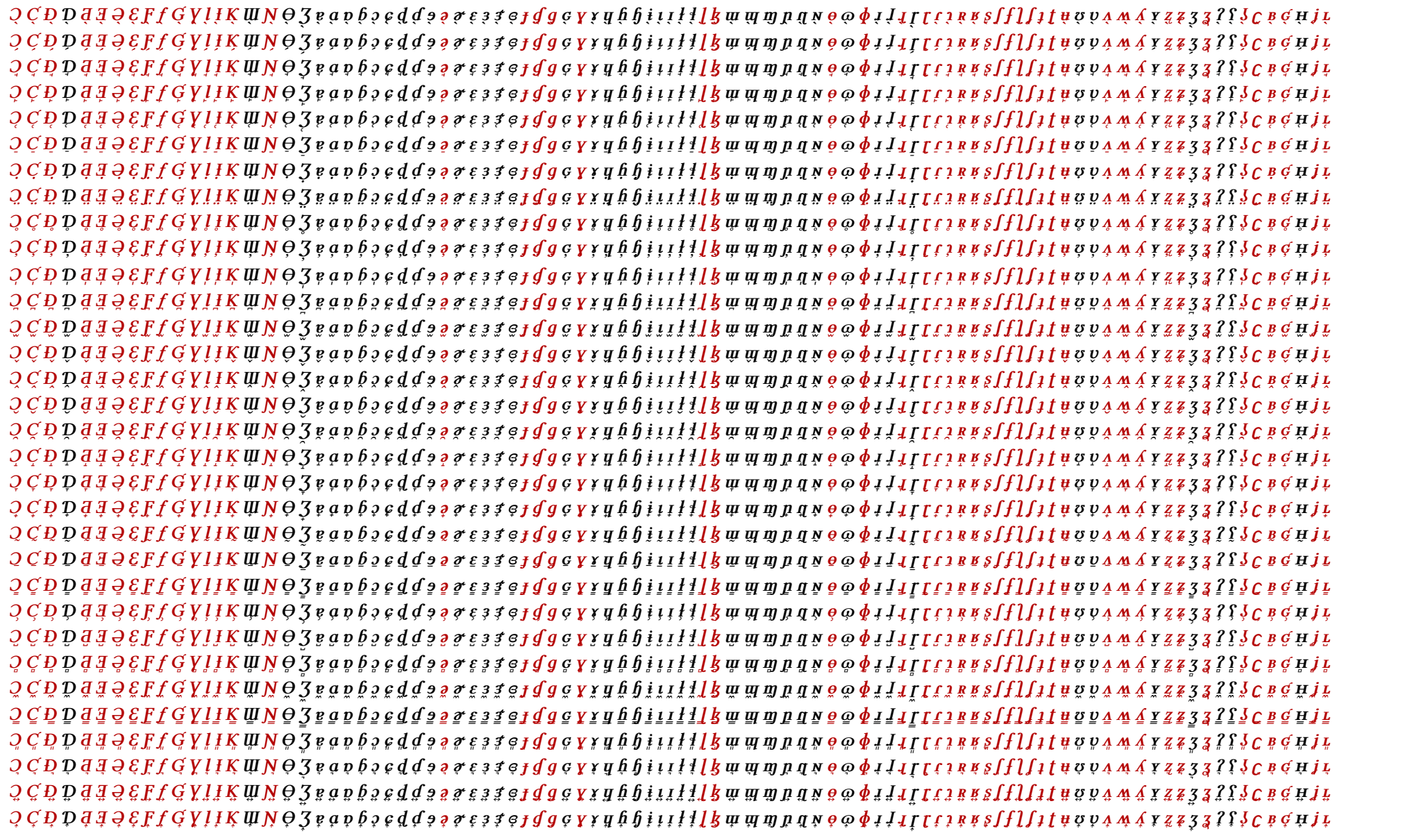
## Semibold patched, IPA, Below



### Semibold italic, IPA, Below



## Semibold italic patched, IPA, Below



# Capital alternates

These grids display capitals with alternate marks.

In the patched fonts, the mark anchor of each capital alternate is set by script (`font_patching_capmarks.py`) from the precomposed capitals A–Z. The $Y$ coordinate is the capital anchor reference (850; 805 in semibold) less the height by which the mark is raised in the precomposed glyphs. The $X$ coordinate of a symmetric mark is the center of its outline on its own axis; for the grave, acute, double acute and double grave it is the median position of the mark in the precomposed glyphs, relative to the midline of the base. In semibold italic both are shifted left by 150 units, as in the semibold italic anchor model. A human-curated anchor overrules the scripted one. Measured over the capitals that have an above-anchor, the median distance of an alternate mark from the midline falls from 142 to 17 units in italic and from 171 to 42 units in semibold italic; regular (16) and semibold (11) are unchanged.

Check these by eye:

- Italic. All ten curated anchors are kept and overrule the script. Their $X$ coordinates differ from the scripted ones by 6 to 62 units: grave $-45$ (script 17), acute $-65$ ($-37$), caron $-50$ ($-29$), inverted breve $-23$ ($-2$), double grave $-47$ ($-34$), breve $-10$ ($-1$), circumflex 16 (24), hook $-27$ ($-19$), dieresis $-75$ ($-69$), double acute $-109$ ($-103$). The curated values fit the designer's capital anchors, which are left of the midline; the scripted values assume an anchor on the midline. If the capital anchors are moved to the midline, the curated values should be dropped.
- Regular. The curated grave $(-175, 723)$ and acute $(-210, 723)$ are kept; the script gives $-162$ and $-218$. The other eight curated anchors are within 4 units of the script and are commented out in `fontanchors_human`.
- Semibold italic. All anchors are scripted. The marks are still about 35 units left of the midline, because the designer's capital anchors are left of the candidates of the semibold italic anchor model; J and L are far off (their base anchors do not conform).
- Semibold. Only J, O, Q, R, S, T, V, W, X, Y, Z have an above-anchor; on the other capitals the alternates are placed by fallback. The hook above is raised by 89 units, and the double acute and double grave are lowered by 30 and 23 units, relative to the designer's anchors.
- The acute, grave and their doubles. The precomposed glyphs place these by eye, letter by letter, so the implied $X$ varies by up to 190 units over the bases (most for the acute, because of L), and one anchor is a compromise.
- `dotaccent.cap` is new in italic and semibold italic, and in regular its anchor is raised by 50 units to match the precomposed dotted capitals.
- Not used as evidence, because the mark outline differs in the precomposed glyph: Ÿ and Ẅ (dieresis), Ľ (comma-style caron), and Ĵ in semibold italic.

## Regular, Capitals, Marks with capital alternates



## Regular patched, Capitals, Marks with capital alternates

À B̀ C̀ D̀ È F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ò P̀ Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Z̀ Ɔ̀ ə̀ ɛ̀ ɵ̀ Ɛ̀ ʊ̀ Ù Ỳ Ƶ̀ Ḛ̀ Ḥ̀ Į̀ Ḛ̀ S̀ T̀ V̀ Ƶ̀ Ạ̀ Ḛ̀ Į̀ Ọ̀ Ụ̀

Á B́ Ć D́ É F́ Ǵ H́ Í J́ Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ú V́ Ẃ X́ Ý Ź Ɔ́ ə́ ɛ́ ɵ́ Ɛ́ ʊ́ Ú Ý Ƶ́ Ḛ́ Ḥ́ Į́ Ḛ́ Ś T́ V́ Ƶ́ Ạ́ Ḛ́ Į́ Ọ́ Ụ́

Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ ə̂ ɛ̂ ɵ̂ Ɛ̂ ʊ̂ Û Ŷ Ƶ̂ Ḛ̂ Ḥ̂ Į̂ Ḛ̂ Ŝ T̂ V̂ Ƶ̂ Ậ Ḛ̂ Į̂ Ộ Ụ̂

Ă B̆ Č D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ L̆ M̆ N̆ Ŏ P̆ Q̆ R̆ Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ɔ̆ ə̆ ɛ̆ ɵ̆ Ɛ̆ ʊ̆ Ŭ Y̆ Ž̆ Ḛ̆ Ḥ̆ Į̆ Ḛ̆ Š T̆ V̆ Ž̆ Ặ Ḛ̆ Į̆ Ọ̆ Ụ̆

Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J̇ K̇ L̇ Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U̇ V̇ Ẇ Ẋ Ẏ Ż Ɔ̇ ə̇ ɛ̇ ɵ̇ Ɛ̇ ʊ̇ U̇ Ẏ Ż̧ Ȩ̰ Ḩ̣ Į̧ Ȩ̰ Ṣ Ṭ V̇ Ż̧ Ạ̇ Ȩ̰ Į̧ Ọ̇ Ụ̇

Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ̈ ə̈ ɛ̈ ɵ̈ Ɛ̈ ʊ̈ Ü Ÿ Z̈ Ḛ̈ Ḧ Ï Ḛ̈ S̈ T̈ V̈ Z̈ Ạ̈ Ḛ̈ Ï Ọ̈ Ụ̈

Å Ḃ C̊ Ḋ Ḛ̊ F̊ Ġ Ḣ Î J̊ K̊ L̊ Ṁ Ṅ Ů Ṗ Q̊ Ṙ Ś T̊ Ů V̊ Ẁ Ẋ Y̊ Ż Ɔ̊ ə̊ ɛ̊ ɵ̊ Ɛ̊ ʊ̊ Ů Y̊ Ż̊ Ḛ̊ Ḥ̊ Į̊ Ḛ̊ Ṣ̊ T̊ V̊ Ż̊ Ạ̊ Ḛ̊ Į̊ Ọ̊ Ụ̊

Ő B̋ C̋ D̋ Ű F̋ G̋ H̋ Ő J̋ K̋ L̋ M̋ N̋ Ő P̋ Q̋ R̋ S̋ T̋ Ű V̋ W̋ X̋ Ÿ Ž̋ Ɔ̋ ə̋ ɛ̋ ɵ̋ Ɛ̋ ʊ̋ Ű Ÿ Ž̋ Ḛ̋ Ḥ̋ Į̋ Ḛ̋ S̋ T̋ V̋ Ž̋ Ạ̋ Ḛ̋ Į̋ Ọ̋ Ụ̋

Ă B̆ Č D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ Ľ M̆ N̆ Ŏ P̆ Q̆ Ř Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ɔ̆ ə̆ ɛ̆ ɵ̆ Ɛ̆ ʊ̆ Ŭ Y̆ Ž̆ Ḛ̆ Ḥ̆ Į̆ Ḛ̆ Š T̆ V̆ Ž̆ Ặ Ḛ̆ Į̆ Ọ̆ Ụ̆

Ȁ B̏ C̏ D̏ Ȅ F̏ G̏ H̏ Ȉ J̏ K̏ L̏ M̏ N̏ Ȍ P̏ Q̏ Ȑ S̏ T̏ Ȕ V̏ W̏ X̏ Ÿ Ž̏ Ɔ̏ ə̏ ɛ̏ ɵ̏ Ɛ̏ ʊ̏ Ȕ Ÿ Ž̏ Ḛ̏ Ḥ̏ Į̏ Ḛ̏ S̏ T̏ V̏ Ž̏ Ạ̏ Ḛ̏ Į̏ Ọ̏ Ụ̏

Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ ə̂ ɛ̂ ɵ̂ Ɛ̂ ʊ̂ Û Ŷ Ƶ̂ Ḛ̂ Ḥ̂ Į̂ Ḛ̂ Ŝ T̂ V̂ Ƶ̂ Ậ Ḛ̂ Į̂ Ộ Ụ̂

## Italic, Capitals, Marks with capital alternates



## Italic patched, Capitals, Marks with capital alternates



## Semibold, Capitals, Marks with capital alternates

À B C D È F G H Ì J K L M Ǹ Ò P Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Ż Ɔ̀ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Á B Ć D É F Ǵ H Í J Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ú V́ Ẃ X́ Ý Ź Ɔ́ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Â B Ĉ D Ê F Ĝ Ĥ Î Ĵ K Ł M N Ô P Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ậ Ệ Ị Ộ Ụ
Ă B C D Ĕ F Ğ H Ĭ J K Ł M N Ŏ P Q̆ R̆ Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ɔ̆ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ặ Ệ Ị Ǒ Ǔ
Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J K Ł Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U V̇ Ẇ Ẋ Ẏ Ż Ɔ̇ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Ä B C D Ë F G̈ Ḧ Ï J K Ł M N Ö P Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Ż Ɔ̈ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Å B C D Ḛ F G̊ Ḥ Ḭ J K Ł M N Ů P Q̊ R̊ S̊ T̊ Ů V̊ W̊ X̊ Y̊ Ż Ɔ̊ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Ő B C D Ẽ F G H Ĩ J K Ł M N Ő P Q̋ R̋ S̋ T̋ Ű V̋ W̋ X̋ Ỹ Ż Ɔ̋ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Ă B Č D Ě F Ğ Ȟ Ĭ J Ǩ Ľ M Ň Ǒ P Q̌ Ř Š Ť Ŭ V̌ W̌ X̌ Y̌ Ž Ɔ̌ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ǒ Ǔ
Ȁ B C D Ȅ F G H Ȉ J K Ł M N Ȍ P Q Ȑ S Ț Ȕ V Ẅ X Ȳ Ż Ɔ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ
Â B C D Ê F G H Î J K Ł M N Ô P Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ Ə Ɛ Ɵ Σ Ʊ Ʊ Ƴ Ƶ Ę Ḥ Į Ṛ Ş Ṭ Ṿ Ẓ Ạ Ẹ Ị Ọ Ụ

## Semibold patched, Capitals, Marks with capital alternates

À B C D È F G H Ì J K L M N Ò P Q R S T Ù V W X Y Z Ɔ Ə Ɛ Ɵ Ɔ̀ Ʊ̀ Ʊ̀ Y Ƶ Ę Ḥ Į Ṛ Ṣ Ṭ Ų̀ Ƶ̀ Ạ Ẹ Ị Ọ Ụ
Á B Ć D É F G H Í J Ḱ Ĺ Ḿ Ń Ó Ṕ Q Ŕ Ś Ť Ú V Ẃ X Ý Ź Ɔ́ Ə́ Ɛ́ Ɵ́ Ɔ́ Ʊ́ Ý Ƶ́ Ę Ḥ Į Ṛ Ṣ Ṭ Ų́ Ƶ́ Ạ Ẹ Ị Ọ Ụ
Â B Ĉ D Ê F Ĝ Ĥ Î Ĵ K Ŀ M N Ô P Q Ŝ Ŝ T Û V Ŵ X Ŷ Ẑ Ɔ̂ Ə̂ Ɛ̂ Ɵ̂ Ɔ̂ Ʊ̂ Ŷ Ƶ̂ Ę Ḥ Į Ṛ Ṣ Ṭ Ų̂ Ƶ̂ Ạ Ệ Ị Ọ Ụ
Ă B C D Ĕ F Ğ H Ĭ J K Ľ M N Ŏ P Q Ř Š Ť Ŭ V Ŵ X Ŷ Ž Ɔ̆ Ə̆ Ɛ̆ Ɵ̆ Ɔ̆ Ʊ̆ Ŷ Ž Ę Ḥ Į Ṛ Ṣ Ṭ Ų̆ Ž Ặ Ẹ Ị Ọ Ụ
Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J K Ŀ Ṁ Ṅ Ȯ Ṗ Q Ṙ Ṡ Ṫ U Ṿ Ẇ Ẋ Ẏ Ż Ɔ̇ Ə̇ Ɛ̇ Ɵ̇ Ɔ̇ Ʊ̇ Ẏ Ż Ę Ḥ Į Ṛ Ṣ Ṭ Ų̇ Ż Ạ Ẹ Ị Ọ Ụ
Ä B C D Ë F G H Ï J K Ľ M N Ö P Q R S T Ü V Ẅ X Ÿ Ž Ɔ̈ Ə̈ Ɛ̈ Ɵ̈ Ɔ̈ Ʊ̈ Ÿ Ž Ę Ḥ Į Ṛ Ṣ Ṭ Ų̈ Ž Ạ Ẹ Ị Ọ Ụ
Å B C D Ḛ F G H Ḭ J K Ḻ Ṃ Ṇ Ǫ P Q Ṛ Ṣ Ṭ Ů V W̊ X Y̊ Ž Ɔ̊ Ə̊ Ɛ̊ Ɵ̊ Ɔ̊ Ʊ̊ Y̊ Ž Ę Ḥ Į Ṛ Ṣ Ṭ Ų̊ Ž Ạ Ẹ Ị Ọ Ụ
Ắ B C D Ḗ F Ḡ Ḫ Ī J Ḱ Ḷ Ḿ Ṉ Ố Ṗ Q Ṛ Ṣ Ṯ Ű V Ẅ X Ӳ Ž Ɔ̋ Ə̋ Ɛ̋ Ɵ̋ Ɔ̋ Ʊ̋ Ӳ Ž Ę Ḥ Į Ṛ Ṣ Ṭ Ų̋ Ž Ạ Ẹ Ị Ọ Ụ
Ă B Č D Ě F Ğ Ȟ Ǐ J Ǩ Ľ M Ň Ǒ P Q Ř Š Ť Ŭ V Ŵ X Ÿ Ž Ɔ̌ Ə̌ Ɛ̌ Ɵ̌ Ɔ̌ Ʊ̌ Ÿ Ž Ę Ḥ Į Ṛ Ṣ Ṭ Ų̌ Ž Ạ Ẹ Ị Ǒ Ǔ
Ȁ B C D Ȅ F G H Ȉ J K Ľ M N Ȍ P Q Ř S T Ȕ V Ẁ X Ȳ Ž Ɔ̏ Ə̏ Ɛ̏ Ɵ̏ Ɔ̏ Ʊ̏ Ȳ Ž Ę Ḥ Į Ṛ Ṣ Ṭ Ų̏ Ž Ạ Ẹ Ị Ọ Ụ
Â B C D Ê F G H Î J K Ŀ M N Ô P Q R S T Û V Ŵ X Ŷ Ẑ Ɔ̂ Ə̂ Ɛ̂ Ɵ̂ Ɔ̂ Ʊ̂ Ŷ Ƶ̂ Ę Ḥ Į Ṛ Ṣ Ṭ Ų̂ Ƶ̂ Ạ Ẹ Ị Ọ Ụ

## Semibold italic, Capitals, Marks with capital alternates



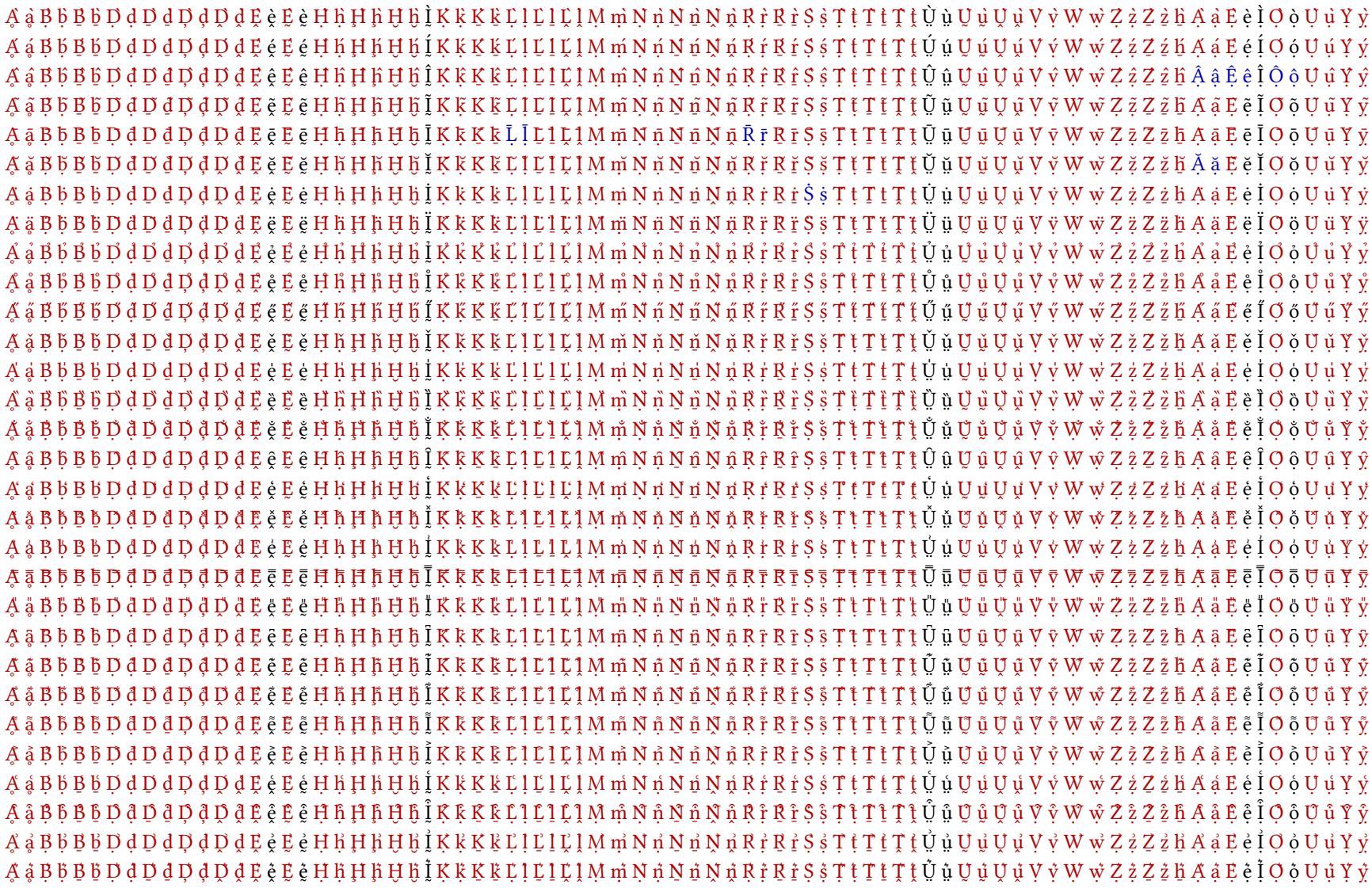
## Semibold italic patched, Capitals, Marks with capital alternates

*À B̀ C̀ D̀ È F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ò P̀ Q̀ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Z̀ Ɔ̀ Ə̀ Ɛ̀ Ɵ̀ Σ̀ Ờ Ừ Ỳ Ƶ̀ Ẹ̀ Ḥ̀ Ị̀ Ṛ̀ Ṣ̀ Ṭ̀ Ṿ̀ Ẓ̀ Ạ̀ Ệ̀ Ị̀ Ọ̀ Ụ̀*
*Á B́ Ć D́ É F́ Ǵ H́ Í J́ Ḱ Ĺ Ḿ Ń Ó Ṕ Q́ Ŕ Ś T́ Ú V́ Ẃ X́ Ý Ź Ɔ́ Ə́ Ɛ́ Ɵ́ Σ́ Ớ Ứ Ý Ƶ́ Ḛ́ Ḥ́ Ḯ Ṛ́ Ṣ́ Ṭ́ Ṿ́ Ẓ́ Ạ́ Ệ́ Ị́ Ọ́ Ụ́*
*Â B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î Ĵ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ Ə̂ Ɛ̂ Ɵ̂ Σ̂ Ơ̂ Ư̂ Ŷ Ƶ̂ Ḛ̂ Ḥ̂ Î Ṛ̂ Ṣ̂ Ṭ̂ Ṿ̂ Ẓ̂ Ậ Ệ̂ Ị̂ Ộ Ụ̂*
*Ă B̆ Č D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ Ǩ L̆ M̆ N̆ Ŏ P̆ Q̆ R̆ Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ɔ̆ Ə̆ Ɛ̆ Ɵ̆ Σ̆ Ơ̆ Ŭ Y̆ Ƶ̆ Ḛ̆ Ḫ̆ Ĭ Ṛ̆ Ṣ̆ Ṭ̆ V̆ Ž̆ Ặ Ệ̆ Ị̆ Ọ̆ Ụ̆*
*Ȧ Ḃ Ċ Ḋ Ė Ḟ Ġ Ḣ İ J̇ K̇ L̇ Ṁ Ṅ Ȯ Ṗ Q̇ Ṙ Ṡ Ṫ U̇ V̇ Ẇ Ẋ Ẏ Ż Ɔ̇ Ə̇ Ɛ̇ Ɵ̇ Σ̇ Ơ̇ Ư̇ Ẏ Ƶ̇ Ḛ̇ Ḣ İ Ṛ̇ Ṡ Ṫ V̇ Ż Ạ̇ Ệ̇ Ị̇ Ọ̇ Ụ̇*
*Ä B̈ C̈ D̈ Ë F̈ G̈ Ḧ Ï J̈ K̈ L̈ M̈ N̈ Ö P̈ Q̈ R̈ S̈ T̈ Ü V̈ Ẅ Ẍ Ÿ Z̈ Ɔ̈ Ə̈ Ɛ̈ Ɵ̈ Σ̈ Ơ̈ Ü Ÿ Ƶ̈ Ḛ̈ Ḧ Ï Ṛ̈ Ṡ̈ Ṫ̈ V̈ Ż̈ Ạ̈ Ệ̈ Ị̈ Ọ̈ Ụ̈*
*Ả B̉ C̉ D̉ Ẻ F̉ Ġ Ḥ Ỉ J̉ K̉ L̉ M̉ N̉ Ỏ P̉ Q̉ Ṛ Ṣ Ṭ Ủ V̉ Ẃ X̉ Ỷ Ż Ɔ̉ Ə̉ Ɛ̉ Ɵ̉ Σ̉ Ở Ủ Ỷ Ƶ̉ Ḛ̉ Ḥ̉ Ỉ Ṛ̉ Ṣ̉ Ṭ̉ V̉ Ẓ̉ Ạ̉ Ệ̉ Ị̉ Ọ̉ Ụ̉*
*Ấ B̋ C̋ D̋ É F̋ G̋ H̋ Í J̋ K̋ L̋ M̋ N̋ Ő P̋ Q̋ R̋ S̋ T̋ Ű V̋ W̋ X̋ Ÿ Z̋ Ɔ́ Ə́ Ɛ́ Ɵ́ Σ́ Ő Ű Ÿ Ƶ́ Ḛ́ Ḧ̋ Ḯ Ṛ̋ Ṣ̋ Ṭ̋ V̋ Ẓ̋ Ạ̋ Ệ̋ Ị̋ Ọ̋ Ụ̋*
*Ắ B̆ C̆ D̆ Ĕ F̆ Ğ H̆ Ĭ J̆ K̆ L̆ M̆ N̆ Ŏ P̆ Q̆ R̆ Š T̆ Ŭ V̆ W̆ X̆ Y̆ Ž Ɔ̆ Ə̆ Ɛ̆ Ɵ̆ Σ̆ Ơ̆ Ŭ Y̆ Ƶ̆ Ḛ̆ Ḫ̆ Ĭ Ṛ̆ Ṣ̆ Ṭ̆ V̆ Ž̆ Ặ Ệ̆ Ị̆ Ọ̆ Ụ̆*
*Ầ B̀ C̀ D̀ Ề F̀ G̀ H̀ Ì J̀ K̀ L̀ M̀ Ǹ Ồ P̀ Q̀ Ṛ̀ Ṣ̀ T̀ Ừ V̀ Ẁ X̀ Ỳ Ż Ɔ̀ Ə̀ Ɛ̀ Ɵ̀ Σ̀ Ờ Ừ Ỳ Ƶ̀ Ḛ̀ Ḥ̀ Ì Ṛ̀ Ṣ̀ Ṭ̀ V̀ Ẓ̀ Ạ̀ Ệ̀ Ị̀ Ọ̀ Ụ̀*
*Ấ B̂ Ĉ D̂ Ê F̂ Ĝ Ĥ Î J̀ K̂ L̂ M̂ N̂ Ô P̂ Q̂ R̂ Ŝ T̂ Û V̂ Ŵ X̂ Ŷ Ẑ Ɔ̂ Ə̂ Ɛ̂ Ɵ̂ Σ̂ Ơ̂ Ư̂ Ŷ Ƶ̂ Ḛ̂ Ḥ̂ Î Ṛ̂ Ṣ̂ Ṭ̂ V̂ Ẓ̂ Ậ Ệ̂ Ị̂ Ộ Ụ̂*

## Precomposed characters

These grids display precomposed characters with referenced glyphs, where a precomposed character can inherit an anchor from the references glyph.

## Regular, Precomposed bases that can inherit an above anchor, Above



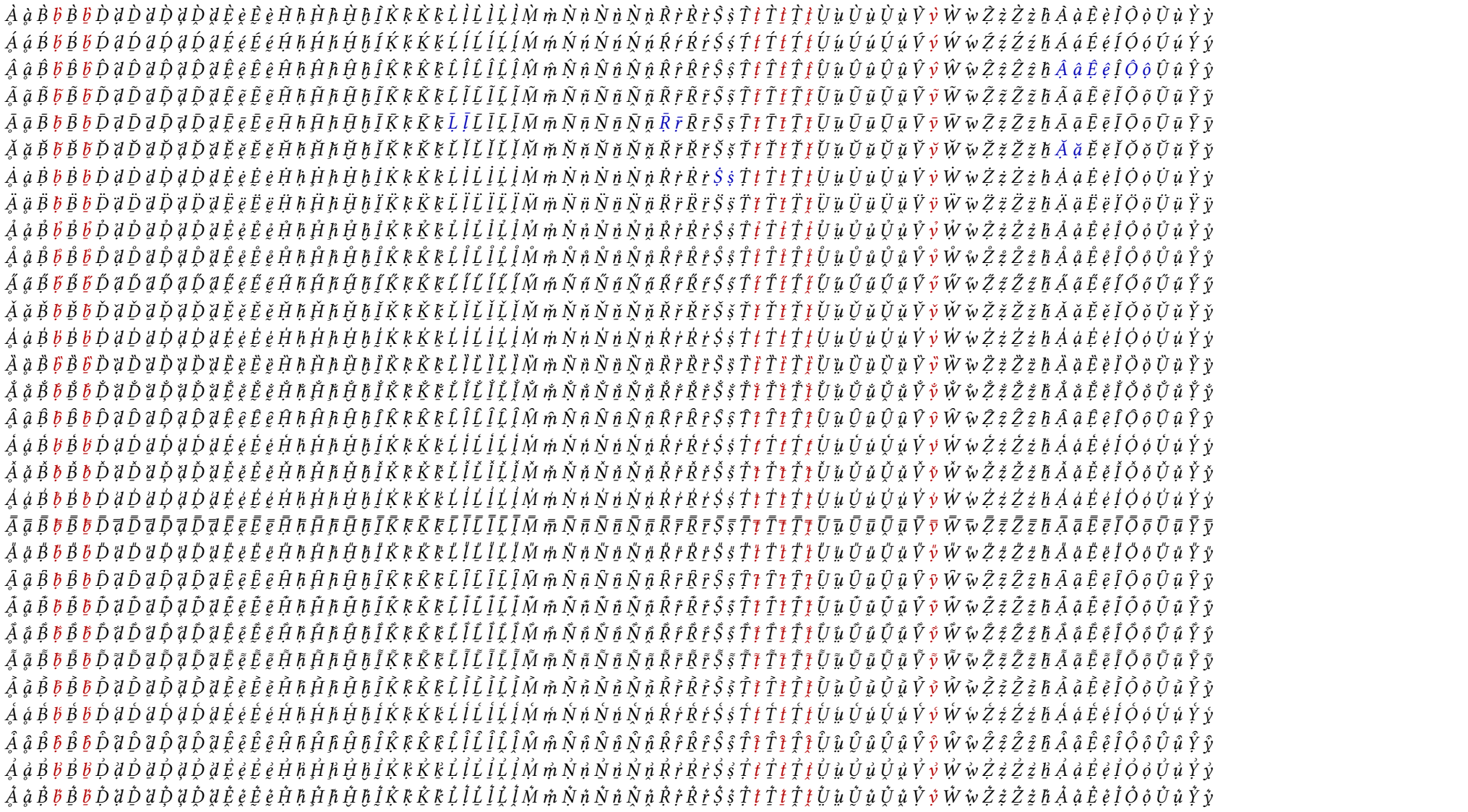
## Regular patched, Precomposed bases that can inherit an above anchor, Above



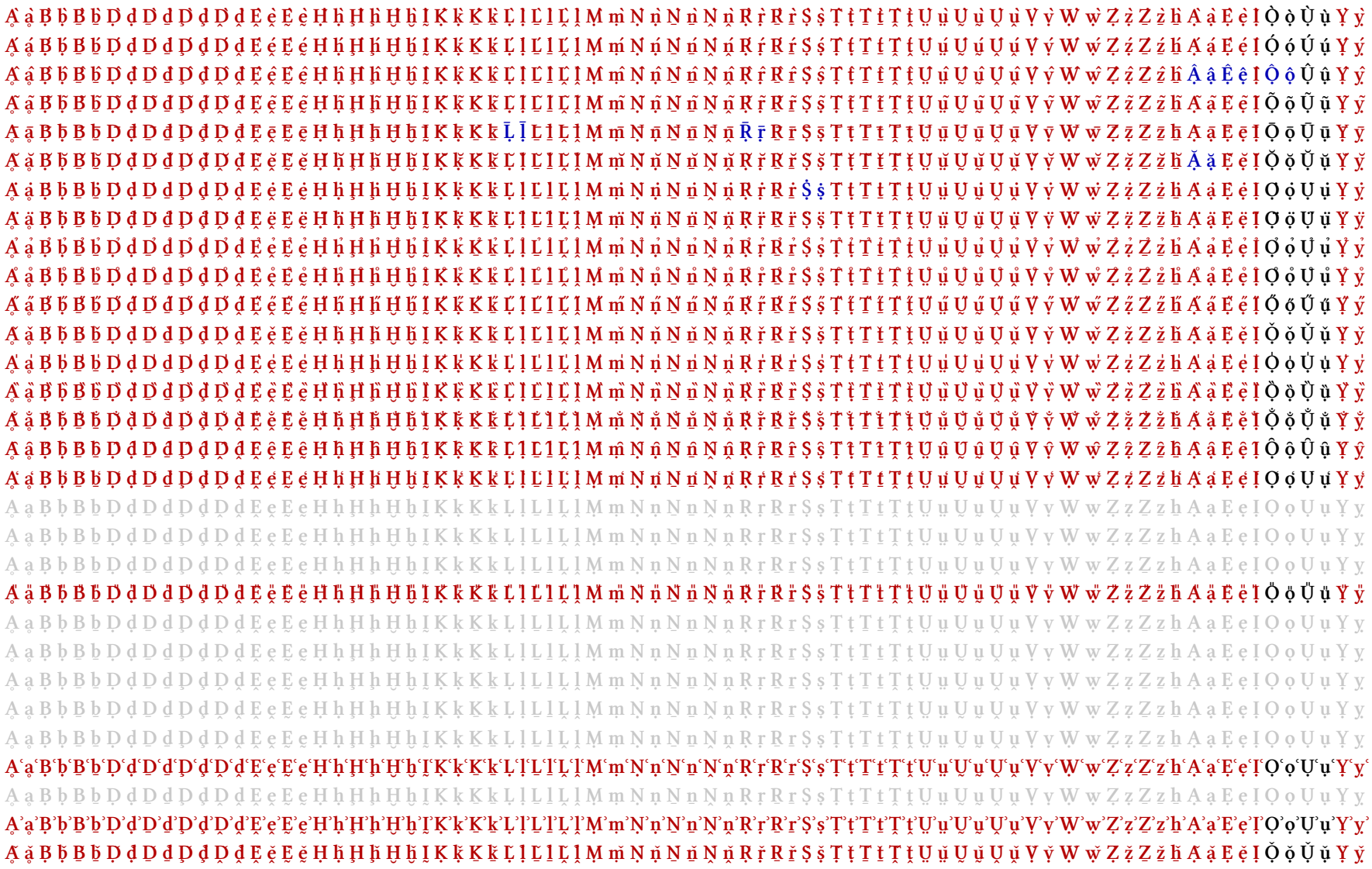
## Italic, Precomposed bases that can inherit an above anchor, Above



## Italic patched, Precomposed bases that can inherit an above anchor, Above



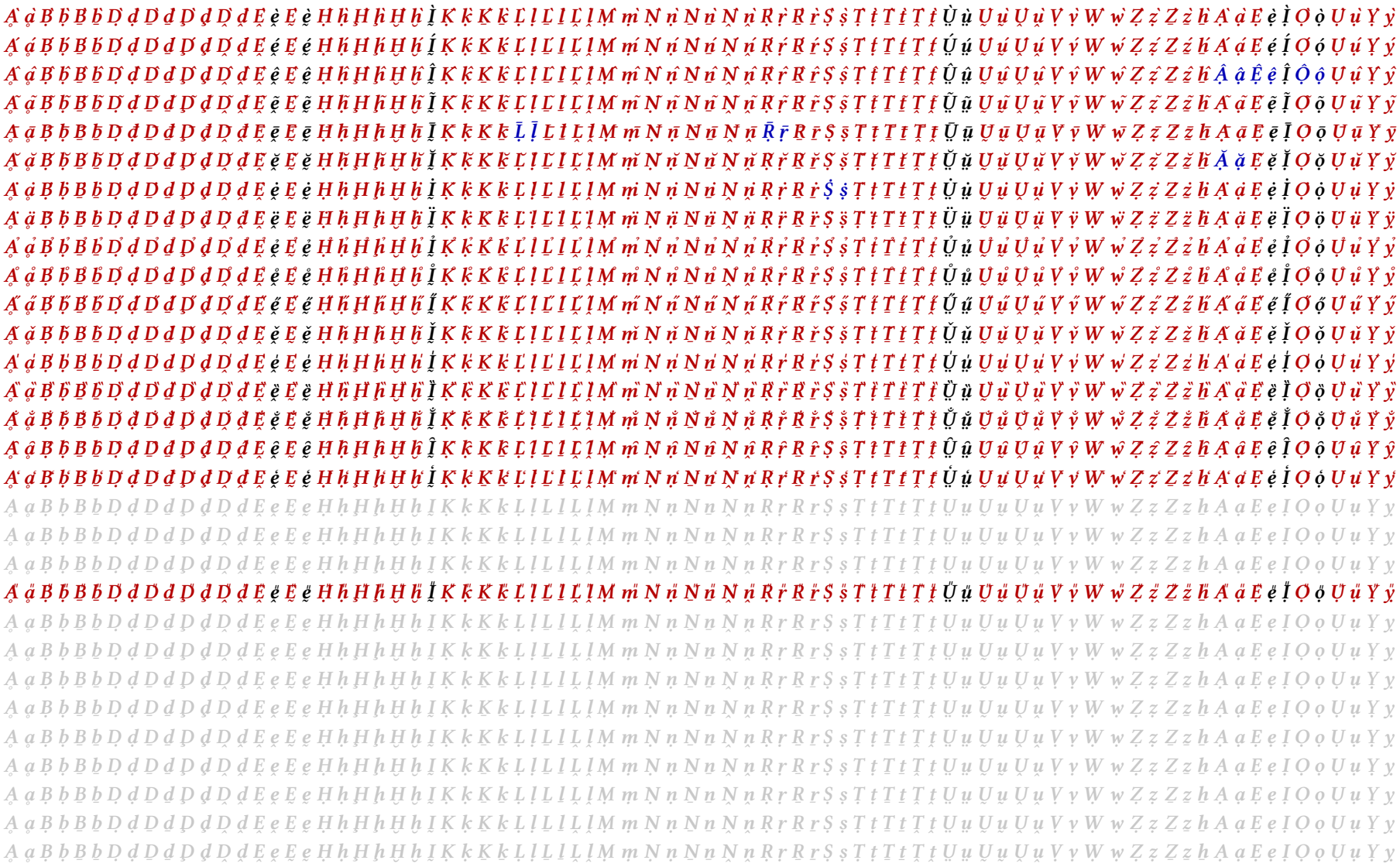
## Semibold, Precomposed bases that can inherit an above anchor, Above



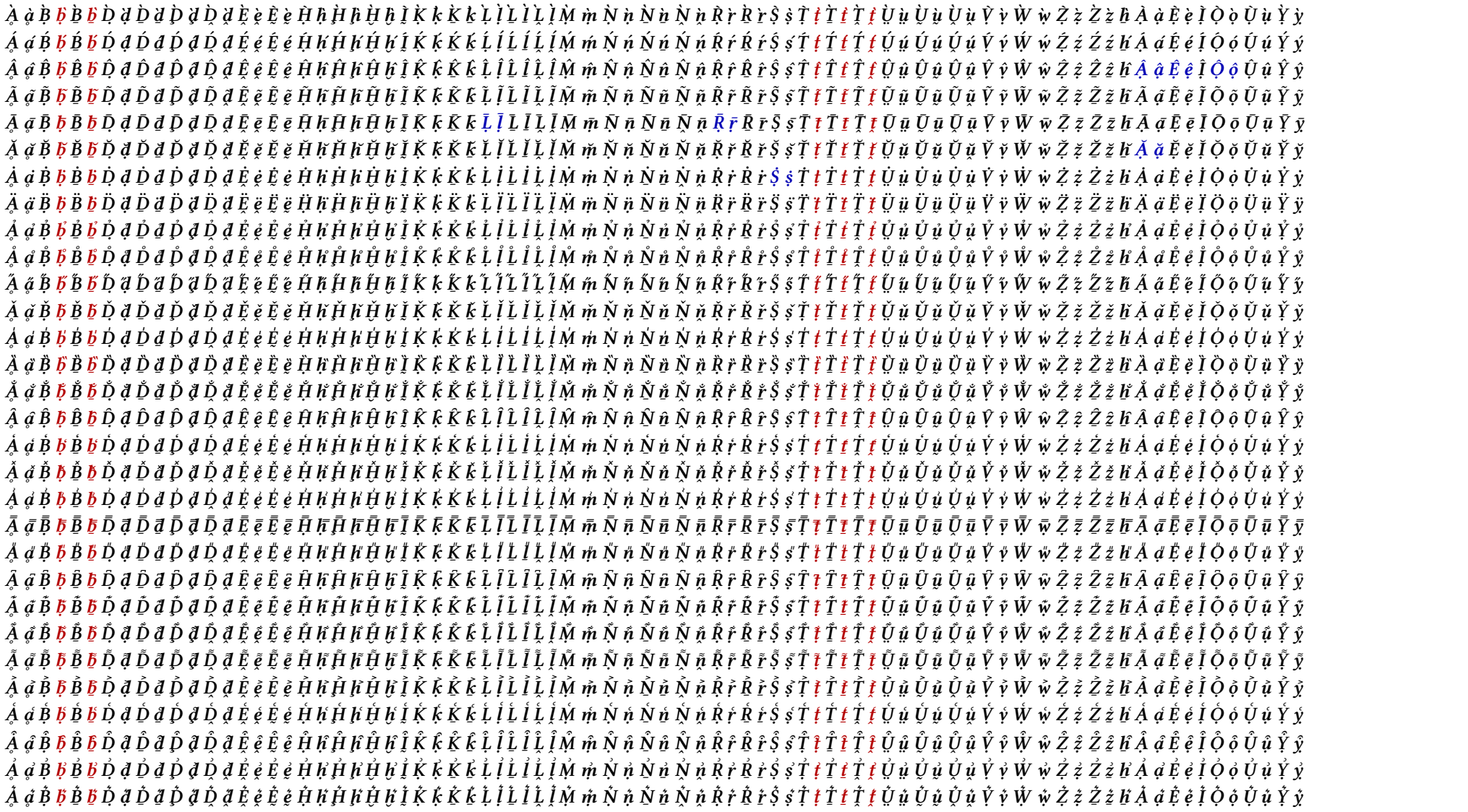
## Semibold patched, Precomposed bases that can inherit an above anchor, Above



## Semibold italic, Precomposed bases that can inherit an above anchor, Above



## Semibold italic patched, Precomposed bases that can inherit an above anchor, Above



## Regular, Precomposed bases that can inherit a below anchor, Below



Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṧ ṧ Ṭ ṭ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ Ⱬ ⱬ Ⱳ ⱳ Ⱶ ⱶ ⱷ ⱷ ⱸ ⱸ ⱹ ⱹ ⱺ ⱺ ⱻ ⱻ ⱼ ⱼ ⱽ ⱽ Ⱦ ⱦ Ⱨ ⱨ Ⱪ ⱪ

## Regular patched, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṯ ṯ Ṳ ṳ Ṷ ṷ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ ⁀

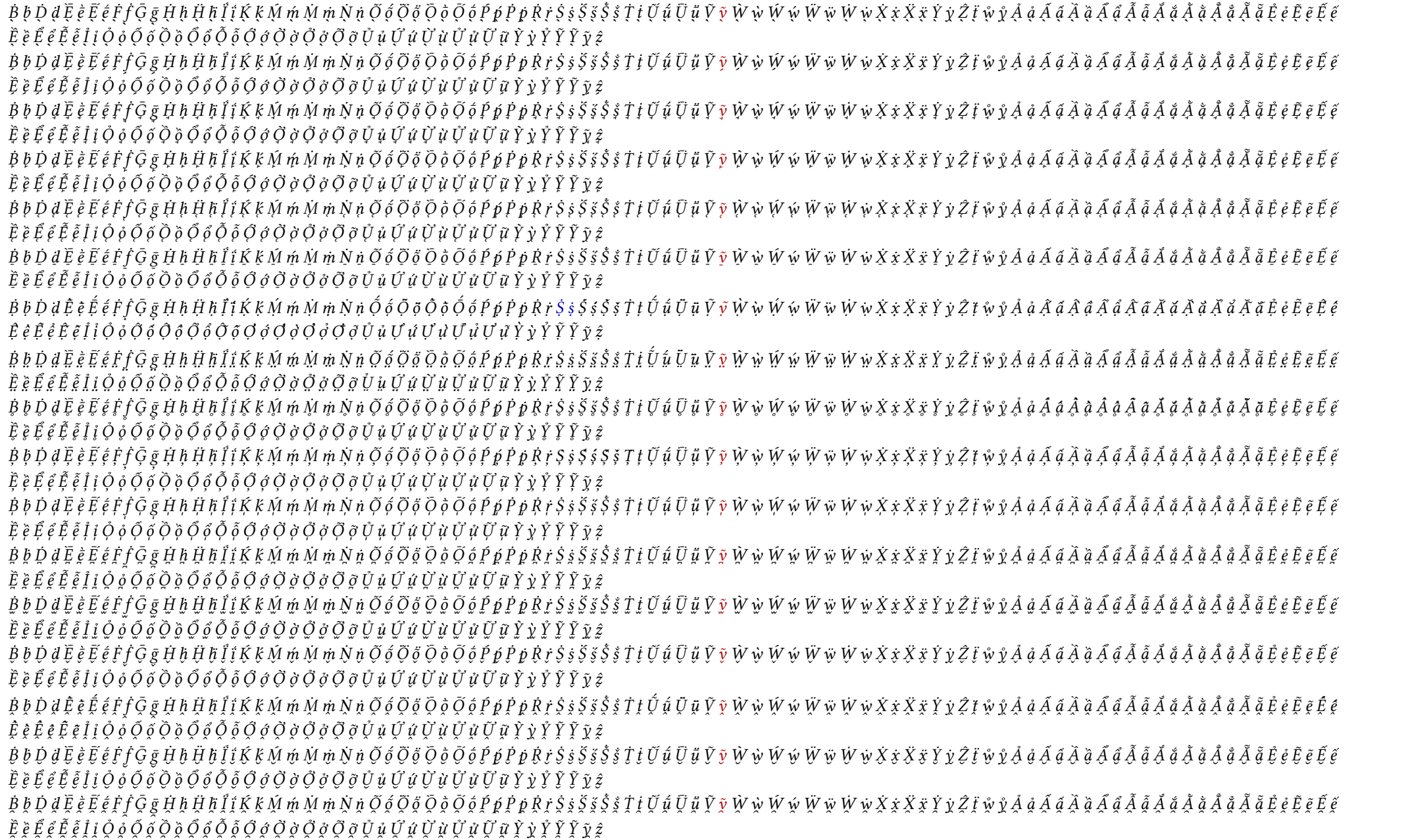
Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṣ ṣ Ṩ ṩ Ṩ ṩ Ṭ ṭ Ṵ ṵ Ṳ ṳ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẁ ẁ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ
Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ ἀ

## Italic, Precomposed bases that can inherit a below anchor, Below



Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṫ ṫ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế
Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

## Italic patched, Precomposed bases that can inherit a below anchor, Below



Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṍ ṍ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṫ ṫ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế
Ề ề Ể ể Ễ ễ Ệ ị Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỿ ỿ

## Semibold, Precomposed bases that can inherit a below anchor, Below



Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṫ ṫ Ṵ ṵ Ṷ ṷ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

## Semibold patched, Precomposed bases that can inherit a below anchor, Below

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṙ ṙ Ṣ ṣ Ṩ ṩ Ṯ ṯ Ṳ ṳ Ṷ ṷ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ểỄ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ
Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṙ ṙ Ṣ ṣ Ṩ ṩ Ṯ ṯ Ṳ ṳ Ṷ ṷ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể 

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḷ ḷ Ḹ ḹ Ṃ ṃ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṙ ṙ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṫ ṭ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

**Semibold italic, Precomposed bases that can inherit a below anchor, Below**

Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḵ ḵ Ḹ ḹ Ṃ ṃ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṭ ṭ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ Ẹ ẹ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ ẑ

Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḣ ḣ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṅ ṅ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṫ ṫ Ṭ ṭ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ṿ ṿ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ Ẓ ẓ Ẕ ẕ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḣ ḣ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṅ ṅ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṩ ṩ Ṫ ṫ Ṭ ṭ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ṿ ṿ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ Ẓ ẓ Ẕ ẕ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỿ ỿ

## Semibold italic patched, Precomposed bases that can inherit a below anchor, Below

Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṩ ṩ Ṯ ṯ Ṻ ṻ Ṹ ṹ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ
Ặ Ẹ ẹ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṫ ṫ Ṻ ṻ Ṳ ṳ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

Ḃ ḃ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḳ ḳ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṫ ṫ Ṻ ṻ Ṳ ṳ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẘ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ Ỵ ỵ

1. ”i” not losing its dot when there is a tone diacritic AND a tilde below

   i̱ ı́ ı̱

   ı̱́

   ı̱̀

2. troubles with capital I

   I̱ I̱ I̱

   Solution: Use combnining grapheme joiner (CGJ)

   I̱ I̱ I̱

3. double acute critic shifted right in bold face

   ***á a̋ ą ą́ ą̋ ą̋***

   Solution: Patch anchor of double acute

   á a̋ ą ą́ ą̋ ą̋ **á a̋ ą ą́ ą̋ ą̋** ***á a̋ ą ą́ ą̋ ą̋***

4. wrong tilde placement together with caron (and grave? and acute? and double grave? OK, it seems it’s rather about vowels...) with some vowels

   ą̌ ą̌ **ą̌ ą̌**

   ų̀ **ų̀ ų ų̀**

   Solution: Use combnining grapheme joiner (CGJ)

   ą̌ ą̌ **ą̌ ą̌**

   ų̀ ų̀ **ų̀ ų̀**

5. wrong acute placement in bold face

   é *é* **é** ***é***

   á *á* **á** ***á***

   ó *ó* **ó** ***ó***

   ú *ú* **ú** ***ú***

   è *è* **è** ***è***

   à *à* **à** ***à***

   ò *ò* **ò** ***ò***

   ù *ù* **ù** ***ù***

   Solution: Patch above-anchor of acute and grave in (semi)bold.

   é *é* **é** ***é*** á *á* **á** ***á***

   ó *ó* **ó** ***ó*** ú *ú* **ú** ***ú***

   è *è* **è** ***è*** à *à* **à** ***à***

   ò *ò* **ò** ***ò*** ù *ù* **ù** ***ù***

6. italic examples with tilde below

*Flą̰*

+ *Flą̰*

*gṵ̀ ą̰*

*gṵ̀ą̰*

+ *gṵ̀ą̰*

*lɛ̰̃ ɔ̰*

+ *lɛ̰̃ ɔ̰*

*mṵ*

*kṵ̃,*

*kṵ̃*

+ *kṵ̃*

*nṵ̃*

+ *nṵ̃*

*nɔ̰̌*

*nṵ̀*

*nṵ̀*

*N-ą̰*

*vɔ̰ṵ̃ṵ̃*

*fɔ̰ṵ̃fɔ̰ṵ̃*

*lɛ̰̃ ɔ̰*

***Flą̰***

***gṵ̀ ą̰***

***gṵ̀ą̰***

***lɛ̰̃ ɔ̰***

***lɛ̰̃ ɔ̰***

***mṵ***

***kṵ̃,***

***kṵ̃***

***nṵ̃***

***nɔ̰̌***

***nɔ̰̌***

***nù̱***

***nù̱***

***N-ą̆***

7. Ɛ̏ bold italics

   Ɛ̏

   Ɛ̀ Ɛ̏ **Ɛ̀ Ɛ̏**

   Solution: Set above-anchor for open-E, patch GSUB to use capital alternate mark glyph ”uni”